



Spec book

SPB-EE

VF MINIVAN Body_Control_Module
Specifcation Book(SPB)

Release

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Review & Approval

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Revision history

Version	Change Description	Responsible person	Date
1.0	2025.04.14 New Release 2025.04.30 1.Vehicle cancellation of Theft-alarm、Immobilizer、Remote keyfob, corresponding to the modification of the relevant description 2.Correspondence to FIP 2025.06.03 1. Modifying milestones in gates and milestones 2.Modify ACC & IG relay status in SF-25-3 ①ACC Relay Status modify Closed ②IG Relay Status modify Closed 3. Add contact information for Vinfast and Hongfa leaders to the project contact list. 4. SF-19-1 Adding a Door Lock Locking Function Condition 5. Modify door status requirements in SF-19-2 Functional Operating Conditions 6. Replace the description OFF functional closure in SF-26-4 from “four doors” to “three doors”. 7. SF-19-7 Remove Tailgate Unlock Description	Zhang Haichao	14.06.2025

	8. Adding Gate Status Signal Descriptions to SF-19-2 Notes 9. Add description of filtering logic for MIST vs. Wash in SF-22-1 10. Modifying the SF-22-2 Front Wiper Service Mode Logic and Signals 11. Modifying 1.1.10 Pin Assignment and Connector 12. Add Mechanical Lock Status Signal Description Note to SF-19-3 Comments 13. Number changed from VFDSADEEP0009 to VFDSANEEP0009 14. Modify wiper maintenance mode startup and shutdown logic to add time monitoring 15. Add ACC relay failure does not affect ON relay operating status in SF-25-3 16. Add the condition "Cancellation during HV on = TRUE" in SF-27-4 2025.06.05 1. .Add signal BCM_SwitchSts_0x109:BCM_STAT_WiperStalk SW2 2. Update Signal Value Description of signal BCM_STS_wiper 3. Rename signal BCM_STS_Driverwindow_switch to BCM_STS_Leftwindow_switch, and update signal value description 4. Rename signal BCM_STS_Passengerwindow_switch to BCM_STS_Rightwindow_switch, and update signal value description 5. Update Signal Value Description of signal BCM_STS_WindowL 6. Update Signal Value Description of signal BCM_STS_WindowR 2025.06.06 1. Adding a mechanical door lock state machine 2. Add Ignition Lock Schematic 3. Modifying the CAN Matrix Diagram 2025.06.12		
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	Determine the V1, T1, and T2 values in the MF-27 LV battery top-up strategy 2025.06.14 Modify the signal value of ID269-VCU_ActualGear in SF-25-3 and add "N" judgment signal in T20.		
2.0	2025.06.23 (blue font) 1.Delete the BCM_STS_SideDoor signal from the MF-19 Output CAN signal clock. 2. Add critical alarm switch control description to SF-21-1 Functional Operation Outputs 3. The STAT_DoorUnlockFL signal is changed to "1" in item (iii) of the functional operation output in SF-19-1. 5.Modify the STAT_DoorUnlockFL signal value in the SF-19-2 Functional Operation Output Description 6. Modification of BCM_STS_mechanicallock_switch to door lock status switch signal in SF-19-3 7. SF-19-5: 1) Delete the description of side doors in SF-19-5 2) Change STAT_DoorUnlockFL to STAT_DoorLockFL in the Function Operation Output 3) Add the center lock working indicator control output signal to the Function Operation Output 4) Delete the VCU_ActualGearValid=0 "Valid" signal in the Function Operation Shutdown 8. Modify the Hazardous Switching Function Definition Description in the SF-21-1 Functional Operation Outputs 8.Adding wiper low speed signal to SF-22-2 function operation outputs 8. Adding Motor Operating Modes to SF-22-2 Functional Maintenance Modes 9. Adding BCM wiper low-speed drive output signal to SF-22-3 function operation outputs 10. Remove the description of the SS Front_Wash signal from the SF-22-3 NOTE. 11. Update voltage mode management table	Zhang Haichao	02.07.2025

	12. Add IG1 (ACC) Relay Status Feedback Signal Description and IG2 (ON) Relay Status Feedback Signal Description to the SF-25-3 note. 13. Modify Ignition Lock and BCM Schematic in SF-25-3 14. Add actual power status source description to SF-25-3 NOTE 15. Add seatbelt latch, SBR valid signal description to MF-28 16. Change BCM_STS_mechanicallock_switch name to BCM_STS_DOOR LOCK STATUS SW 17. LV battery top-up serial number adjusted, logic unchanged 18. Add detailed functional description explanation 19. Add "Prevent Key Abuse" logic in SF-25-3 Note		

Component summary

[\[Insert of table the contents here\]](#)

SpecBook Number: VFDSANEEP0009	Component DVP Number: DVPRANEEP0009	
SOR Number: VFSRANEEP0009	Full vehicle DVP Number: NA	
Part Number / Rev: EEP64011100AA	Component: BCM	
Written By	Zhang Haichao	Zhang_haichao_tj@launchdesign.com.cn
Authorised By	Li Xiaobin	Li_xiaobin_tj@launchdesign.com.cn
Project	Mini VAN	
Module Group	EE	
Build to Print or Full Service Supplier	FSS	
Supplier	NA	
Material	Refer to drawing 参考图纸	
Component Weight	450g	
Manufacturing Process	NA	
Surface	NA	
Colour / Type of grain(颜色/纹理类型)	NA	

Specification summary 规范总结

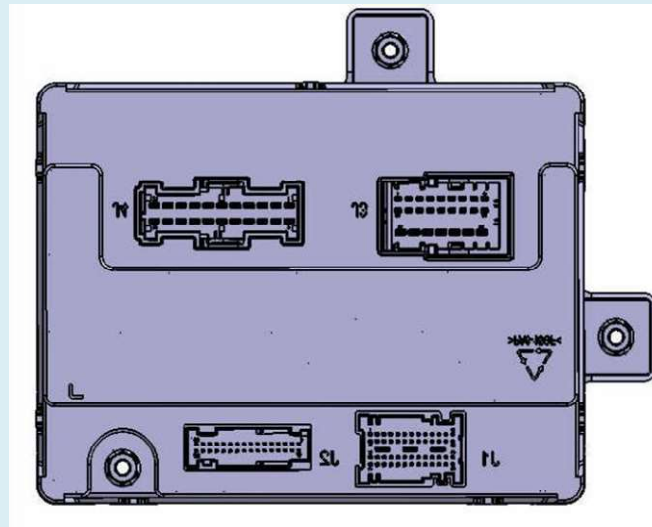
Technical requirements summary (1)技术要求概述 <i>Refer to DVP for Targets and acceptance criteria 参考 DVP 目标和验收标准</i>	
Material 材料	<i>Refer to drawing 参考图纸</i>
Functional 功能 / Performance 性能	<i>Refer to drawing</i>
Part quality 零部件质量	<i>Refer DVP</i>
Strength 强度	<i>NR</i>
Durability 耐用 性/Fatigue 疲劳	<i>Refer DVP</i>
Durability /Impact 冲击	<i>Refer DVP</i>
Durability /Corrosion 腐蚀	<i>Refer DVP</i>
Safety – Security 安全保护	<i>NR</i>
Thermal and aero 热分析	<i>NR</i>
EMC	<i>Refer DVP</i>
NVH	<i>NR</i>
Safety 安全	<i>NR</i>
Serviceability 适 用性	<i>NR</i>

Scope and purpose 范围及目的

VinFast (VinFast Trading and Production Limited Liability Company) have started to design and develop new vehicles MiniVAN Environmental impact and sustainability of the program are taken into account as essential target features. This SPB shall provide details for BCM components for mass production.

Number of SPB follow VFDST00010202.

BCM(Body_Control_Module)



Responsible Contact: [Zhang Haichao](#)

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1. General Document Guidelines 一般文件指南

a. Purpose and Legal Binding. 目的和法律约束力

This version is legally binding. 此版本具有法律约束力

This document is based on the VDA component requirement specification structure and VF standard.

本文件基于 VDA 组件需求规格结构和 VF 标准。

Requirement documents are divided into three levels, which are complementary to each other. All three levels are legally binding. If there is a conflict with regard to content, the higher level is applicable (e.g. level 3 vs level 2 □ level 2 is applicable).

需求文件分为三个层次，相互补充。所有三个级别都具有法律约束力。如果在内容上有冲突，则适用较高级别的要求（例如，第 3 级与第 2 级 □ 第 2 级适用）。

b. Language 语言

This document is written in English. 本文件以英文撰写。

All official project documentations have to be done in English. If documents are bilingual, the English text is binding. 所有正式项目文件必须使用英文。如果文件为中英文对照，则以英文文本为准。

c. Obligation of Requirement Wording.要求措辞的义务

The following wording rules are used throughout this document. 本文件通篇使用以下措辞规则。

Shall

The word “shall” is used to state a binding requirement. Such requirements are externally verifiable by manipulation or observation.

“Shall”一词用于说明具有约束力的要求。此类要求可通过操作或观察进行外部验证。

Should

The word “should” is used for preferences or desires in implementation that are not strictly binding.

“Should”用于表示实施中的偏好或愿望，但不具有严格的约束力。

May

The word “may” is used to denote options that are left to the implementer.

“May”用于表示留给实施者的选择。

i. Vehicle profile 车辆概况

Project No	Mini VAN	Front Trunkcapacity - BEV	NA
Body style	Mini commercial car	Minimum turning Circlediameter (m)	9.8
Curb mass	830kg	Payload (Kg)	650
Passenger	2	Roof load (Kg)	NA
Tire	≥ 165 R14	Maxpower(kW)	30
Wheel base(mm)	2520	Max.torque(Nm)	110
Range(km)	200		

ii. Sales Markets

Vietnam 越南

Production Volume 生产量

Production volume 生产量	10 years cycle and total 289,813 vehicles 周期为 10 年，共 289,813 辆
	1st=6800,2nd=8840,3rd=11492,4th=14940,5th=19421,6th=25248,7th=32822,8th=42669,9th=55470,10th=72111

2. General Project Guidelines 一般项目指南

a. Development Scope Abstract 开发范围摘要

This technical specification is a written summary of requirements to be taken into account in the development of the vehicle control unit (BCM).本技术规范是对开发车辆控制单元（BCM）时应考虑的要求的书面总结。

The requirements described in this technical specification relate primarily to functional characteristics.本技术规范中描述的要求主要涉及功能特性。

The contractor ensures that the development of the components respect all applied standards, characteristics and project requirements.

承包商应确保组件的开发符合所有适用的标准、特性和项目要求。

Any possible cost savings as a result of improvements in design, production or validation details have to be reported immediately.任何因设计、生产或验证细节的改进而可能带来的成本节约都必须立即报告。

The contractor is a 'Full-Service Supplier', with complete responsibility of the supplied system within the system boundaries.

承包商是“全方位服务供应商”，在系统范围内对所提供的系统负全部责任。

b. Initial Situation and Objective 初步情况和目标

The vehicle control unit is a major trend in the design of electric vehicle EE structures today. With this trend, how to safely, effectively and reasonably control the various high-pressure systems of the entire vehicle is a very critical factor.

车辆控制单元是当今电动汽车 EE 结构设计的一大趋势。在这种趋势下，如何安全、有效、合理地控制整车的各种高压系统是一个非常关键的因素。

c. Classification of the System / Component 系统/组件分类

NO.	Part Number	Part Name	Number	Make/buy	Remark
1	EEP64011100AA	BCM	1	Buy	
2	EEP64011001AA	BCM HW	1	Buy	
3	SOW64011001AA	BCM FLASHDRIVER	1	Buy	
4	SOW64011002AA	BCM XML	1	Buy	
5	SOW64011003AA	BCM SW	1	Buy	

d. Execution Regulations and Standards 执行条例和标准

The supplier must ensure that the developed parts meet of the specified requirements of VinFast, obtain the type certification of the parts and submit all necessary documents to support the type certification of the whole vehicle. 供应商必须确保所开发的部件符合 VinFast 的规定要求，获得部件的型式认证，并提交所有必要文件以支持整车的型式认证。

When synchronous development is completed, suppliers need to provide the necessary documentation to prove that their products have met the required regulatory requirements, including but not limited to: 同步开发完成后，供应商需要提供必要的文件，以证明其产品符合所需的法规要求，包括但不限于

FMVSS 101 standard for Controls and Displays 控制和显示标准

FMVSS 208 standard for Occupant crash protection 乘员碰撞保护标准

N o	Regulations and Standards	Type	Regulations and Standards Name	Remark
1	VFDSXVEEP0047		CES_VINFAST_EE_CAN_Specification-V2.0.0	
2	VFDSP50000012		CES_VINFAST_EE_LIN_Specification_V3.1.2	
3	VFDSP50000017		Coding Specification_V1.0.4	
4	VFDSVAEEP0003		EE_2_0__Flash_Specification_V3.3	
5	VFDSP50000020		CES_VINFAST_EE _Security_Access_Algorithm_V1.0.3	
6	VFDSVAEEP0002		Basic DID PID Specification_V2.0.1	
7	VFCSCMCSS00110		VinFast-Cybersecurity-Requirements-v4.2.1	
8			VinFast Addendum Flash Specification for Cybersecurity_v1.1	
9	VFDSVAEEP0001		Generic_Diagnostic_Specification_V2.0	

10	VFDSXVEEP0013		ECU Identifier Specification_V2.0.2	
11	VFDST00010202		VF_PTSP_TCOT08_Technical Document Numbering Standard_V21	
12	VFDST00026307		EMC Requirements and Testing for Component_V3	
13	VFDST00028121		Electrical and electronic components in motor vehicles - Electrical requirements and testings_V2	
14	VFDST00028131		Electrical and electronic components in vehicles - Environmental requirements and testing_V4	
15	VFDST0009000		Mounting/dismounting 安装/拆卸	

e. Development and Supply Scope 开发及供应范围

i. Work Interface Chart (RASI).工作界面图 SS

Work Interface Chart (RASI)

The main supplier obligations are described in the work interface chart (RASI).工作界面图（RASI）描述了供应商的主要义务。

The status of the supplier ("Built to Print" or "System Development Supplier") is defined below
System / Component Specification.供应商的状态（"按印刷生产"或"系统开发供应商"）在"系统/组件规格"中定义。

RASI is defined as the following RASI 的定义如下：

R: Responsible

Responsible for proper execution of the activities to fulfil the requirements. She/he is entitled to request the necessary support from other project team members

A: Approval

Obligation to decide / responsible for release. If no decider is defined, the person who is responsible for execution R is the decider

S: Support

Obligated to active support (even if this requires capacity, expenses and/or transfer of know-how).

This project team member must proactively assist the person responsible for execution R in achieving the targets.

I: Information

Right to information. This team member is entitled to demand information from R

Details deliverables and RASI

Here the detailed responsibilities for all relevant parties can be described using a RASI matrix.

这里使用 RASI 矩阵来描述所有相关方的详细责任。

No	Task Description任务描述	LD	Supplier	VF
1	Product Function Definition 产品功能定义	R	S	A

2	Design assignment release 设计任务发布	R	S	A
3	Performance Indicators and Acceptance Methods 性能指标及验收方法	R	S	A
4	Product Dimension Data and Boundary Conditions 产品尺寸数据和边界	R	S	A
5	Technical Proposal (system analysis/hardware and design validation/software design, integration and testing/product validation, etc.) 技术建议书（系统分析/硬件和设计验证/软 件设计、集成和测试/产品验证等）	S	R	A
6	Project Development Plan (work plan related to the whole development life cycle) 项目开发计划（与整个开发生命周期相关 的工作计划）	S	R	A
7	Material list (configuration information and BOM list) 材料清单（配置信息和 BOM 清单）	S	R	A
8	Wiring harness information input form 线束信息输入表	S	R	A
9	Sample requirements list 样品要求清单	R	S	A
10	2D/3D data production and analysis 2D/3D 数据制作和分析	S	R	A
11	Peripheral parts and interface circuit confirmation 外围部件和接口电路确认	S	R	A
12	Hardware circuit design and verification 硬件电路设计与释放	S	R	A
13	DFMEA	S	R	A
14	DBC file release DBC文件释放	R	S	A
15	DVP test plan DVP测试计划	S	R	A
16	DV sample DV样品	S	R	A

17	DV test report DV测试报告	S	R	A
18	Data freezing and mold opening review 数据冻结及开模评审	S	R	A
19	PFMEA	S	R	A
20	PV sample PV样品	S	R	A
21	PV test report PV测试报告	S	R	A
22	OTS approved data submission OTS批准数据提交	S	R	A
23	OTS approved OTS批准	R	S	A
24	Summary of development experience 开发经验总结	S	R	A

ii. Development Interface Agreement (DIA)

According to functional safety requirements. 符合功能安全要求

f. Quotation Scope

Refer to SOR and RFQ. 参考 SOR 和 RFQ。

g. Development Process. 开发过程

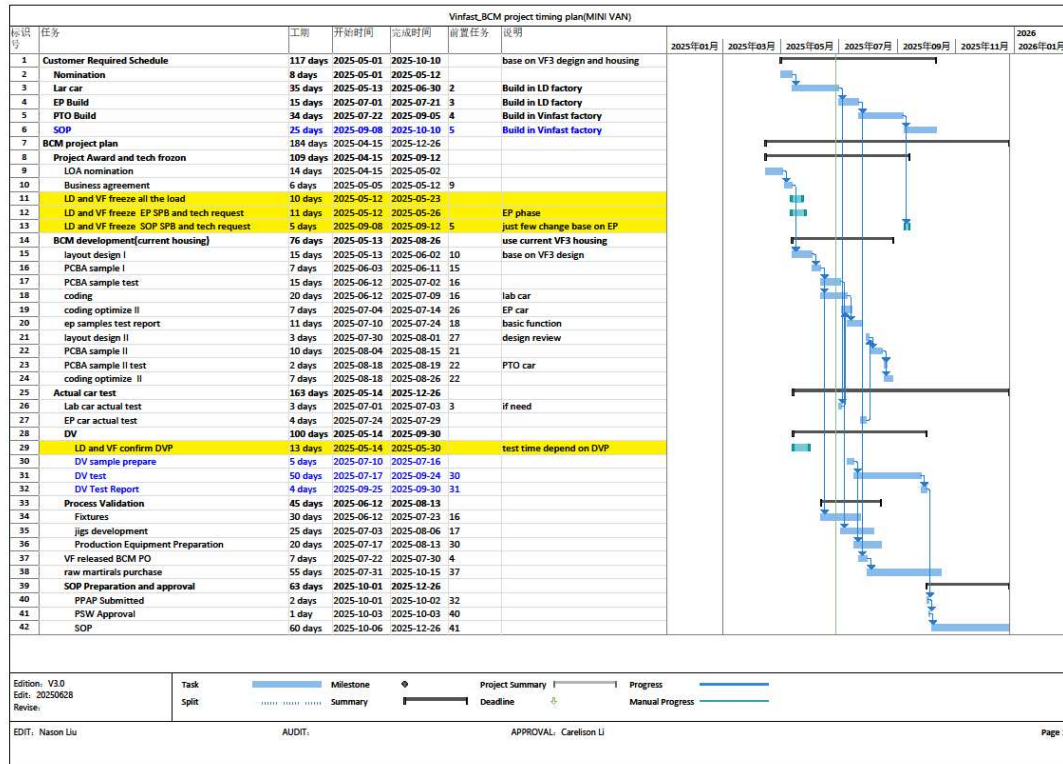
N/A

i. Gates and Milestones. 节点和里程碑

Gates

N/A

Milestones



ii. Prototype Statuses 样机状态

Sample No. 样品序号	Sample Name 样品名称	Sample Status 样品状态	Request Date 日期	Amount 数量
1	A sample Labcar	CAN:100% Diagnostic:0% Basic Functions:100%	Follow Vehicle Build Plan	Subject to purchase order
2	B sample EP	DV sample; sample CAN :100% Diganositc:100% Basic Functions+ Basic Functions+A Bug:100%,	Follow Vehicle Build Plan	Subject to purchase order
3	C sample PTO	PV sample; CAN and Diganostic:100% Software:100%, (NO A/B level bug)	Follow Vehicle Build Plan	Subject to purchase order
4	PTR	SOP sample Software:100%, (No bug)	Follow Vehicle Build Plan	Subject to purchase order

Please refer to Mini VAN-FIP for specific software maturity levels.

请参考 Mini VAN-FIP 文件中规定的软件成熟度。

iii. Prototype Quantities 样机数量

Subject to purchase order. 以采购订单为准。

iv. Acceptance Procedure 验收程序

N/A

v. Acceptable Criteria 接受标准

N/A

vi. Type Approval and Certification 型号核准和认证

VinFast is responsible for the integration and certification of the vehicle grade system. Vehicle level certification projects include urban simulation road test and enhanced road test. Suppliers of spare parts/subsystems will be responsible for timely response to failure reports found in the process of integration and certification of vehicle level system. All countermeasures to resolve issues addressed in the failure report must be executed by the supplier and approved by PPAP.

VinFast 负责车辆级系统的集成和认证。车辆级认证项目包括城市模拟道路测试和增强型道路测试。备件/子系统供应商将负责对车辆级系统集成和认证过程中发现的故障报告做出及时响应。所有解决故障报告中所述问题的对策必须由供应商执行，并经 PPAP 批准。

The integration and certification of components/subsystems is the responsibility of the supplier of components/subsystems. The supplier must complete all parts/subsystems tests. 部件/子系统的集成和认证由部件/子系统供应商负责。供应商必须完成所有部件/子系统的测试。

The certification of parts/subsystems can be implemented by the supplier of parts/subsystems or by the supplier appointing a lower supplier or a third party after VinFast approval. 部件/子系统的认证可由部件/子系统供应商实施，也可由供应商在 VinFast 批准后指定下级供应商或第三方实施。

However, in any case, the supplier is obliged to complete all specified test requirements.

但无论如何，供应商都有义务完成所有规定的测试要求。

The supplier of parts/subsystems must provide the VinFast Product Release Engineer with a complete test certification plan for parts/subsystems and obtain the approval of the Engineer. The plan should include at least the number of specimens, test sequence, place and time of each test.

部件/子系统供应商必须向 VinFast 产品发布工程师提供完整的部件/子系统测试认证计划，并获得工程师的批准。该计划至少应包括试样数量、测试顺序、每次测试的地点和时间。

h. Quality and Reliability 质量与可靠性**i. Quality Concepts 质量概念**

Supplier must satisfy the quality commitment made to VinFast, that is, to guarantee 100 waste products per 1, 000, 000 pieces of spare parts sent to VinFast. $PPM \leq 100$, $CPK \geq 1.33$.

供应商必须满足向 VinFast 做出的质量承诺，即保证每 1,000,000 件发往万快的备件中产生 100 件废品。 $ppm \leq 100$, $cpk \geq 1.33$ 。

ii. Risk Management 风险管理

The supplier shall notify the VinFast Product Release Engineer in writing of any potential technical matters, such as problems, accidents, latest information, project changes, etc. which may lead to

increased costs, delays in project nodes and plans, and delays in delivery dates. The supplier written notice should include a specific repair plan and a risk assessment and description of the overall project schedule.

供应商应将任何可能导致成本增加、项目节点和计划延误以及交货日期延迟的潜在技术事项，如问题、事故、最新信息、项目变更等，书面通知文快产品发布工程师。供应商的书面通知应包括具体的维修计划和风险评估，以及对整个项目进度的描述。

iii. Validation of the SW Development 软件开发验证

The supplier shall regularly report to VinFast the relevant design input factors and provide engineering support. At the same time, suppliers should attend supplier development meetings and complete necessary design validation.

供应商应定期向 VinFast 报告相关的设计输入因素并提供工程支持。同时，供应商应参加供应商开发会议并完成必要的设计验证。

If there are inconsistencies with the specified specification requirements, the supplier should work with VinFast to solve the problem. Suppliers also need to provide VinFast with competent on-site service personnel to support VinFast in resolving problems associated with supplier parts in vehicle assembly during vehicle start-up phase.

如果出现与指定规格要求不一致的情况，供应商应与 VinFast 合作解决问题。供应商还需向 VinFast 提供称职的现场服务人员，以支持 VinFast 解决在车辆启动阶段与供应商零部件有关的问题。

3. Project Management and Organization 项目管理和组织

a. Responsibilities in the Project and Project Management Plan 项目和项目管理计划中的职责

- Development plan approved by VinFast (including quality plan)

VinFast 批准的开发计划（包括质量计划）：

- 1) Mold Development Plan 模具开发计划
- 2) DVP Test Plan for Parts and Components 零部件的 DVP 测试计划
- 3) Other (including quality plan) 其他（包括质量计划）

- Product data (3D, 2D) 产品数据（3D、2D）
- List of materials 材料清单
- DVP test report DVP 报告
- Test Report of Prohibited and Restricted Substances Approved by VinFast VinFast 批准的禁用和限用物质测试报告

The supplier will be responsible for 供应商将负责：

Contact list of project (VF & Supplier) 项目联系清单（VF 和供应商）

/	contacts	contact information
Vinfast	Dhanush	v.dhanushs@vinfast.vn
LD	Zhang Haichao	zhang_Haichao_tj@launchdesign.com.cn

Hongfa	Nason (Liu Zhongcheng)	1901815@hongfa.cn
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b. Documentation.文档

i. Hardware Documentation 硬件文档

Supplier should be required to documents the following pieces of hardware at various stages

应要求供应商在不同阶段记录一下硬件文档：

1.2D, 3D Data drawings.2D、3D 数据图纸。

2.Report of banned substances.禁用物质报告。

3.If designed by VinFast, supplier shall provide PFMEA of the product. If supplier designs, both PFMEA and DFMEA shall be provided.

如果由 VinFast 设计，供应商应提供产品的 PFMEA。如果由供应商设计，则应同时提供 PFMEA 和 DFMEA。

4.BOM of parts. 零部件 BOM。

5.Provide DV/PV type test report according to the technical requirements of product development

根据产品开发的技术要求提供 DV/PV 型式测试报告

ii. Software Documentation 软件文档

Supplier should be required to documents the following pieces of software at various stages

应要求供应商在不同阶段提供以下软件文档：

1.Software unit test report.软件单元测试报告。

2.Software integration test report.软件集成测试报告。

3.Software system test report.软件系统测试报告。

iii. Conformity Requirements 需求整合

All homologation of the vehicle relevant legal requirements shall be fulfilled for the planned sales markets Vietnam. All corresponding drafts, proposals and all series of amendments/supplements have to be considered during the development.在越南计划的销售市场，应满足所有与车辆相关的法律要求。在开发过程中，必须考虑所有相应的草案、建议和所有系列的修正/补充。

In order to obtain the whole vehicle type approval, the supplier must provide all required documents, certificates and test reports for separate technical units, parts and systems together with all applicable legal markings in accordance with the regulations of the planned sales markets.为了获得整车的型式认证，供应商必须根据计划销售市场的规定，提供单独技术单元、部件和系统所需的所有文件、证书和测试报告，以及所有适用的法律标识。

If separate technical units, parts and systems for the planned sales markets required to be certified or be tested by a Technical Service or Authority, the supplier is solely responsible for the timely and complete provision of the certificates and/or test reports. The selection of the Technical Service must be agreed with the homologation department of VinFast.如果计划销售市场的独立技术单元、部件和系统需要由技术服务机构或权威机构进行认证或测试，则供应商应自行负责及时、完整地提供证书和/或测试报告。技术服务机构的选择必须征得 VinFast 认证部门的同意。

If requirements of a planned sales market are subject to conformity of production for separate technical units, parts and systems, the supplier is responsible for compliance. All information regarding test performance and test results must be delivered to the homologation department of VinFast. 如果计划销售市场的要求受制于独立技术单元、部件和系统的生产符合性，则供应商负责遵守。有关测试性能和测试结果的所有信息必须提交给 VinFast 公司的认证部门。

4. System Environment 系统环境

a. Functional System Environment. 系统功能环境

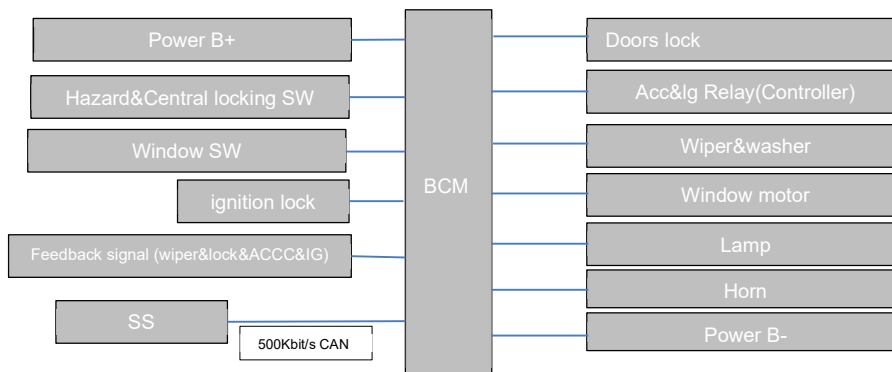
Temperature Environment Term	Temperature (°C)		Note
	The lowest	The highest	
Working temperature	- 40	85	Ambient temperature of external space where the BCM works normally
Storage temperature	- 40	90	Ambient temperature of external space when the BCM is statically stored in the state of unpowered

b. Physical System Environment

Refer to the f Electrical Interfaces 参考电器接口

See 1.1.10 for details

c. System Diagram

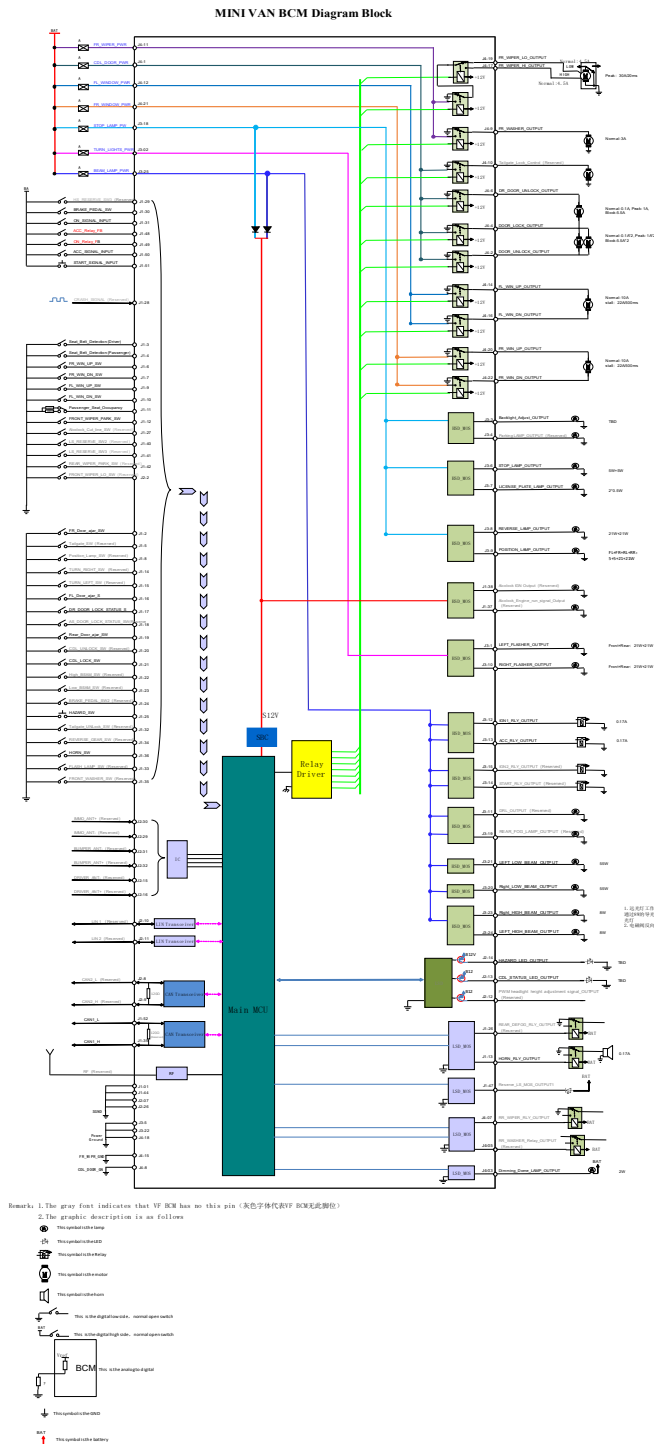


5. Functional and Technical Requirements 功能与技术要求

a. Technical requirement 技术要求

NA

b. Block and Principle Diagram.原理框图



i. Elements of Systems 系统要素
BCM:

Body control module is an electronic control unit that manages the main body functions of the car thus performing gateway functions between high and low speed CAN and between LIN and CAN networks. 车身控制模块是一个电子控制单元，用于管理汽车的主要车身功能，从而在高速和低速 CAN 之间以及 LIN 和 CAN 网络之间执行网关功能。

ABS:

ABS achieves control of wheel lockup by controlling the retraction and retraction of brake oil pressure, ensuring that the vehicle is always in a critical locked gap rolling state.

BMS:

Monitor the voltage, current, temperature of the high voltage battery pack, collect the protocol voltage pack current from the BCM, HVIL state insulation status and other necessary information. The BMS should also control the high relay according to the BCM command. 监控高压电池组的电压、电流和温度，收集来自 BCM 的协议电压组电流、HVIL 状态绝缘状态和其他必要信息。BMS 还应根据 BCM 的指令控制高压继电器。

2IN1(DCDC+PDU):

DCDC: Voltage conversion, The output voltage of the battery is converted by a DC/DC converter and then supplied to the motor driver. DCDC: 电压转换，电池的输出电压经 DC/DC 转换器转换后供应给电机驱动器。

Stable voltage: The output voltage of the battery is unstable, and it is regulated by the DC/DC converter closed-loop control system.

稳定电压：电池输出电压不稳定，由 DC/DC 转换器闭环控制系统进行调节。

PDU It is electrically connected with high-voltage components to provide charging and discharging control, high-voltage component power-on control, circuit overload short-circuit protection, high-voltage sampling, low-voltage control, etc. for the high-voltage system of the vehicle to protect and monitor the operation of the high-voltage system. PDU 与高压组件电连接，为车辆高压系统提供充放电控制、高压组件通电控制、电路过载短路保护、高压采样、低压控制等功能，以保护和监控高压系统的运行。

EDS:

The drive motor control system is a device that controls the energy transfer between the main traction power source and the motor. Its main functions include idle control of the vehicle, vehicle advancement (control motor forward rotation), vehicle reversing (control motor reversal), DC/AC, etc. 驱动电机控制系统是控制主牵引动力源与电机之间能量传递的装置。其主要功能包括车辆怠速控制、车辆前进（控制电机正转）、车辆倒车（控制电机反转）、直流/交流等。

EAC:

EAC is a CAN Node and controls the speed of compressor with the commands of BCM. EAC 是一个 CAN 节点，根据 BCM 的指令控制压缩机的转速。

SS:

The Straw switch sends a driver shift request to the main control module through a CAN signal frame. 牵引开关通过 CAN 信号帧向主控制模块发送驱动器移位请求。

Switches:

Central control switch: provide the signal of connecting and locking the door inside the vehicle and indicate the locking state of the door.

中控开关：提供车内车门接通和锁紧信号，并指示车门的锁紧状态。

hazard SW: Provide the signal function of turning on the left and right turn signals when the vehicle is in emergency.

危报开关：当车辆遇到紧急情况时，提供打开左右转向灯的信号功能。

Window SW: used to output door glass control signal to realize the opening and closing function of doors and windows.

门窗开关：用于输出门玻璃控制信号，实现门窗的打开与关闭功能。

VCU:

1. Vehicle control unit VCU (Vehicle control unit) as the new energy vehicle central control unit is the core of the entire control system. The VCU collects the motor and battery status, accelerator pedal signal, brake pedal signal and other actuator sensor controller signals. After comprehensive analysis and corresponding judgment based on the driver's driving intention, it monitors the actions of the controllers of the lower layers. Responsible for the normal driving of the car, energy management of the vehicle's engine and power battery, network management, fault diagnosis and processing, vehicle status monitoring, etc., In order to ensure the normal and stable work of the vehicle under the conditions of better power, higher economy and reliability

整车控制单元 VCU (Vehicle control unit) 作为新能源汽车的中央控制单元，是整个控制系统的核心。VCU 收集电机和电池状态、油门踏板信号、刹车踏板信号和其他执行器传感器控制器信号。根据驾驶员的驾驶意图进行综合分析和相应判断后，对下层控制器的动作进行监控。负责汽车的正常行驶、车辆发动机及动力电池的能量管理、网络管理、故障诊断与处理、车辆状态监控等工作，以保证车辆在动力性更好、经济性和可靠性更高的条件下正常、稳定地工作

2. The gateway controls data interaction between the vehicle networks. It routes between CAN, LIN and other networks.

网关控制车辆网络之间的数据交互。它在 CAN、LIN 和其他网络之间进行路由。

c. Functions 功能

i. Overview Functions Description 功能介绍

The supplier shall provide BCM module, which can be applied to vinfast vehicle model (Mini VAN); BCM shall include hardware and software information; It can receive switch information, and realize the control functions of lamps, wiper&washing, doors lock, power management and so on.

供应商应提供 BCM 模块，该模块可应用于 Mini VAN 车型；BCM 应包括硬件和软件信息；可接收开关信息，实现灯具、雨刷清洗、门锁、电源管理等控制功能。

BCM unit shall communicate on CAN to the rest of the vehicle ECUs. The unit hardware supports up to 2 CAN channel. CAN communication rate requires 500Kbit/s.

BCM 单元应通过 CAN 与其他车辆 ECU 通信。设备硬件最多支持 2 个 CAN 通道。CAN 通信速率要求 500Kbit/s

The BCM outputs the vehicle status (SLEEP, OFF, ACC, IGN, READY) through the startup process, and performs relevant functions based on the vehicle status and other component information.

BCM 通过启动过程输出车辆状态 (SLEEP, OFF, ACC, IGN, READY)，并根据车辆状态和其他部件信息执行相关功能。

ii. Detailed Functions Description 详细功能描述

Fun ID	Feature 特性	Function 功能					
			Level 1	Level 2	Level 3	Base version (标配)	High version (选配)
	EE, BCM	Window Control function	System				
MF-18	EE	Window Control function	System				
SF-18-1	EE	Window Up&Down function				×	√
	EE	System		Up		×	√
	EE	System		Down		×	√
SF-18-2	EE	The windows stop				×	√
	EE	System		Stop		×	√
MF-19	EE	Door lock&Unlock function	System following functions: central locking, central unlocking				
SF-19-1	EE		Central control switch unlock two doors			×	√
SF-19-2	EE		Central control switch lock two doors			×	√
SF-19-3	EE		Lock with mechanical key (Emergency case or Forced lock)			×	√
SF-19-4	EE		Unlock with mechanical key (Emergency case or Forced unlock)			×	√
SF-19-5	EE		Auto speed lock			×	√
SF-19-6	EE		Auto-relock			×	√
SF-19-7	EE		Motor lock thermal protection			×	√
MF-21	EE	External and interior Lighting Control	System				
SF-21-1	EE		Position light and license plate light			√	√
SF-21-2	EE		Low beam			√	√
SF-21-3	EE		High beam			√	√
SF-21-4	EE		Flash light			√	√
SF-21-5	EE		Brake light			√	√
SF-21-6	EE		Reversing light			√	√
SF-21-7	EE		Turn light			√	√
SF-21-8	EE		Lane change function			√	√
SF-21-9	EE		Hazard warning light			√	√
SF-21-10	EE		Emergency braking double flash-ABS			√	√
SF-21-11	EE		Reading lamp			×	√
MF-22	EE	Wiper & Wash Control	System				
SF-22-1	EE		Front Wiper control			√	√
	EE		System	low speed		√	√
	EE		System	high speed		√	√
	EE		System	intermission		√	√
SF-22-2	EE		Front Wiper service mode			√	√
SF-22-3	EE		Front Wash mode			√	√
SF-22-4	EE		Wiper motor protection			√	√
MF-23	EE	Accessory control function	System				
SF-23-1	EE		Horn control-BCM			√	√
SF-23-2	EE		Horn control-Manual			√	√
MF-25	EE	PDU function	System				
SF-25-1	EE		Voltage Mode management			√	√
	EE		System and discriminates and operates therelevant functions according to the voltage detection results. BCM根据电压高低控制相关电源电压	Stop Voltage Mode (SVM)		√	√
	EE		System	Low voltage mode		√	√
	EE		System	Normal voltage mode		√	√
	EE		System	High voltage mode		√	√
	EE		System	Overvoltage mode		√	√
SF-25-2	EE		Bus sleep and Wake-up conditions.			√	√
SF-25-3	EE		Vehicle BCM Power model			√	√
SF-25-4	EE		Limp home mode			√	√
MF-26	EE	IPC & BCM Display function	System				
SF-26-1	EE		Pos.lamp working instruction display			√	√
SF-26-2	EE		High beam working instruction display			√	√
SF-26-3	EE		Turn & Hazard lamp working instruction display			√	√
SF-26-4	EE		Open door warning			√ (driver)	√
SF-26-6	EE		ODO Storage backup			√	√
MF-27	EE	LV battery top-up strategy	System				
SF-27-1	EE		Enable top-up			√	√
SF-27-2	EE		Disable top-up			√	√
SF-27-3	EE		Cancellation during HV ON			√	√
SF-27-4	EE		BCM OFF function activation			√	√
SF-27-5	EE		The vehicle is at sleep			√	√
MF-28	EE	SBR (Seat Reminder)	System				
SF-28-1	EE		SBR (Seat Reminder Function)			√	√
MF-70	Common Function		System				
SF-70-3	Diagnosis flash	02_BCM	Diagnosis flash			√	√
SF-70-14	Diagnosis UDS	02_BCM	Diagnosis UDS			√	√
SF-70-26	AutoSAR NM	02_BCM	Network			√	√
SF-70-34	COM Layer	02_BCM	Network			√	√
SF-70-45	E2E	02_BCM	Network			√	√
SF-70-57	Physical Layer	02_BCM	Network			√	√
	Network Type		System				
			System				
			System				
		C-APP	Alternates service schedule notification				
			Suggest maintenance schedule				
			Car statistics (remaining battery, coolant level etc.)				
			Locate vehicle location remotely				
			Basic navigation				

Note:

1. BCM has only one version of the software BCM 只有一版软件
2. For lower models, the BCM will not set the DTCs if there is no input signal. The absence of an input signal indicates that the vehicle is not equipped with window lift switches, center switches, or door lock feedback signals, and the BCM will not operate these loads, and therefore will not detect the appropriate diagnostic information.

MF-18 Window Control Function 窗口控制功能

The window up&down control function involves components: BCM, window switch, window motor and other components. 车窗升降控制功能涉及多个组件：BCM、车窗开关、车窗电机和其他部件。

The window switch has three gears: manual up, stop, manual descending,. 车窗开关有三个档位：手动上升、停止、手动下降。

Note 注释:

1. Set the two gear of the switch. 设置开关的两个档位。

The status signals of part are shown in the table below: 部件的状态信号如下表所示:

Input signal: 输入信号

Signal name	out type	Wake source	Signal description	Remark
FL window up 左前上升	Internal relay 内部继电器	H/L	Up:High Down:Low	Simultaneous work 同时进行的工作
FL window down 左前下降	Internal relay 内部继电器	H/L	Down:High Up:Low	
FR window up 右前上升	Internal relay	H/L	Up:High Down:Low	Simultaneous work 同时进行的工作
FR window down 右前下降	Internal relay	H/L	Down:High Up:Low	

Output CAN signal 输出 CAN 信号

	Signal Name	Signal Description	Value Matrix
0x109			
	Checksum_0x109	Checksum	
	Rolling_Counter_0x109	Rolling_Counter	15 "Invalid"
	BCM_STS_Leftwindow_switch	BCM STS Driverwindow switch	3 "Signal invalid" 2"left window switch down" 1"left window switch up" 0"left window switch off"
	BCM_STS_Rightwindow_switch	BCM STS Passengerwindow switch	3 "Signal invalid" 2"right side window switch down" 1"right side window switch up" 0"right side window switch off"
	BCM_STS_WindowL	BCM STS WindowL	3"Signal invalid" 2"left side window down" 1"left side window up" 0"left side window off"

	BCM_STS_WindowR	BCM STS WindowR	3"Signal invalid" 2"right side window down" 1"right side window up" 0"right side window off"
--	-----------------	-----------------	---

schematic diagram (Auto CAD) 原理图

NA

SF-18-1 Window Up&Down function 1 窗口上下功能

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	×	√	√

(1) Operating conditions 运行条件

① BCM activates the window function when all of the following conditions are met: 当满足以下所有条件时，BCM 将激活窗口功能：

a. Vehicle state as above table, or within 2min after ign leaves the ON gear; 车辆状态如上表，或切换到 OFF 的 2 分钟内；

(2) Function operation input 功能操作输入

BCM receives the window switch signal. BCM send BCM_STS_Leftwindow_switch & BCM_STS_Rightwindow_switch; BCM 接收车窗开关信号；

(3) unction operation output: 功能操作输出：

① Manual UP: When the switch is hit to UP, the window , when the switch is OFF, the window stops up 手动上升：当开关打到 UP 时，车窗上升，当档位 OFF 时，车窗停止上升

① Manual Down: When the switch is hit to Down, the window , when the switch is OFF, the window stops down

手动下降：当开关打到 Down 时，车窗下降，当档位 OFF 时，车窗停止下降

(4) Note 注释：

① Window motor output can last for up to 8 seconds 车窗电机输出可持续长达 8 秒钟

② When the BCM receives the main driver switch up || down signal, the BCM send BCM_STS_Leftwindow_switch = 1"left window switch up" || 2"left window switch down"

BCM 接收到主驾开关上升||下降信号时。

② When the BCM receives the main passenger switch up || down signal, the BCM send BCM_STS_Rightwindow_switch = 1"right side window switch up" || 2"right side window switch down"

BCM 接收到乘客开关上升||下降信号时。

- ③ When the BCM drives the main driver's door window up||down, BCM send BCM_STS_WindowL= 1" left side window up"|| 2" left side window down"
BCM 驱动主驾门窗上升||下降时
- ④ When the BCM passenger the door window up||down, BCM send BCM_STS_WindowR= 1" right side window up"|| 2" right side window down "
BCM 驱动副驾门窗上升||下降时
- ⑤ When the driver & passenger door window switches are not operated.BCM send BCM_STS_Leftwindow_switch= 0" left window switch off"& BCM_STS_Rightwindow_switch= 0" right side window switch off"
主驾&副驾门窗开关无操作时,
- ⑥ When BCM does not drive the main & passenger door windows, BCM send BCM_STS_WindowL= 0" left side window off" & BCM_STS_WindowR= 0" right side window off"
BCM 不驱动主驾&副驾门窗时
- ⑦ The main & passenger door windows can be operated simultaneously. 主驾&副驾门窗可以同时操作

SF-18-2 The windows stop 窗口停止

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	x	x	x	√	√

(1) Operating conditions 运行条件

① When any of the following conditions are met, the BCM will stop the window action: 满足以下任何条件时, BCM 将停止窗口操作:

a. Vehicle state as above table

b. Stop output after window overload or open circuit time reaches (within $400 \pm 50\text{ms}$); 车窗电机过载或开路时间达到 ($400 \pm 50\text{ms}$ 内) 后停止输出;

② After ign changes from ON to OFF or ACC exceeds 2min: 点火开关从 ON 切换到 OFF 或 ACC 超过 2 分钟内

③ If the window is in manual operation mode, stop the window immediately. 如果车窗处于手动操作模式, 请立即停止车窗。

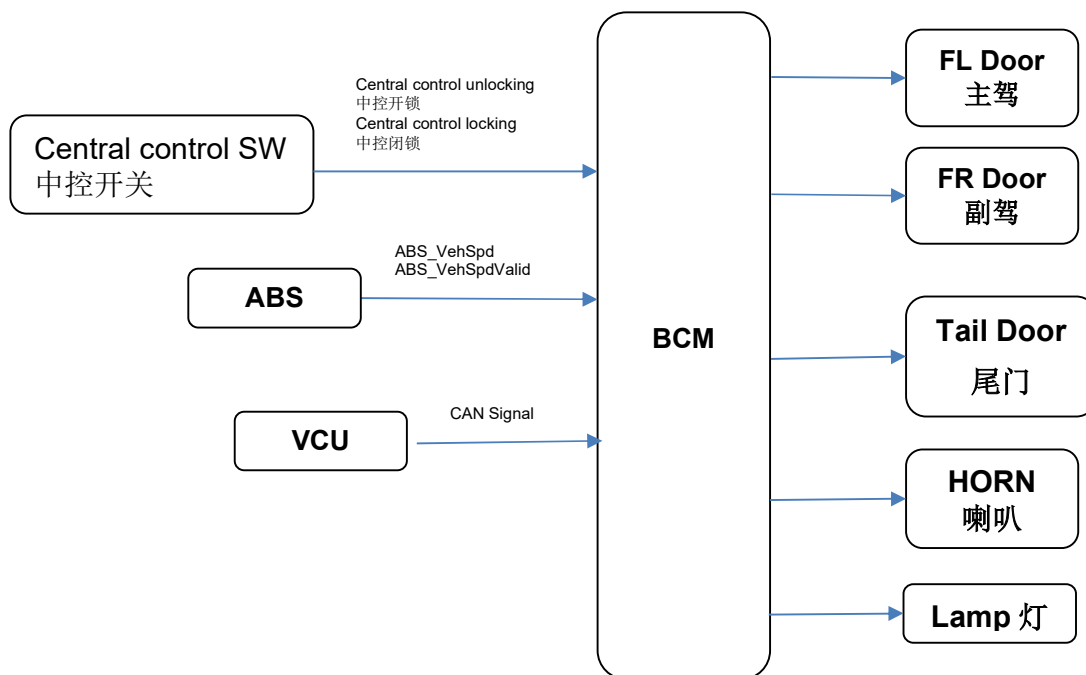
④ The maximum single driving time of the window is 8s. If the locked rotor is not reached after 8s, stop the window driving. 车窗的最大单次运行时间为 8 秒。如果 8 秒后仍未到达想要的位置, 则停止窗口驱动。

MF-19 Door lock&Unlock function 门锁解锁功能

Door lock function related components: BCM,ABS, Mechanical key, Central locking SW and indicator light, two door and tail door lock, Horn, Hazard warning light, etc.

门锁功能相关部件：BCM、ABS、机械钥匙、中控开关及开关指示灯、双门及尾门锁、喇叭、危险警示灯等。

The status signals of part are shown in the table below 部件的状态信号如下表所示：



Output CAN signal 输出 CAN 信号

ID Name ID 名称	Signal Name 信号名称	Signal Description 信号的描述	Value Matrix 矩阵
BCM_STAT_DOOR_FLAP_0x105			
	Checksum_0x105	Checksum	
	Rolling_Counter_0x105	Rolling_Counter	15 "Invalid"
	BCM_STAT_DriverDoor	Driver's door status 司机车门状态	3 "Signal invalid" 2 "Reserved" 1 "Opened" 0 "Closed"

	BCM_STAT_PassDoor	Passenger's door status 乘客门状态	3 "Signal invalid" 2 "Reserved" 1 "Opened" 0 "Closed"
	BCM_STS_TailDoor	尾门开关状态 Switch status of the tail door	3 "Signal invalid" 2 "Reserved" 1 "Opened" 0 "Closed"
BCM_STAT_CENTRAL_LOCK_0x107			
	Checksum_0x107	Checksum 校验码	
	Rolling_Counter_0x107	Rolling_Counter 滚动计数器	15 "Invalid"
	STAT_ATWS	Locking/UnLocking Status Anti theft Invalid value = 7 锁定/解锁状态防盗无效值=7	7 "Signal invalid" 6 "Reserved" 5 "Reserved" 4 "Reserved" 3 "Partly secured" 2 "Alarm" 1 "Unsecured" 0 "Secured"
	STAT_DoorUnlockFL	Real time feedback on door lock status	3 "Signal invalid" 2 "Reserved" 1 "Unlocked" 0 "Locked"
	STAT_DoorLockFL	Lock Status Driver Door used as central lock information (lock/unlock the vehicle by the fob/PESW/CentralLock button) 驾驶员门锁状态反馈控闭锁 (解锁车辆)	3 "Signal invalid" 2 "Reserved" 1 "Locked" 0 "Unlocked"
BCM_Comfort_Confirm_status_0x41d			
	BCM_STAT_DoorSpeedLock	Status feedback from BCM for Setting Door speed lock mode	7 "Signal invalid" 5 "25KMH" 4 "20KMH" 3 "15KMH" 2 "10KMH" 1 "5KMH" 0 "OFF"
BCM_LIGHT_0x106			
	Checksum_0x106	Checksum	
	Rolling_Counter_0x106	Rolling_Counter	15 "Invalid"
	BCM_TurnIndicatorSts	Turn light output Status	0 "OFF" 1 "Left On" 2 "Right On" 3 "Left and Right On"

	BCM_horn_relay_Sts	喇叭继电器控制指令 horn relay output status	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_Doorlock_ledlight_Sts	门锁 led 灯控制（车内面板按键） Door lock led light output status(Interior panel button)	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
0x109			
	BCM_STS_centrallock_switch	BCM STS centrallock switch（中控锁）	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_STS_DOOR_LOCK_STATUS_SW	BCM STS DOOR LOCK STATUS SW 车门锁状态反馈信号	3 "Signal invalid" 2 "Reserved" 1 "Unlocked" 0 "Locked"

Input CAN signal 输入 CAN 信号

ID Name	Signal Name 信号名称	Signal Description 信号的描述	Value Matrix
ABS_WheelSpdR_0x342			
	Checksum_0x342	Check sum	
	RollingCounter_0x342	Rolling Counter	15 "Invalid"
	ABS_VehSpd	Current Longitudinal Vehicle Speed 当前车辆前进速度	32767 "Invalid"
	ABS_VehSpdValid	Speed Valid Flag 速度有效	1 "Valid" 0 "Invalid"
ABS_Status_0x142			
	Checksum_0x142	check sum 校验码	
	RollingCounter_0x142	Rolling counter 滚动计数器	15 "Invalid"
	ABS_ESS_Active	Emergency Stop Signaling Active,when emergency stop, this signal will set true 紧急停止信令活动，当紧急停止时，此信号将设置为“真”	1 "active" 0 "inactive"
VCU_Pedals_0x269			

	Checksum_0x269	check sum 校验码	
	RollingCounter_0x269	Rolling counter 滚动计数器	15 "Invalid"
	VCU_VehicleHVStatus1	Vehicle High voltage Ready State 车辆高压就绪状态	7 "Reserved" 6 "Reserved" 5 "Reserved" 4 "Error" 3 "Charging" 2 "ReadyToDrive" 1 "HV On" 0 "Not ready"
	VCU_ActualGear	actual gear 实际的档位	7 "Fault" 6 "Reserved" 5 "Reserved" 4 "Reserved" 3 "D" 2 "N" 1 "R" 0 "P reserved"
	VCU_ActualGearValid	Indicate the current gear position 显示当前档位	1 "invalid"无效 0 "Valid"有效
	VCU_BrakePedalSts	Brake Pedal Sts 刹车踏板信号	3 "Invalid" 2 "Reserved" 1 "pressed" 0 "Not pressed"

schematic diagram(Auto CAD)

SF-19-1 Central control switch unlock 中控解锁

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	awaken	Accessory	Driver Present	Ready
Applicable	√	√	√	√	√

(1)Functional operating conditions (①&②&④||①&③&④) 功能操作条件

①Vehicle state as above table 车辆状态如上表所示;

②When the vehicle power gear is ON or Ready, The Vehicle speed < 5km/h (ABS_VehSpd < 5Km/h or ABS_VehSpdValid=0 or lost CAN sign ABS_VehSpd = 0);

当车辆动力装置处于 HV ON 或 REDRY 时，车速 < 5km/h (ABS_VehSpd < 5Km/h 或 ABS_VehSpdValid=0 或 CAN 信号 ABS_VehSpd = 0 信号丢失)

- ③ When the vehicle power gear is not ON or Ready, the vehicle speed signal will not be detected

当车辆动力装置未处于 HV ON 或 READY 状态时，将无法检测到车速信号

- ④ Door locks are locked 门锁处于闭锁状态

(2) Function operation input 功能操作输入：

- ① Central switch button – Press , BCM send BCM_STS_centrallock_switch = 1 "ON"

中控开关按钮-按下.

(3) Function operation output (①&②&③) 功能操作输出：

- ① BCM drive the unlock output to perform a three-door unlock , BCM send STAT_DoorLockFL= 0 "Unlocked".

BCM 驱动解锁输出，执行三门解锁，BCM 发送 STAT_DoorLockFL = 0 "Unlocked"

- ② Central locking SW work indicator light off, BCM send BCM_Doorlock_ledlight_Sts= 0 "OFF"
中控开关工作指示灯熄灭.

- ③ BCM receives real time feedback on door lock status, BCM send STAT_DoorUnLockFL = 1 "Unlocked";

BCM 接收门锁状态实时反馈，BCM 发送 STAT_DoorUnLockFL= 1 "Unlocked")

(4) Note 注释:

- ① Through the central control switch button, the whole car can be unlocked.

通过中控开关按钮，可以实现全车解锁

- ② Repeat the operation to execute the central control lock function 重复操作，执行中控闭锁功能。

- ③ The lock is in the blocked state, perform the unlock function 锁处于闭锁状态，执行解锁功能

SF-19-2 Central control switch lock 中控闭锁

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	awaken	Accessory	Driver Present	Ready
Applicable	√	√	√	√	√

(1) Functional operating conditions 功能操作条件

- ① Vehicle state as above table 车辆状态如上表所示;

- ② Door lock is Unlocked. 门处于解锁状态.

(2) Function operation input: 功能操作输入

① Central control SW lock input - Button press, BCM send BCM_STS_centrallock_switch= 1 "ON" 中控开关按钮-按下。.

(3) Function operation output (①&②&③&④) 功能运行输出:

① The BCM drives the lockout output and performs a **three**-door lockout while sending STAT_DoorLockFL= "Locked"

BCM 驱动闭锁输出, 执行三门闭锁, 同时发送 STAT_DoorLockFL = 1 "Locked"

② Central locking SW indicator light on, BCM send BCM_Doorlock_ledlight_Sts= 1 "ON"
中控开关指示灯亮;

③ All the doors can't be opened. 所有车门不能被打开

④ BCM receives real time feedback on door lock status, also send STAT_DoorUnLockFL= 0 "Locked"
BCM 接收门锁状态实时反馈, 同时发送 STAT_DoorUnLockFL = 0 "Locked"

(4) Note 注释:

① Through the central control switch, you can lock all the doors

通过中控开关, 可以闭锁全部车门;

② Repeat the operation to execute the central control unlock function

重复上述操作, 即可执行中控解锁功能;

③ If the one or more doors are not closed, The BCM will lock the doors first and then unlock the doors (an interval of 500ms). 如果有一扇或多扇门没有关闭, BCM 会先执行闭锁, 然后执行开锁 (间隔 500ms)。

④ Door status signal: door closed status signal changes from low level to suspended, door open status signal is opposite 门状态信号: 门关状态信号由低电平变为悬空, 门开状态信号相反

SF-19-3 Lock with mechanical key 机械钥匙闭锁

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	√	√	√	√	√

(1) Functional operating conditions 功能操作条件

① Vehicle state as above table 车辆状态如上表所示;

② The Driver Door、side door、tail door In the off state, BCM send BCM_STAT_DriverDoor= 0 "Closed"& BCM_STAT_PassDoor= 0 "Closed"& BCM_STS_TailDoor= 0 "Closed"

司机门、乘客门、尾门处于关闭状态

(2) Function operation input 功能操作输入:

① When the BCM detects a “lock” request signal from the driver door latch , BCM send BCM_STS_DOOR LOCK STATUS SW= 0 "Locked"

当 BCM 检测到来自驾驶员门锁的“lock”请求信号。

(3) Function operation output 功能操作输出:

① BCM performs full door locking , while sending STAT_DoorLockFL= 1 "Locked"

BCM 执行全车门闭锁，同时发送 STAT_DoorLockFL= 1 "Locked"

② BCM receives real-time feedback on the door lock status while sending BCM_STS_DOOR LOCK STATUS SW= 0 "Locked"&STAT_DoorUnLockFL=0 "Locked"

BCM 接收门锁状态实时反馈，同时发送 BCM_STS_DOOR LOCK STATUS SW= 0 "Locked"&STAT_DoorUnLockFL=0 "Locked"

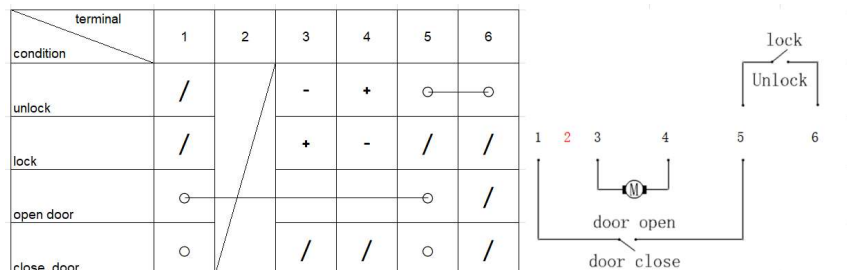
(4) Note 注释:

① If the vehicle power supply is in ACC or ON or Ready state, the BCM performs the “Central switch locks all doors” function 如果车辆电源处于 ACC 或 ON 或 Ready 状态，则 BCM 执行“中控开关闭锁全车门”功能;

② Door status signal: door closed status signal changes from low level to suspended, door open status signal is opposite 门状态信号：门关状态信号由低电平变为悬空，门开状态信号相反

③ Mechanical lock status signal: the locking status signal changes from low level to suspended, and the door unlocking status signal is opposite (as shown in the figure)

机械锁状态信号：闭锁状态信号由低电平变为悬空，门解锁状态信号相反（如图）



④ If the one or more doors are not closed , The BCM will lock the doors first and then unlock the doors (an interval of 500ms). 如果有一扇或多扇门没有关闭，BCM 会先执行闭锁，然后执行开锁（间隔 500ms）。

SF-19-4 Unlock with mechanical key 机械钥匙解锁

Mechanical key unlock (Emergency case or Forced unlock)n availability in Vehicle States

机械钥匙解锁（紧急情况下或强制解锁）在车辆状态可用

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready

Applicable	√	√	√	√	√
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(1) Functional operating conditions 功能操作条件

① Vehicle state as above table; 车辆状态如上表所示;

(2) Function operation input 功能操作输入

When the BCM detects an “unlock” request signal from the driver door lock latch, BCM send BCM_STS_DOOR LOCK STATUS SW= 1 "Unlocked"当 BCM 检测到来自驾驶员门锁的“unlock”请求信号.

(3) Function operation output 功能操作输出:

① BCM performs full door unlocking, BCM send STAT_DoorLockFL= 0 "Unlocked" BCM执行全车门解锁;

SF-19-5 Auto speed lock 自动落锁

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	awaken	Accessory	Driver Present	Ready
Applicable	x	x	x	x	√

(1) Functional operating conditions 功能操作条件

① Vehicle state as above table 车辆状态如上表所示;

② The Driver Door、 Pass Door、 Tail door In the off state, BCM send BCM_STAT_DriverDoor= 0 "Closed"& BCM_STAT_PassDoor= 0 "Closed"& BCM_STS_TailDoor= 0 "Closed"

司机门、乘客门、尾门处于关闭状态;

③ Vehicle speed signal is valid, vehicle speed $\geq 15\text{km/h}$, BCM reception ABS_VehSpd $\geq 15\text{km/h}$ & ABS_VehSpdValid=1 "Valid"; 车速信号有效, 车速 $\geq 15\text{km/h}$.

(2) Function operation input 功能操作输入:

① The central lock is unlocked, BCM send STAT_DoorUnLockFL= 1 "Unlocked" 门锁处于解锁状态;

(3) Function operation output 功能操作输出:

① BCM drives all doors to lock , Concurrent BCM send STAT_DoorLockFL = 1 "Locked"

BCM 驱动所有门闭锁,同时发送 (STAT_DoorLockFL = 1 "Locked") ;

② BCM drives the Central lock SW work indicator to turn on, BCM send BCM_Doorlock_ledlight_Sts= 1 "ON" BCM 驱动中控锁工作指示灯亮.

(4) Function operation-close 功能运行关闭:

- ① The Vehicle speed < 15km/h (ABS_VehSpd < 15Km/h or ABS_VehSpdValid=0 or lost can sign ABS_VehSpd = 0);
- ② The vehicle is in "lock" status, BCMsend STAT_DoorUnLockFL= 0 "Locked"
车辆处于“锁定”状态

(5) Note 注释:

- ① The vehicle speed information follow the speed value send out from
车辆速度信息根据 ABS 发出的速度值:
- ② BCM default speed is 15km/h. BCM 默认车速是 15km/h
- ③ If the vehicle speed is greater than the set value and the automatic locking function is triggered, the unlocking function (unclusing the central locking function) will be shielded
如果车速大于设定值, 触发自动锁定功能, 解锁功能 (不包括中控开关功能) 被屏蔽;
- ④ If the vehicle speed is lower than the set value and the automatic locking function is not triggered, the central locking switch can perform the unlocking and locking function normally
若车速低于设定值且未触发自动锁定功能, 则中控锁可正常执行解锁和锁定功能;
- ⑤ If the door is not closed properly and the vehicle speed increases by more than Set value, the system will not lock, but when the door is closed again, the system will add locking
如果车门未正确关闭, 车速增加超过设定值, 系统将不上锁, 但当车门再次关闭时, 系统将加锁.

SF-19-6 Auto-relock 自动再闭锁

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	awaken	Accessory	Driver Present	Ready
Applicable	√	√	×	×	×

(1) Functional operating conditions 功能操作条件

- ① Vehicle state as above table 车辆状态如上表所示;
- ② No door is opened.没有门被打开
- ③ No change in vehicle power state 车辆电源状态无变化;

(2) Function operation input 功能操作输入:

Trigger the mechanical key to unlock the button, start the 30s timing, when the 30s timing ends, and during the timing period (the door status signal is not detected to turn into the open state signal ||

mechanical key locking button is triggered) BCM drives the door lock motor to lock.

触发机械钥匙开锁按键，开始 30s 计时，当 30s 计时结束，且计时期间内（未检测到门状态信号变为未关状态信号||机械钥匙闭锁按键触发）BCM 驱动门锁电机闭锁。

(3) Function operation output 功能操作输出:

1. BCM drive lock motor lock , BCM send STAT_DoorLockFL =1 "Locked"

BCM 驱动门锁电机闭锁

SF-19-7 Motor lock thermal protection 电机锁紧热保护

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	√	√	√	√	√

(1) Functional operating conditions 功能操作条件

Vehicle state as above table 车辆状态如上表所示;

(2) Function operation input 功能操作输入:

The BCM shall allow the central control lock to complete a maximum of 4 cycles (lock, unlock) within 10s

BCM 应允许中控锁在 10 秒内完成最多 4 个循环（闭锁，解锁）。

(3) Function operation output 功能操作输出:

The central Driver Door、Pass Door、Tail door shall be inactive for the following 10s. but if the 9th cycle command is unlock command, BCM needs to take action to this command, then the next command must be inhibited. 在接下来的 10 秒内，司机门、乘客门、后背门应该处于非活动状态。但是如果第 9 个周期命令是 unlock 命令，BCM 需要对这个命令采取动作，那么下一个命令必须被抑制。

(4) Note 注释:

The “Motor lock thermal protection” function is not applicable to collision signals
“电机锁热保护”功能不适用于碰撞信号。

MF-21 Light Control 灯光控制

The lamp control function involves relevant components: BCM, light swrelay, lamps, BCM, etc.

灯控功能涉及相关组件：BCM、SS、灯、BCM 等。

List the priority of turn signals 列出转向灯的首选次序:

- ① Emergency automatic 紧急制动
- ② Danger alarm 危险报警

③ cornering lamp 转向灯

④ Lane changing light 变道灯

The status signals of part are shown in the table below: 部件的状态信号如下表所示:

Output CAN signal 输出 CAN 信号

ID Name	Signal Name	Signal Description	Value Matrix
BCM_LIGHT_0x106			
	Checksum_0x106	Checksum	
	Rolling Counter_0x106	Rolling Counter	15 "Invalid"
	BCM_HighBeamSts	High beam output Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_LowBeamSts	Low beam output Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_PosLmpSts	Position light output Status 位置灯输出状态	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_TurnIndicatorSts	Turn light output Status 转向灯输出状态	0 "OFF" 1 "Left On" 2 "Right On" 3 "Left and Right On"
	BCM_Overtaking_lamp	Overtaking lamp output Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_ReverseLightSts	Reverse light output Status 倒车灯输出状态	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_StopLightSts	Stop light output Status 停止灯输出状态	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"

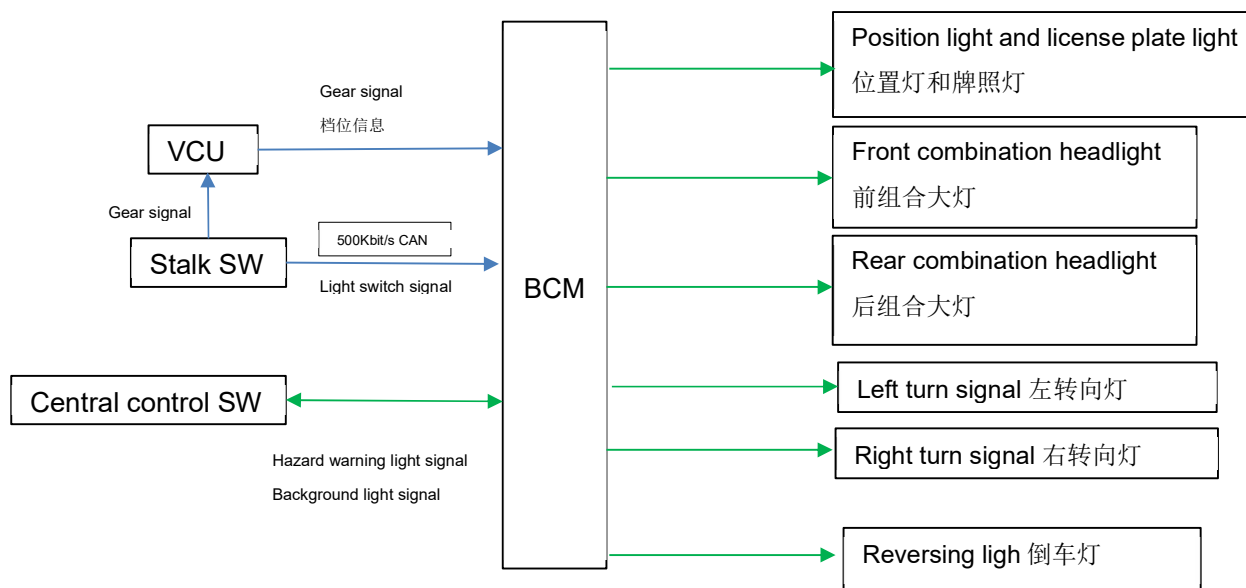
	BCM_Fr_Dimming_dome_lampSts	Dimming dome lamp Front Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_Emergency_brake_light_Sts	双闪灯控制指令 (车内面板按键) Emergency brake light output status(Interior panel button)	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_number_plate_light_Sts	牌照灯控制指令 number plate light output status	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_Doorlock_ledlight_Sts	门锁 led 灯控制 (车内面板按键) Door lock led light output status(Interior panel button)	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
BCM_SwitchSts_0x109			
	Checksum_0x109	Checksum	
	Rolling_Counter_0x109	Rolling_Counter	15 "Invalid"
	BCM_STAT_HazardWarn	Hazard Warning switch activation	3 "Signal invalid" 2 "Reserved" 1 "Active" 0 "Not active"
	BCM_STS_brake_pedal	BCM STS brake pedal	3 "Invalid" 2 "Reserved" 1 "pressed" 0 "Not pressed"
BCM_PEPSFun_0x211			
	BCM_Positionlight_reminder	reminde driver forgot turn OFF the position light when vehicle go to off 提醒驾驶员在车辆熄火时忘记关闭位置灯	0 "Inactive" 1 "Active" 2 "Reserved" 3 "Signal Invalid"

Input CAN signal 输入 CAN 信号

ID Name	Signal Name	Signal Description	Value Matrix
GW_ABS_Status_0x142			
	Checksum_0x142	Check sum	
	Rolling_Counter_0x142	Rolling Counter	15 "Invalid"
	ABS_ESS_Active	Emergency Stop Signaling Active,when emergency stop, this signal will set true 紧急停止信令活动，当紧急停止时，此信号将设置为 true	1 "active" 0 "inactive"
SS_DriveReq_0x108			
	Checksum_0x108	Check sum	
	Rolling_Counter_0x108	Rolling Counter	15 "Invalid"
	SS_HighBeamSts	High beam output Status	0 "OFF" 1 "ON"
	SS_LowBeamSts	Low beam output Status	0 "OFF" 1 "ON"
	SS_PosLmpSts	Position light output Status	0 "OFF" 1 "ON"
	SS_TurnIndicatorSts	Turn light output Status	0 "OFF" 1 "Left On" 2 "Right On" 3 "Reserved"
	SS_Overtaking_lamp	Overtaking lamp output Status	0 "OFF" 1 "ON"
	SS_lane_change_lampSts	lane change light output Status 变道灯输出状态	0 "OFF" 1 "Left lane change" 2 "Right lane change" 3 "Reserved"
	BCM_number_plate_light_Sts	牌照灯控制指令 number plate light output status	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
VCU_Pedals_0x269			
	Checksum_0x269	Check sum	
	Rolling_Counter_0x269	Rolling counter	15 "Invalid"
	VCU_VehicleHVStatus1	Vehicle High voltage Ready State 车辆高压就绪状态	7 "Reserved" 6 "Reserved" 5 "Reserved" 4 "Error" 3 "Charging"

			2 "ReadyToDrive" 1 "HV On" 0 "Not ready"
	VCU_ActualGear	actual gear 实际档位	7 "Fault" 6 "Reserved" 5 "Reserved" 4 "Reserved" 3 "D" 2 "N" 1 "R" 0 "P"
	VCU_ActualGearValid	Indicate the current gear position 显示当前档位	1 "nvalid" 0 "Valid"
	VCU_BrakePedalSts	Brake Pedal Sts	3 "Invalid" 2 "Reserved" 1 "pressed" 0 "Not pressed"

schematic diagram(Auto CAD)



SF-21-1 Position light and license plate light 位置灯和车牌灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
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Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	√	√	√

(1) Functional operating conditions 功能操作条件

① Vehicle state as above table 车辆状态如上表所示.

(2) Function operation - open 功能操作-打开:

① BCM reception SS CAN message SS_PosLmpSts= 1 "ON"; (BCM 接收 SS CAN 信号输入)

(3) Function operation output 功能操作输出:

① BCM drives the position lamp, license plate lamp and switch backlight to light up , BCM send BCM_PosLmpSts=1 "ON"& BCM_number_plate_light_Sts =1 "ON"

BCM 驱动位置灯、车牌灯及车内开关背光亮起;

② BCM drive hazard switch backlight on (high variant: backlight and warning light are the same light, warning light has high priority, low variant: backlight and warning light are not the same light, when working, they are on at the same time, backlight is driven by the position light, BCM software does it by high variant, low variant harness does the treatment)

BCM 驱动危险开关背光灯亮 (高配版:背光灯与警告灯同灯, 警告灯优先级高, 低配版: 背光灯与警报灯不同灯, 工作时, 同时亮, 背光灯由位置灯驱动, BCM 软件按高变体做, 低变体线束做处理)

(4) Function operation-close 功能运行关闭:

① BCM reception SS CAN message SS_PosLmpSts= 0 "OFF"; (BCM 接收 SS CAN 信号输入)

② BCM turns off the position lamp, license plate lamp,switch backlight and hazard switch indicator(hazard warning lamp does not work) , BCM send BCM_PosLmpSts=0 "OFF" &BCM_number_plate_light_Sts =0 "OFF" BCM 关闭位置灯、车牌灯、开关背光和危险开关指示灯 (危险警告灯不工作);

(5) Note:

① When the position lamp is working, the vehicle Power into OFF, and the BCM will drive the POS. lamp turn on 60s and sent the reminder signal that the POS.lamp is not turned off on the IPC " BCM_Positionlight_reminder=1 "Active"; . (In this 60s, if SS_PosLmpSts != "ON", BCM shall turn OFF Positon lamp immediately)

位置灯工作时, 车辆电源进入 OFF 状态, BCM 将驱动位置灯打开 60s 并在 IPC 上发送位置灯未关闭的提醒信号 " BCM_Positionlight_reminder=1 "Active"" (在此 60s 内, 如果 SS_PosLmpSts != "ON", BCM 应立即关闭位置灯)。

② When the hazard lamp works, the indicator of the hazard SW flashes synchronously with the turn lamp of the IPC.

当危险指示灯工作时, 危险开关 指示灯与 IPC 转向灯同步闪烁。

SF-21-2 Low beam 近光灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	×	√	√

(1) Functional operating conditions 功能操作条件

- ① Vehicle state as above table 车辆状态如上表所示.

(2) Function operation – open 功能操作-打开:

- ① BCM reception SS CAN message SS_LowBeamSts= 1 "ON"; (BCM 接收 SS CAN 信号输入)

(3) Function operation output 功能操作输出:

- ① BCM drives the low beam to light up, BCM send BCM_LowBeamSts=1 "ON"
BCM 驱动近光灯点亮;

(4) Function operation-close 功能运行关闭:

- ① BCM power change to not ON, or BCM reception SS CAN message SS_LowBeamSts= 0 "OFF"

BCM 电源切换到非 ON 模式, 或 BCM 接收 SS CAN 信号输入 SS_LowBeamSts= 0 "OFF";

- ② BCM turns off the low beam , DCM send d BCM_LowBeamSts=0 "OFF"
BCM 停止驱动近光灯;

(5) Note:

- ① When the power mode of the body control module changes to off, the position light will remain on for 60 seconds and then turn off.

当车身控制模块的电源模式变为 OFF 时, 位置灯将保持 60 秒, 然后关闭。

- ② The CAN message BCM_LowBeamSts= 1 "ON" can only be sent when the BCM has completed the left & right low beam driver both output successfully.

BCM 完成左&右近光灯驱动都输出成功时, 才能发送 CAN 信息 BCM_LowBeamSts= 1 "ON".

- ③ Low beam operation with position lamps on 近光灯工作时, 位置灯处于打开状态

SF-21-3 High beam 远光灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	×	√	√

(1) Functional operating conditions 功能操作条件

- ① Vehicle state as above table 车辆状态如上表所示;

(2) Function operation - open 功能操作-打开:

- ① BCM reception SS CAN message SS_HighBeamSts= 1 "ON"; (BCM 接收 SS CAN 信号输入)

(3) Function operation output 功能操作输出:

- ① BCM drives the high beam Solenoid valve operation; The vehicle is in high beam mode , BCM send BCM_HighBeamSts= 1 "ON"

BCM 驱动远光灯电磁阀工作; 车辆处于远光灯模式

(4) Function operation-close 功能运行关闭:

- ① BCM power change to not NO, or BCM reception SS CAN message SS_HighBeamSts= 0"OFF"

BCM 电源切换到非 ON, 或 BCM 接收 SS CAN 信号输入 SS_HighBeamSts= 0"OFF"

- ② BCM turns off the hight beam, BCM send BCM_HighBeamSts= 0 "OFF" BCM 关闭远光灯输出;

(4) Note 注释:

- ① When the high beam lamp works, the low beam position lamp works at the same time
远光灯工作时, 近光灯位置灯同时工作.
- ② The CAN message BCM_hightBeamSts= 1 "ON" can only be sent when the BCM has completed the left & right hight beam driver both output successfully.
BCM 完成左&右远光灯驱动都输出成功时, 才能发送 CAN 信息 BCM_hightBeamSts= 1 "ON".

SF-21-4 Flash light 超车灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	×	√	√

(1) Functional operating conditions 功能操作条件

- ① Vehicle state as above table 车辆状态如上表所示.

(2) Function operation - open 功能操作-打开:

- ① BCM reception SS CAN message SS_Overtaking_lamp= 1 "ON"; BCM 接收 SS CAN 信号输入

② BCM drives the high beam lamp solenoid valve work. The low beam lamps work at the same time, BCM send BCM_LowBeamSts=1 "ON"&BCM_HighBeamSts=1 "ON";

BCM 驱动远光灯电磁阀工作。低光灯同时工作;

(3) Function operation-close 功能操作-关闭:

- ① BCM reception SS CAN message SS_Overtaking_lamp= 0 "OFF"; BCM 接收 SS CAN 信号输入

② BCM turn off the high beam lamp and the low beam lamp, BCM send BCM_LowBeamSts= 0 "OFF" & BCM_HighBeamSts= 0 "OFF"; BCM 关闭远光灯和近光灯;

(4) Note:

① When the position lamp or low beam lamp works, turn on the Overtaking light function. If Overtaking is turned off, the working state of the position lamp or low beam lamp remains unchanged.

当位置灯或近光灯工作时, 开启超车灯功能, 如果 “超车 ” 功能关闭, 位置灯或近光灯的工作状态保持不变。

SF-21-5 Brake light 刹车灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	√	√	√	√

(1) Functional operating conditions 功能操作条件

① Vehicle state as above table 车辆状态如上表所示。.

(2) Function operation input 功能操作输入:

① BCM receives CAN signal sent by VCU VCU_BrakePedalSts= 1 "pressed" or BCM detects brake switch pressed, BCM sends BCM_STS_brake_pedal= 1 "pressed"

BCM 接收由 VCU 发送 CAN 信号 VCU_BrakePedalSts= 1 "pressed"或 BCM 检测到制动开关按下,BCM 发送 BCM_STS_brake_pedal= 1 "pressed"

(3) Function operation output 功能操作输出:

① BCM drives the left and right brake lamps 。 BCM send BCM_StopLightSts=1"ON"

BCM 驱动左右刹车灯, 同时发送 BCM_StopLightSts=1"ON"

(4) Closing conditions 关闭条件:

① BCM does not receive CAN signal from VCU BCM_StopLightSts=1 "ON" or signal is lost or verified incorrectly

BCM 未接收到 VCU 发送 CAN 信号 BCM_StopLightSts=1"ON"或信号丢失或验证错误

② BCM did not detect brake switch pressed BCM 未检测到刹车开关按下

SF-21-6 Reversing light 倒车灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
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Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	×	×	√

(1) Functional operating conditions 功能操作条件

① Vehicle state as above table 车辆状态如上表所示.

(2) Function operation input 功能操作输入:

1. BCM receives R gear signal and valid signal from VCU , VCU_ActualGear= 1 "R"& VCU_ActualGearValid= 1 "nvalid"

BCM 接收到 VCU 发送的 R 档信号及有效信号

(3) Function operation output 功能操作输出:

1. BCM drives the Reversing lamp works.BCM send BCM_ReverseLightSts= 1 "ON"

BCM 驱动倒车灯工作.

SF-21-7 Turn light 转向灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	√	√	√

(1) Functional operating conditions 功能操作条件

① Vehicle state as above table 车辆状态如上表所示.

(2) Function operation input 功能操作输入:

① BCM receives turn signal switch signal (left or right) SS_TurnIndicatorSts = 1 "Left On"||2"Right On"

BCM 接收组合开关信号（左或右）SS_TurnIndicatorSts.

(2) Function operation output 功能操作输出:

① BCM drives the turn signal lamp to work (left turn signal lamp or right turn signal lamp , BCM send BCM_TurnIndicatorSts= 1 "Left On"||2"Right On"

BCM 驱动转向灯工作（左转向灯或右转向灯）

(3) Note 注释:

① The flashing frequency of turn signal lamp is 75 times / min(400ms on/400ms off), and the duty cycle is 50 ± 5%

转向灯闪烁频率为 75 次/ min（400ms 开/400ms 灭），占空比为 50±5%.

② When the turn signal lamp (front or rear bulb) are damaged, The BCM drives the corresponding side turn signal lamp to flash, and the flashing frequency is 200ms on and 200ms off, which has prompted the fault information 当转向灯（前灯泡或后灯泡）损坏时，BCM 驱动对应侧转向灯闪烁，闪烁频率为 200ms 亮，200ms 灭，提示故障信息；

③ When the left turn signal lamp is activated, the corresponding bulb is intact, the BCM sends " BCM_TurnIndicatorSts=1 "Left On" , The flashing frequency of turn signal lamp is 75 times / min(400ms on/400ms off)

左转向灯激活时，对应侧灯泡完好，BCM 发送“BCM_TurnIndicatorSts= 1 "Left On"”，转向灯闪烁频率为 75 次/ min（400ms 开/400ms 灭）

④ When the Right turn signal lamp is activated, the corresponding bulb is intact, the BCM sends " BCM_TurnIndicatorSts=2 "Right On" , The flashing frequency of turn signal lamp is 75 times / min(400ms on/400ms off)

右转信号灯激活时，对应灯泡完好，BCM 发送“BCM_TurnIndicatorSts=2"Right On"”，转向灯闪烁频率为 75 次/ min（400ms 开/400ms 灭）

⑤ When the left / right turn signal is off, BCM sends " BCM_TurnIndicatorSts=0" signals.

当左/右转信号关闭时，BCM 发送“bcm_turnindicators =0”信号。

SF-21-8 Lane change function 变道功能

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	√	√	√

(1) Functional operating conditions 功能操作条件

① Vehicle state as above table 车辆状态如上表所示。.

(2) Function operation input 功能操作输入:

①BCM receives SS sending lane change CAN signal: SS_lanechange_lampSts= 1 “Lane change to left”|| 2 “Lane change to right”

BCM 接收到 SS 发送变道 CAN 信号: SS_lane_change_lampSts= 1 "Left lane change"|| 2 "Right lane change"

(3) Function operation output 功能运行输出:

① The corresponding side turn signal lamp flashes 3 times 400msON/OFF , BCM send“BCM_TurnIndicatorSts= 1 "Left On"||2"Right On"

对应的侧转信号灯闪烁 3 次 400msON/OFF

(4) Note 注释:

1. When the lane change function is activated, if the steering switch is turned again, the BCM will flash 3 times.

当变道功能启动时，如果再次转动转向开关，BCM 将闪烁 3 次。

2. During the flashing process of the turn signal activated by the lane change function, if the other side's lane change is inputted again, the previous lane change output will stop. If the other side's lane change flashing function is immediately activated, the previous lane change output will stop.

在变道功能激活的转向灯闪烁过程中，如果再次输入对方的变道，则停止之前的变道输出。对方的变道闪烁功能立即被激活，之前的变道输出将停止

3. When the turn signal lamp (front or rear bulb) are damaged, The BCM drives the corresponding side turn signal lamp to flash, and the flashing frequency is 200ms on and 200ms off, which has prompted the fault information;

当转向信号灯（前灯泡或后灯泡）损坏时，BCM 驱动相应的侧转向信号灯闪烁，闪烁频率为 200ms 亮，200ms 灭，提示故障信息；

SF-21-9 Hazard warning light 危险警示灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	√	√	√	√	√

(1) Functional operating conditions 功能操作条件

- ① Vehicle state as above table 车辆状态如上表所示;
- ③ Hazard warning lights off 危险警示灯熄灭.

(2) Function operation input 功能操作输入:

1. BCM receives hazard warning lamp switch input , BCM send BCM_STAT_HazardWarn =1 "Active"

BCM 接收危险警告灯开关输入，BCM 发送 BCM_STAT_HazardWarn=1 "Active"。.

(4) Function operation output 功能操作输出:

① All turn signals flash at the same time(75 times / min(400ms on/400ms off)) , BCM send BCM_TurnIndicatorSts= 3 "Left and Right On"所有转向灯同时闪烁（75 次/分钟（400ms 开/400ms 关））；

- ② The BCM drives the hazard switch work indicator to work and the BCM sends BCM_Emergency_brake_light_Sts= 1 "ON"
BCM 驱动危险开关工作指示灯工作，BCM 发送 BCM_Emergency_brake_light_Sts= 1 "ON"

(5) Note 注释:

- ① The hazard switch work indicator flashes simultaneously with the external lights ,
危险开关工作指示灯与外部灯同时闪烁。
- ② The working frequency and fault are the same as that of turn signal lamp, but can signal is "BCM_TurnIndicatorSts=3 "Left and Right On"

. 工作频率和故障与转向灯相同，但可以发出“BCM_TurnIndicatorSts=3"Left and Right On"的信号。

③ The change of power gear status will not affect the working state of hazard warning lamp-Synchronous operation with turn signal lamp. 电源档位的变化不影响危险警示灯的工作状态——与转向灯同步运行。

④ When the turn signal lamp function and the hazard warning lamp function are effective at the same time, the rear operation takes priority. 当转向灯功能和危险警示灯功能同时有效时，后方操作优先。

⑤ When the hazard warning lamp switch is pressed again, the hazard warning lamp, instrument and switch work indicator stop working. 再次按下危险警示灯开关时，危险警示灯、仪表、开关工作指示灯停止工作。

⑥ When the turn signal lamp (front or rear bulb) are damaged, The BCM drives the corresponding side turn signal lamp to flash, and the flashing frequency is 200ms on and 200ms off, which has prompted the fault information;.

当转向信号灯（前灯泡或后灯泡）损坏时，BCM 会驱动相应的侧转向信号灯闪烁，闪烁频率为 200ms 亮 200ms 灭，提示故障信息；。

⑦ The work indicator on the hazard switch is also used as the backlight of the switch 危险开关上的工作指示灯也用作开关的背光；

a. When the position light works, the hazard light is not turned on, and the BCM drive work indicator is always on; 位置灯工作时，危险灯不亮，BCM 驱动器工作指示灯亮

b. When the hazard light is turned off, the BCM stops flashing and changes to the current position light working state again; 当危险灯关闭时，BCM 停止闪烁并重新切换到当前位置灯工作状态；

SF-21-10 Emergency braking double flash-ABS 紧急制动双闪- abs

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	×	×	√

(1) Functional operating conditions 功能操作条件

① Vehicle state as above table 车辆状态如上表所示；

(2) Function operation input 功能操作输入：

① BCM receives the emergency braking signal from ABS (ABS_ESS_Active= 1 "active")

BCM 接收到 ABS 发出的紧急制动信号 (ABS_ESS_Active= 1“active”)

(3) Function operation output 功能操作输出：

① BCM Driven Hazard Warning Lights (300ms ON - 300ms OFF), BCM send BCM_TurnIndicatorSts=3 "Left and Right On";

BCM 驱动危险警示灯。（300ms ON - 300ms OFF）；

(4) Function OFF 关闭条件

When the vehicle meets any of the following conditions, this function will be off: 当车辆满足以下任一条件时，此功能将关闭：

- ① The power gear is from READY to ACC or OFF; 电源档位从 READY 到 ACC 或 OFF;
- ② BCM receives CAN signal (ABS_ESS_Active= 0 "inactive"); BCM 接收到 CAN 信号 (ABS_ESS_Active= 0 "inactive") ;

- ② Press the hazard warning light switch; 按下危险警示灯开关;

(4) Note 注释:

① When the emergency braking function is activated, the hazard warning lamp is activated for at least 3 seconds, and the flashing frequency of the hazard warning lamp is 300ms on-300ms off. BCM sends "BCM_TurnIndicatorSts=3", and the step is consistent with the flashing frequency of turn signal lamp.

当紧急制动功能激活时，危险警示灯至少激活 3 秒，危险警示灯闪烁频率为 300ms 开 300ms 灭。BCM 发送“BCM_TurnIndicatorSts=3”，频率与转向灯闪烁频率一致。

② When the emergency braking double flash function is activated within 3S, press the hazard alarm switch and keep the emergency braking double flash function activated at least 3S. 当紧急制动双闪功能在 3S 内激活时，按下危险报警开关，并保持紧急制动双闪功能激活至少 3S。

③ after the emergency braking double flash function is activated for 3S, press the danger alarm switch to close immediately. 紧急制动双闪功能激活 3S 后，按下危险报警开关立即关闭。

④ When the brake signal is valid, the brake lamp function is also valid. 当刹车信号有效时，刹车灯功能也有效。

SF-21-11 Reading lamp 阅读灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	√	√	√	√

(1) Functional operating conditions 功能操作条件

- ① Vehicle state as above table 车辆状态如上表所示;
- ② The reading lamp door switch is on 阅读灯 DOOR SW 打开了

(2) Function operation input 功能操作输入:

- ① The BCM receives an open signal from either the main driver's door or the passenger's door.

BCM 接收主驾门与副驾门中任意一扇的打开信号

(3) Function operation output 功能操作输出:

- ① BCM drive reading lamp on , .BCM send BCM_Fr_Dimming_dome_lampSts =1 "ON"

BCM 驱动阅读灯亮.

(4) Note 注释:

① When the manual switch of the reading lamp is turned on, the reading lamp is not affected by the state of the door 当阅读灯的手动开关打开时，阅读灯不受门的状态影响;

② When the reading light is turned on manually, any door (driver door and passenger door) is turned on, the instrument displays the door opening warning 当手动打开阅读灯，打开任意一扇门（主驾门与副驾门）时，仪表显示开门警告.

③ Within 15 seconds after the reading light is activated, turn the mechanical key to ON or HV ON and the ceiling light will immediately fade out. 阅读灯激活后 15 秒内，转动机械钥匙 ON 或 HV ON，阅读灯会立即逐渐熄灭。

④ When the manual switch of the reading lamp is on, when the door is closed, the BCM reading lamp control output needs to be in the suspended state. 当阅读灯的手动开关打开时，当门关闭时，BCM 阅读灯控制输出需要处于暂停状态。

⑤ When the vehicle power supply turns OFF and Deactivating anti-theft status, the reading lamp will turn off automatically after 15s 当车辆电源由非 OFF 变为 OFF 并处于未设防盗状态时，阅读灯将在 15s 后自动熄灭;

⑥ When the vehicle is locked in OFF and the mechanical key is latched, the reading light will go out immediately; 当车辆锁定在 OFF 状态，机械钥匙闭锁后，阅读灯会立即熄灭;

⑦ When the door is closed in IGN ON/ready mode, it will immediately turn off, and when the door is closed in other power modes, it will delay and light up for 15s

在 IGN ON/ready 模式下关门时立即关闭，在其他电源模式下关门时延时点亮 15s

MF-22 Wiper & Wash Control 雨刷和冲洗控制

The wiper washing control function involves components: BCM, front wipers, washing, wiper washing switch , etc. 雨刮器洗涤控制功能包括：BCM、前雨刮器、洗涤、雨刮器洗涤开关等组件。

The status signals of part are shown in the table below 部件的状态信号如下表所示：：

Output CAN signal 输出 CAN 信号

ID Name	Signal Name	Signal Description	Value Matrix
0x109			
	BCM_STS_wiper	BCM STS wiper	7 "Signal invalid" 6 Reserved 5 "Reserved" 3 "low spd" 4 "high spd" 2 Reserved 1 Reserved 0 "OFF"

	BCM_STAT_WiperStalkSW2	Status of Wiper stalk SW2(rear wiper and washer)	7 "Invalid" 6 "Reserved" 5 "Reserved" 4 "wash(Front wash)" 3 "wiper&wash" 2 "wiper (rear wiper)" 1 "OFF(rear wiper)" 0 "wiper&wash(rear wiper and rear wash)"
BCM_Comfort_Confirm_sts_0x41D			
	BCM_STAT_WiperRepairMode	Wiper Repair Mode (feedback to ss)	0 "Off" 1 "On" 2 "no action" 3 "Invalid"

Input CAN signal

ID Name	Signal Name	Signal Description	Value Matrix
SS_DriveReq_0x108			
	Checksum_0x108	Checksum	
	Rolling_Counter_0x108	Rolling_Counter	15 "Invalid"
	FR_Wipercontrol_switch	Front Wiper control switch 前雨刷控制开关	7 "Invalid" 6 "reserved" 5 "reserved" 4 "High" 3 "Int2" 2 "Int1" 1 "Low" 0 "OFF"
	Front_Wash	Front Wash Status 前洗状态	3 "Invalid" 2 "Wash" 1 "Mist" 0 "OFF"
	SS_TargetGearReq	Target gear request	7 "Reserved" 4 "Invalid" 3 "D" 2 "N" 1 "R" 0 "RightSwitch_Pressed"

schematic diagram (Auto CAD)

SF-22-1 Front Wiper control 前雨刷控制

The front wiper switch gears are divided into: OFF, MIST, INT1/ INT, Low speed and High speed.

See the following table for the gear and input signal of front wiper switch:

前雨刮器开关档位分为：OFF， MIST, INT1/ INT， 低速和高速。

前雨刷开关档位及输入信号见下表：

Level position on Column Switch 开关上的档位位置	Selected Input Status 选择输入状态		Front Wiper State 前雨刷状态
	Wiper 1 刮水器 1	Wiper 2 刮水器 2	
Mist 洗涤	OPEN 打开	CLOSE 关闭	Front Wiper Low 前雨刮低速
OFF 关闭	OPEN	OPEN	Front Wiper OFF 前雨刮低速
Interval 间隙	CLOSE	OPEN	Front Wiper Interval 前雨刷间歇
Low speed 低速	OPEN	CLOSE	Front Wiper Low 前雨刮低速
High speed 高速	CLOSE	CLOSE	Front Wiper High 前雨刮高速

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	×	√	√

A. Mist and low modes 洗涤和低速（洗涤模式）

(1) Functional operating conditions 功能操作条件

① When all the following conditions are met, BCM drives the front wiper to operate in low Speed:

当满足以下所有条件时，BCM 驱动前雨刷低速运行：

a.Vehicle state as above table 车辆状态如上表所示；

(2) Function operation input 功能操作输入：

① BCM receives CAN message from SS FR_Wipercontrol_switch = 1 “Low” or Front_Wash=2 “Wash”

BCM 接收到 SS 发送的 CAN 报文 FR_Wipercontrol_switch=1 “Low” or Front_Wash=2 “Wash”

(3) Function operation output 功能操作输出：

①BCM drives the wiper motor at low speed , BCM send BCM_STS_wiper= 3 "low spd" &BCM_STAT_WiperStalkSW2=3 "wiper&wash"

BCM 驱动雨刮电机低速运转

(4) Note 注释:

① When the low-speed wiper is activated: when the wiper switch switches to other modes(not OFF), the wiper will work in other modes immediately; 低速雨刮启动时：当雨刮开关切换到其他模式（非 OFF）时，雨刮将立即在其他模式下工作

② When the wiper switch is turned from low speed to off, the wiper will automatically work at low speed until it returns to the wiper stop position (whether ign on or not). 当雨刷开关从低速转到关闭状态时，雨刷将自动以低速工作，直到回到雨刷停止位置（无论是否点火）。

③ When there is a Mist signal in the wiper gear OFF, int1, and int2 states, prioritize executing the Mist action. Does not respond to Mist signals at low and high speeds. 当雨刷齿轮 OFF、int1 和 int2 状态下有洗涤信号时，优先执行洗涤动作。在低速和高速行驶时不响应洗涤信号

④SS shall trigger MIST and send a short time mist signal when pressing wash, BCM is required to block the time of triggering MIST, the time is set to: 150ms

SS 在按下 wash 时，须触发 MIST，发送短时间的 mist 信号，需要 BCM 屏蔽触发 MIST 的时间，时间设置为：150ms

B. High modes 高速模式

(1) Functional operating conditions 功能操作条件

① When all the following conditions are met, BCM drives the front wiper to operate in high speed:

满足下列所有条件时，BCM 驱动前雨刷高速运转：

a. Vehicle state as above table 车辆状态如上表

(2) Function operation input 功能操作输入:

① BCM receives CAN message from SS FR_Wipercontrol_switch =4 "High"

BCM 接收到 SS 发送的 CAN 报文 FR_Wipercontrol_switch =4 "High"

(3) Function operation output 功能操作输出:

①BCM drives the wiper motor at high speed , BCM send BCM_STS_wiper= 4 "high spd"

(4) Note 注释:

① When the high-speed wiper is activated; the wiper switch switches to other modes(not OFF or mist), the wiper will work in other modes immediately; 当启动高速雨刮器时：雨刮开关切换到其他模式（非 OFF 或洗涤），雨刮将立即在其他模式下工作

② When the high-speed wiper is started, washing will not affect the operation of the high-speed wiper

启动高速雨刷时，洗涤不会影响高速雨刷的运行

③ When the wiper switch is turned from high speed to off, the wiper will automatically work at low speed until it returns to the wiper stop position (whether ign on or not). 当雨刮开关从高速转到时，雨刮将自动以低速工作，直到回到雨刮停止位置（无论是否点火）

④ Lost and invalid wiper signals, the wiper will automatically work at low speed until it returns to the wiper stop position 雨刮器信号丢失或无效，雨刮器将自动低速工作，直到回到雨刮器停止位置

B. Interval modes 间隔模式

(1) Functional operating conditions 功能操作条件

① When all the following conditions are met, BCM drives the front wiper to operate in intermittent gear:

当满足下列所有条件时，BCM 将驱动前雨刷以间歇挡运行：

a. Vehicle state as above table 车辆状态如上表

(2) Function operation input 功能操作输入：

① BCM receives CAN message from SS FR_Wipercontrol_switch = 2 "Int1" or 3 "Int2"

BCM 接收到 SS 发送的 CAN 报文 FR_Wipercontrol_switch = 2 "Int1" or 3 "Int2"

(3) Function operation output 功能操作输出：

① BCM drives the wiper motor at Wiper Interval , BCM send BCM_STS_wiper=3 "low spd"

BCM 驱动雨刮电机运转

(4) Note 注释：

① When the intermittent wiper is activated: when the wiper switch switches to other modes, the wiper will work in other modes immediately; 间歇式雨刮器启动时：当雨刮器开关切换到其他模式时，雨刮器将立即在其他模式下工作；

② When the wiper switch is turned from intermittent to OFF, the wiper will automatically work at low speed until it returns to the wiper stop position (whether ign ON or not); 当雨刮开关从间歇转到关闭时，雨刮将自动以低速工作，直到回到雨刮停止位置（无论是否点火 ON）；

③ The interval of intermittent gear of wiper can be selected according to the sensitivity switch of wiper. The sensitivity switch is divided into two gears: 4s(int1) and 2s (int2) . 可根据雨刮灵敏度开关选择雨刮间歇档的间隔时间。灵敏度开关分为两档：4 秒（int1）和 2 秒（int2）。

④ When the intermittent gear of the wiper is activated, the intermittent time is switched to other gears, and the activation status of the wiper is executed as follows: 当启动雨刮的间歇档位时，间歇时间会切换到其他档位，雨刮的启动状态如下：

a. The new time interval is less than the original time interval: 新的时间间隔小于原来的时间间隔：

➤ If the wiper is suspended, the wiper will be activated at a new interval immediately;

如果雨刮暂停，雨刮将立即以新的间隔启动；

➤ If the wiper is in operation, the wiper will be activated at a new time interval at the next pause time;

如果雨刮器处于运行状态，则在下一次暂停时将以新的时间间隔启动雨刮器；

b. The new time interval is greater than the original time interval:

新的时间间隔大于原来的时间间隔：

➤ If the wiper is suspended, the wiper will be activated at a new interval after completing the current cycle;

如果雨刷暂停，雨刷将在完成当前周期后以新的间隔启动；

- If the wiper is in operation, the wiper will be activated at a new interval when it is suspended next time;

如果雨刮器处于工作状态，下次暂停时将以新的间隔启动雨刮器；

SF-22-2 Front Wiper service mode 前雨刮维修模式

Function description 功能说明

The wiper arm is located under the hood. Turn on the wiper service mode and stop the front wiper in the proper position. 雨刮刮臂位于机盖下方。打开雨刮服务模式，前雨刮停在适当位置

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	√	×	×

(1) Functional operating conditions 功能操作条件

- ① Vehicle state as above table 车辆状态如上表
- ② Front wiper is inoperative 前雨刷未进行工作
- ③ Front wiper is at parking position 前雨刷处于初始位置
- ④ Wiper service mode is inactive 雨刮服务模式处于非激活状态

(2) Function operation input 功能操作输入

- ① BCM receives CAN signal from SS : SS_TargetGearReq=0"RightSwitch_Pressed
BCM 接收 SS 发出的 CAN 信号: SS_TargetGearReq=0"RightSwitch_Pressed
- ② BCM monitor SS_TargetGearReq=0 "RightSwitch_Pressed time>1.5s, BCM execute maintenance mode
BCM 监控 SS_TargetGearReq=0"RightSwitch_Pressed 的时间>1.5s,BCM 执行维修模式

(3) Function operation output: 功能操作输出:

- ① BCM shall operate to drive wiper to "service position" 716ms (BCM calibrate for best position) and send **BCM_STAT_WiperRepairMode** = 1 "ON" and **BCM_STS_wiper=3** "low spd".
- ② BCM 应执行操作，将刮水器驱动至 716ms 的“维修位置”（BCM 校准最佳位置），并发送 BCM_STAT_WiperRepairMode= 1 "ON" 和 **BCM_STS_wiper=3** "low spd".

(4) Function OFF: 功能关闭:

Receive BCM monitor SStargetGearReg=0 again for 1.5s "RightSwitch_Pressed time >1.5s, or Power state switching to non-ACC (STAT_Terminal≠1 "ACC (Park)") BCM performs exit maintenance mode.

再次接收 BCM 监控 SStargetGearReg=0 再次“右开关按下时间 >1.5 秒，或电源状态切换为非 ACC (STAT_Terminal≠1 "ACC (Park)")”。BCM 执行退出维护模式。

(5) Note 注释:

① When vehicle state change to above ACC, BCM shall drive Front Wiper back to park position. BCM confirms current wiper service mode via 0x41D::BCM_STAT_WiperRepairMode = 0 "OFF"

当车辆状态变为 ACC 以上时，BCM 应将前雨刮调回初始位置。BCM 通过 0x41D::BCM_STAT_WiperRepairMode = 0 "OFF" 确认当前的雨刮维修模式。

② When vehicle STATE changes to OFF, BCM shall turn OFF wiper service mode immediately, BCM shall drive Front Wiper back to park position, BCM confirms current wiper service mode via 0x41D::BCM_STAT_WiperRepairMode = 0 "OFF"

当车辆 STATE 状态变为 OFF 时，BCM 应立即关闭雨刮服务模式，BCM 应将前雨刮回位至初始位置，BCM 通过 0x41D::BCM_STAT_WiperRepairMode = 0 "OFF" 确认当前的雨刮服务模式。

③ When the maintenance mode is activated, the BCM drives the wiper in low-speed mode and the CAN sends the corresponding signal value.

维修模式启动时 BCM 驱动雨刮低速模式，同时 CAN 发送对应信号值

SF-22-3 Front Wash mode 前洗模式

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	×	√	√

(1) Functional operating conditions 功能操作条件

① When all the following conditions are met, the cleaning motor before BCM drive works:

满足以下所有条件时，BCM 驱动器前洗涤将工作：

a. Vehicle state as above table; 车辆状态如上表；

(2) Function operation input 功能操作输入

BCM receives a CAN message from SS: Front_Wash= 2 "Wash"

BCM 接受到 SS 发送 CAN 信息：Front_Wash= 2 "Wash"

(3) unction operation output: 功能操作输出：

① BCM drives wiper low speed and washer motor operation, BCM send BCM_STG_wiper = 1 "3 "low spd "& BCM_STAT_WiperStalkSW2= 3 wiper&wash" 洗涤

(5) Note 注释

① BCM receives SS CAN signal Front_Wash= 1 "Mist", BCM drives washing motor to work, BCM sends BCM_STAT_WiperStalkSW2=4"wash(Front wash)

BCM 接收 SS CAN 信号 Front_Wash= 1 "Mist", BCM 驱动洗涤电机工作，BCM 发送 BCM_STAT_WiperStalkSW2=4"wash(Front wash) 喷水"

③ Driver operated wash switch for more than 150ms duration, SS sends CAN message : Front_Wash= 1 "Mist" (SS)

驾驶员操作洗涤开关在持续时间超过 150ms 时，SS 发出 CAN 信息：Front_Wash= 1 "Mist"

② At the end of pre purge activation: 在预清除激活结束时：

a. When the wiper switch is in off gear, the wiper will wipe for 3 cycles at low speed; If BCM receives a new pre cleaning activation request within 3 cycles, it will execute a new action; 当雨刮开关处于关闭挡位时，雨刮将以低速擦拭 3 个周期；如果 BCM 在 3 个周期内收到新的预清洁激活请求，它将执行新的操作；

SF-22-4 Wiper motor protection 雨刮电机保护

(1) The function is defined to protect the front wiper motor. 该功能用于保护前雨刷电机。

(2) When the wiper function is being activated, if the BCM can not detect the parking signal for 5s ,wiper output STOP,. 当启动雨刮功能时，如果 BCM 在 5 秒内无法检测到回位信号，则雨刮输出停止。

(3) Unprotect 取消保护：

⑧ Wiper mode switch switching (no ACC/OFF) 切换雨刮模式开关（无 ACC/OFF）

② Power state switching 电源状态切换

③ wash mode open 洗涤模式打开

MF-23 Accessory control function 附件控制功能

The status signals of part are shown in the table below: 部件的状态信号如下表所示：

Output CAN signal 输出 CAN 信号

ID Name	Signal Name	Signal Description	Value Matrix
BCM_SwitchSts_0x109			
	BCM_STAT_Horn_Reserved	Status switch horn 状态开关喇叭	"3 ""Signal invalid"" 2 ""Reserved"" 1 ""ON"" 0 ""OFF""
0x106			
	BCM_horn_relay_Sts	喇叭继电器控制指令 horn relay output status	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"

Input CAN signal

N/A

schematic diagram (Auto CAD)

SF-23-1 Horn control-BCM 喇叭控制

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	√	√	×	×	×

(1) Note 注释:

- ① When the mechanical key successfully unlocks all doors, The horn sounds twice (200ms On/600ms OFF); 当机械钥匙成功解锁全车门时，喇叭会响两声（200 毫秒开/600 毫秒关）；
- ② When the mechanical key successfully locks all doors, The horn sounds once (200ms On/600ms OFF); 当机械钥匙成功闭锁全车门时，喇叭会响一次（200 毫秒开/600 毫秒关）；

SF-23-2 Horn control-Manual 喇叭控制-手动

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	√	√	√	√	√

(1) Functional operating conditions 功能操作条件

- ① Vehicle state as above table 车辆状态如上表所示;

(2) Function operation input 功能操作输入:

- ① BCM detects horn switch input , BCM send BCM_STAT_Horn_Reserved = 1 "ON"

BCM 检测到喇叭开关输入

(3) Function operation output 功能操作输出:

- ① BCM drives horn relay operation BCM send BCM_horn_relay _Sts = 1 "ON" BCM 驱动喇叭继电器工作

(4) Note 注释:

- ① When the horn switch is pressed, the horn will sound. Release the horn switch and the horn stops working.

按下喇叭开关时，喇叭会响起。

- ② when the horn switch is not operated, BCM sends BCM_STAT_Horn_Reserved=0 "OFF", BCM does not drive the horn relay to work, BCM sends BCM_horn_relay _Sts=0 "OFF"

不操作喇叭开关时,BCM 发送 BCM_STAT_Horn_Reserved=0 "OFF", BCM 不驱动喇叭继电器工作, BCM 发送 BCM_horn_relay_Sts=0 "OFF"

MF-25 PDU function PDU 功能

Wake 唤醒

The status signals of part are shown in the table below: 部件的状态信号如下表所示:

Input CAN signals 输入 CAN 信号

ID Name	Signal Name	Signal Description	Value Matrix
VCU_Pedals_0x269			
	VCU_ActualGear	actual gear 实际档位	7 "Fault" 6 "Reserved" 5 "Reserved" 4 "Reserved" 3 "D" 2 "N" 1 "R" 0 "P"Reserved"
	VCU_ActualGearValid	Indicate the current gear position 显示当前档位位置	1 "invalid" 0 "Valid"
	VCU_VehicleHVStatus	Vehicle High voltage Ready State (VCU_VehicleHVStatus to be received by all modules,Used as a flash programming pre-condition (replacement for engine running) 车辆高压 RREADY 状态 (所有模块都将收到的 VCU_VehicleHVStatus, 用作闪存编程的前提条件 (替代发动机运转)。	3 "Reserved" 2 "ReadyToDrive" 1 "HV On" 0 "Not ready"
	VCU_BrakePedalSts	Brake Pedal Sts 制动踏板	3 "Invalid" 2 "Reserved" 1 "pressed" 0 "Not pressed"

Output CAN signals 输出 CAN 信号

ID Name	Signal Name	Signal Description	Value Matrix
BCM_VOLTAGE_0x10a			
	Ubatt	Battery reference Voltage(UBatt to be received by all	

		modules,Required for DTC snapshots) 电池参考电压（所有模块接收的 UBatt，DTC 快照所需的电压）	
BCM_CLAMP_STAT_0x112			
	STAT_Terminal	Represent the status of physical terminal(STAT_Terminal to be received by all modules,Required for DTC snapshots and for FOTA) 表示物理终端的状态 (STAT_Terminal 将被所有模块接收，DTC 快照和 FOTA 均需使用)	7 "Signal invalid" 6 "Reserved" 5 "Reserved" 4 "SW update OTA (Not used in VFe34)" 3 "Engine start (eMotor running)" 2 "ON (Driver Present)" 1 "ACC (Park)" 0 "OFF(Comfort Enable)"
	STAT_CL30s_L_Reserved	Status Clamp 30s Logical ON/OFF 电源 30 秒 逻辑 ON/OFF	3 "Signal invalid" 2 "Failure" 1 "ON" 0 "OFF"
BCM_Power_0x11b			
	BCM_ACCRelayOutputStatus	ACCRelays Output Status ACC 继电器 输出状态	0 "Inactive" 1 "Active" 2 "Reserved" 3 "Signal Invalid"
	BCM_IGN1RelayOutputStatus	IG1 Relays Output Status	0 "Inactive" 1 "Active" 2 "Reserved" 3 "Signal Invalid"
	BCM_ACCRelayFalutSts	ACCRelays Fault Status	0 "No Fault" 1 "Fault" 2 "Reserved"
	BCM_IGN1RelayFalutSts	IG1 Relays Fault Status	0 "No Fault" 1 "Fault" 2 "Reserved"
0x109			
	BCM_STS_ACC	BCM STS ACC	3 "Signal invalid" 2 "Reserved"

			1 "ON" 0 "OFF"
	BCM_STS_IG	BCM STS IG	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_STS_Start	BCM STS Start	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"

SF-25-1 Voltage Mode management 电压模式管理

The BCM detects the power supply voltage according to its own results, and discriminates and operates the relevant functions according to the voltage detection results. BCM 根据自身结果检测电源电压，并根据电压检测结果判别和操作相关功能。

The vehicle voltage is distinguished as follows: 车辆电压区分如下

Voltage Describe 电压描述	Voltage 电压(±0.5)	Duration 持续时间	Function description 功能说明	Remark 备注
Over voltage 过电压	> 18V	1.5s≥t	All functions stop	OVM
High voltage 电压 过高	16V<U≤18V	1.5s≥t	Few functions work	HVM
Normal voltage 正 常电压	9V≤U≤16V	1.5s≥t	All functions work	NVM
Low voltage 电压 过低	6.5V≤U<9V	1.5s≥t	Some functions work	LVM
Ultra low voltage 超低电压	U<6.5V	1.5s≥t	All functions stop	SVM

PIN NO.	Function	SVM U<6.5V	LVM 6.5V≤U <9V	NVM 9V≤U≤ 16V	HVM 16V<U≤ 18V	OVM >18V
J4-17	FR_WIPER_HI_OUTPUT	FSC C	FSC C	FSC A	FSC C	FSC C
J4-19	FR_WIPER_LO_OUTPUT	FSC C	FSC C	FSC A	FSC C	FSC C
J4-6	DR_DOOR_UNLOCK_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC A
J4-4	DOOR_LOCK_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC A
J4-2	DOOR_UNLOCK_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC A
J4-14	FL_WIN_UP_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C

J4-16	FL_WIN_DN_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C
J4-20	FR_WIN_UP_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C
J4-22	FR_WIN_DN_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C
J3-3	Backlight_Adjust_OUTPUT	FSC C	FSC C	FSC A	FSC C	FSC C
J3-6	STOP_LAMP_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C
J3-7	LICENSE_PLATE_LAMP_OUT PUT	FSC C	FSC C	FSC A	FSC C	FSC C
J3-8	REVERSE_LAMP_OUTPUT	FSC C	FSC C	FSC A	FSC C	FSC C
J3-9	POSITION_LAMP_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C
J3-1	LEFT_FLASHER_OUTPUT	FSC C	FSC A	FSC A	FSC A	FSC C
J3-10	RIGHT_FLASHER_OUTPUT	FSC C	FSC A	FSC A	FSC A	FSC C
J3-12	IGN1_RLY_OUTPUT	FSC C	FSC C	FSC A	FSC C	FSC C
J3-13	ACC_RLY_OUTPUT	FSC C	FSC C	FSC A	FSC C	FSC C
J3-21	LEFT_LOW_BEAM_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C
J3-20	Right_LOW_BEAM_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C
J3-23	Right_HIGH_BEAM_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C
J3-24	LEFT_HIGH_BEAM_OUTPUT	FSC C	FSC C	FSC A	FSC A	FSC C
J2-14	HAZARD_LED_OUTPUT	FSC C	FSC C	FSC A	FSC C	FSC C
J2-13	CDL_STATUS_LED_OUTPUT	FSC C	FSC C	FSC A	FSC C	FSC C
J1-13	HORN_RLY_OUTPUT	FSC C	FSC A	FSC A	FSC A	FSC C
J4-03	Dimming_Dome_LAMP_OUTP UT	FSC C	FSC C	FSC A	FSC C	FSC C

Function Availability vs Voltage Mode 功能可用性与电压模式的关系

* No deactivation allowed, it has to work as long as possible 不允许停用，必须尽可能长时间地工作

Note 注释:

- Some functional work: it refers to the functions of some electrical parts in operation: door and window motors, etc 一些功能性工作：指一些电气部件在运行中的功能：如门窗电机等
- Real-time monitoring of the whole vehicle battery voltage threshold after the BCM wakes up, sending the voltage to the CAN network in real time
BCM 唤醒后实时监测整车蓄电池电压阈值，将电压实时发送到 CAN 网络中
- Functional status A**
All functions of the DUT must be performed as specified during and after exposure to the test parameters.
Note: The functional status A must also be achieved in the case of deviations of the designed performance (function limitation at high and low temperatures) when the operating voltage is 、 fallen below or exceeded, if this is described in the Component Performance Specifications as permissible (derating)
- Functional status C**

One or more functions of the DUT are not performed as specified during exposure to the test parameters, but automatically continue to fulfill all functions as specified after the end of exposure. Undefined functions are not permitted.

SF-25-2 Bus sleep and Wake-up conditions. 总线休眠和唤醒条件。

(1) BUS sleep conditions 总线睡眠条件

① Sleep conditions for CAN communication. CAN 通信的休眠状态

② Body-CAN will go to sleep after all the following conditions are met:

满足以下所有条件后，Body-CAN 将进入睡眠状态：

a-Vehicle goes to OFF mode(STAT_Terminal=OFF);车辆进入 OFF 模式

b-ACC Power not ON; ACC 电源关闭

c-IG Power not ON; IG 电源关闭

d-horn OFF 喇叭无操作

e- door not open or close;门没有打开或关闭

f-drive's door lock status signal not change; 驾驶室门锁状态信号不变;

g-Hazared light SW not operation; 危报开关不工作

h-Central lock SW not operation; 中控锁开关 不工作;

i- No operation of the brake pedal 制动踏板无操作

j-No network management requests from other ECUs for a running CAN bus.

运行中的 CAN 总线没有来自其他 ECU 的网络管理请求。

③ Note:

a. BCM enters hibernation after 2 minutes when all hibernation conditions are met

BCM 所有休眠条件达到时，2 分钟后进入休眠状态

b. BCM enters sleep mode without low voltage charging function

BCM 进入睡眠模式，无智能补电功能

c.The above sleep is network stop sleep; 上述休眠是网络休眠要求;

d. When the vehicle is in the OFF state, the vehicle is not locked, and the doors are open, BCM enters network sleep after the indoor lights are turned off; 当车辆处于 OFF 状态、车辆未闭锁且车门打开时，BCM 会在关闭室内灯后进入网络休眠状态;

(2) BUS Wake-up conditions 总线唤醒条件

① Wake-up conditions for CAN CAN 的唤醒条件

② BCM and Body-CAN will wake up, if any of the following conditions is met:

如果满足以下任何条件，BCM 和 Body-CAN 将被唤醒:

a-ACC Power ON;

b-IG Power ON;

c-Break pedal is pressed down -踩下制动踏板

d-horn on 喇叭

e-two door open or close; 两扇门打开或关闭;

f-drive's door lock status signal change(Mechanical key unlocking, locking and other operations); 驾驶室门锁信号变化（机械钥匙开锁、闭锁和其他操作）;

j- hazard sth light SW operation; 危险开关操作

h-Central lock SW operation; 中控锁开关 操作

i-network management requests from other ECUs for a running CAN bus. 其他 ECU 对运行中的 CAN 总线提出的网络管理请求。

③ Note:

Since the wake-up source does not include the high beam switch, the high beam switch, overtaking light switch and turn light SW cannot wake the BCM after deep sleep.

由于唤醒源不包括远光灯开关，因此远光灯开关、超车灯开关和转向灯 开关 不能在睡眠后唤醒 BCM。

SF-25-3 Vehicle BCM Power model 车辆 BCM 电源模型

Output CAN signal 输出 CAN 信号

ID Name	Signal Name	Signal Description	Value Matrix
0x109			
	BCM_STS_ACC	BCM STS ACC	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_STS_IG	BCM STS IG	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_STS_Start	BCM STS Start	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_STS_brake_pedal	BCM STS brake pedal	3 "Invalid" 2 "Reserved" 1 "pressed" 0 "Not pressed"
BCM_Power_0x11b			
	BCM_ACCRelayOutputSts	ACCRelays Output Status	0 "Inactive" 1 "Active" 2 "Reserved" 3 "Signal Invalid"
	BCM_IGN1RelayOutputSts	IG1 Relays Output Status	0 "Inactive" 1 "Active"

			2 "Reserved" 3 "Signal Invalid"
	BCM_ACCRelayFalutSts	ACCRelays Fault Status	0 "No Fault" 1 "Fault" 2 "Reserved" 3 "Signal Invalid"
	BCM_IGN1RelayFalutSts	IG1 Relays Fault Status	0 "No Fault" 1 "Fault" 2 "Reserved" 3 "Signal Invalid"

Input CAN signal 输入 CAN 信号

ID Name	Signal Name	Signal Description	Value Matrix
ABS_WheelSpdR_0x342			
	Checksum_0x342	Check sum	
	Rolling_Counter_0x342	Rolling Counter	15 "Invalid"
	ABS_VehSpd	Current Longitudinal Vehicle Speed	32767 "Invalid"
	ABS_VehSpdValid	Speed Valid Flag	1 "Valid" 0 "Invalid"
VCU_Pedals_0x269			
	Checksum_0x269	Check sum	
	Rolling_Counter_0x269	Rolling counter	15 "Invalid"
	VCU_VehicleHVStatus1	Vehicle High voltage Ready State	7 "Reserved" 6 "Reserved" 5 "Reserved" 4 "Error" 3 "Charging" 2 "ReadyToDrive" 1 "HV On" 0 "Not ready"
	VCU_ActualGear	actual gear	5-7 "Reserved" 4 "Invalid" 3 "D" 2 "N" 1 "R" 0 RightSwitch_Pressed

	VCU_ActualGearValid	Indicate the current gear position	1 "Invalid" 0 "Valid"
	VCU_BrakePedalSts	Brake Pedal Sts	3 "Invalid" 2 "Reserved" 1 "pressed" 0 "Not pressed"

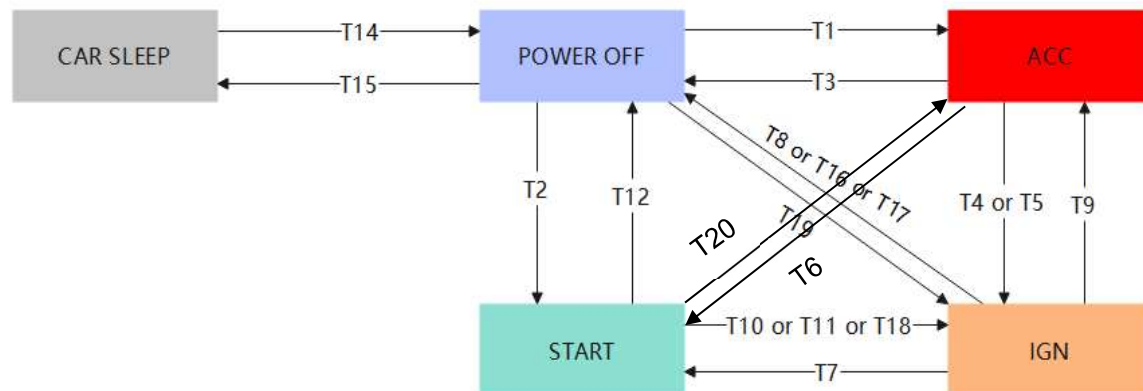
Vehicle State	Battery Disconnected 电池未连接	Car Sleep	OFF (car awake)	ACC	IGN ON (Driver Present)	START (E-motor Running)	OFF (car awake)	Car Sleep	Battery Disconnected
Value of BCM_Power Mode 电源模式	N/A	N/A	OFF	ACC	ON	START	OFF	N/A	N/A
ACC Relay Status 继电器	Open	Open	Open	Closed	Closed	Closed	Open	Open	Open
IGN Relay Status	Open	Open	Open	Open	Closed	Closed	Open	Open	Open

For the purposes of determining the E-motor status, the BCM shall interpret the high voltage status from the BCM as per the below table. 为确定电动机状态，BCM 应按下表解释来自 BCM 的高压状态

Value of BCM_VehicleHVStatus BCM 车辆 HV 状态的价值	BCM Interpreted E-motor Status BCM 解读电机状态
"Ready"	Running
Any other value	Not Running

When the START power mode is entered the BCM shall send **BCM_PowerMode** = "START" keep it in START if receive the feedback signal from VCU **BCM_VehicleHVStatus**= "Ready". If not back to IGN ON.

当进入 START 电源模式时，如果收到来自 VCU 的反馈信号 BCM_VehicleHVStatus= "Ready"，BCM 应发送 BCM_PowerMode = "START"，使其保持在 START 状态。如果没有，则返回到 IGN ON（接通）



The vehicle power mode switch conditions as follow description: 车辆电源模式切换条件说明如下：

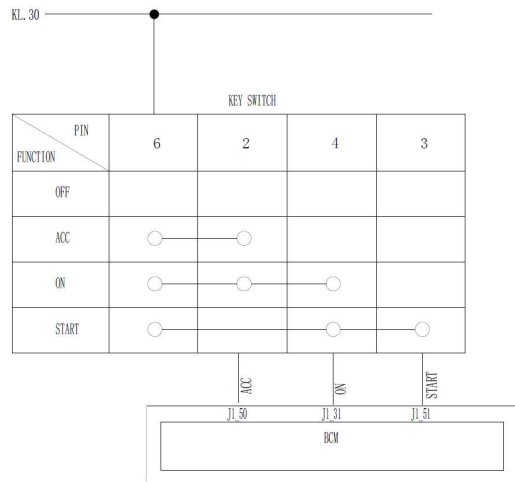
T1	The key switch to ACC gear 钥匙切换到 ACC 档;
T2	1.The key switches to START gear 钥匙切换到 START 挡. 2.The brake pedal is pressed down 踩下制动踏板. 3.The Gear is N, and the Gear signal is valid 齿轮为 N, 齿轮信号有效
T3	witch the key to OFF if the power status is in ACC and more than 5 minutes; If there is no power distribution operation within 5 minutes, the power will stay at ACC and the BCM controller will not enter into sleep. 如果电源状态处于 ACC 且超过 5 分钟, 则将钥匙切换至 OFF; 如果在 5 分钟内 有配电操作, 则电源将保持在 ACC, BCM 控制器不会进入休眠状态
T4	The key switch to ON gear 将钥匙切换到 ON 档.
T5(reserved)	
T6	1.The key switches to START gear 钥匙切换到启动档. 2.The brake pedal is pressed down 踩下制动踏板. 3.The Gear is N, and the Gear signal is valid 档位为 N, 档位信号有效
T7	1.The key switches to START gear 钥匙切换到 START 挡. 2.BCM receives brake CAN signal (The brake pedal is pressed down). BCM 接收 到制动 CAN 信号 (制动踏板踩下) 3.The Gear is N, and the Gear signal is valid.档位为 N, 档位信号有效
T8	1.The key switches to OFF gear.钥匙切换到 OFF 档 2.The speed is <= 3km/h and speed signal valid 速度 <= 3km/h, 速度信号有效
T9	1.The key switch to ACC gear. 钥匙切换到 ACC 档
T10(reserved)	
T11(reserved)	

T12	1.The key switches to OFF gear.钥匙切换到 OFF 档 2.The Gear is N, and the Gear signal is valid.档位为 N, 档位信号有效 3.The speed is <= 3km/h and speed signal valid. 速度 <= 3km/h, 速度信号有效
T14	Meets wake-up conditions(SF-25-2 BUS Wake-up conditions) 满足唤醒条件 (SF-25-2 总线唤醒条件)
T15	Meets bus sleep conditions(SF-25-2 BUS sleep conditions) 满足总线休眠条件 (SF-25-2 总线休眠条件)
T16(reserved)	
T17(reserved)	
T18(reserved)	
T19	1.The key switch to ON gear OR Switch key to Start without press brake.
T20	1.The key switches to ACC gear.钥匙切换到 ACC 档 2. The Gear is N, and the Gear signal is valid.档位为 N, 档位信号有效 3.The speed is ≤ 3km/h and speed signal valid. 速度 ≤ 3km/h, 速度信号有效

Note:注释

- ① The above vehicle speed signals are ABS output signals; The gear signal is the output gear signal Driver door ed by BCM.
以上车速信号为 ABS 输出信号; 档位信号是由 BCM 驱动的输出档位信号。
- ② In the whole vehicle entering the ACC state, the BCM will drive the ACC relay to close and at the same time issue STAT_Terminal =1 "ACC (Park)" and BCM_ACCRelayOutputSts= 1 "Active"
在整车进入 ACC 状态中, BCM 会驱动 ACC 继电器闭合, 同时发出 STAT_Terminal =1 "ACC (Park)"与 BCM_ACCRelayOutputSts= 1 "Active"
- ③ IG1 (ACC) relay status feedback changes from a suspended signal to a high level, BCM sends BCM_ACCRelayFalutSts= 0 "No Fault" IG1 (ACC) 继电器状态反馈从悬浮信号变为高电平, BCM 发送 BCM_ACCRelayFalutSts= 0 "No Fault"
- ④ In the case that the whole vehicle enters the HV ON state, the BCM will drive the ON relay to close and send STAT_Terminal = 2 "ON (Driver Present)" and BCM_IGN1RelayOutputSts= 1 "Active" at the same time
在整车进入 HV ON 状态中, BCM 会驱动 ON 继电器闭合, 同时发送 STAT_Terminal = 2 "ON (Driver Present)"与 BCM_IGN1RelayOutputSts= 1 "Active"
- ⑤ IG2 (ON) relay status feedback changes from a suspended signal to a high level, BCM send BCM_IGN1RelayFalutSts= 0 "No Fault"
IG2 (ON) 继电器状态反馈从悬浮信号变为高电平时, BCM 发送 BCM_IGN1RelayFalutSts= 0 "No Fault"
- ⑥ During ACC relay closure, the BCM detects a relay fault and issues the CAN message BCM_ACCRelayFalutSts= 0 "No Fault"
ACC 继电器闭合过程中, BCM 会检测继电器故障, 并发出 CAN 信息 BCM_ACCRelayFalutSts= 0 "No Fault"
- ⑦ During ON relay closure, the BCM detects a relay fault and sends a CAN message BCM_IGN1RelayFalutSts= 0 "No Fault"
ON 继电器闭合过程中, BCM 会检测继电器故障, 并发出 CAN 信息 BCM_IGN1RelayFalutSts= 0 "No Fault"

- ⑧ When the BCM receives the ignition switch gear signal, the BCM sends BCM_STS_ACC=1 "ON"||BCM_STS_IG=1 "ON"||BCM_STS_Start=1 "ON"
BCM 在收到点火开关齿轮信号时, BCM 会发送 BCM_STS_ACC=1 "ON"||BCM_STS_IG=1 "ON"||BCM_STS_Start=1 "ON"
- ⑨ When the ignition is ACC and the ACC relay is not closed, the ignition lock jumps to ON and the ON relay should be closed, the ignition lock schematic is shown below:
在点火为 ACC 时, ACC 继电器未闭合情况下, 点火锁跳转为 ON, ON 继电器应闭合, 点火锁原理图如下图:



- ⑩ Low-voltage power status of the entire vehicle, sent by BCM, the signal is STAT_Terminal=0 "OFF (Comfort Enable)"||"ACC (Park)"||"ON (Driver Present)"
整车的低压电源状态, 由 BCM 发送, 信号是 STAT_Terminal=0 "OFF(Comfort Enable)"||"ACC (Park)"|| "ON (Driver Present)"
- ⑪ High voltage power status of the entire vehicle, sent by the VCU, is signaled by VCU_VehicleHVStatus1=0 "Not ready"|| 1 "HV On"|| 2 "ReadyToDrive"|| 4 "Error"
整车高压电源状态, 由 VCU 发送, 信号是 VCU_VehicleHVStatus1=0 "Not ready"|| 1 "HV On"|| 2 "ReadyToDrive"|| 4 "Error"
- ⑫ Add this to prevent this abusing of a key:

Within 10sec, if the driver do continuous acc-on the key for more than 6 times, keep the car in ACC for 10sec(even if the key if on), after 10sec it behaves normally based on the key position.

Explanation:

- When key switch from OFF/ACC to ON/START, start a timer and counter.
- Every time the HV ON/OFF, counter +1, if during timer<10s, counter>6 (3 cycles for HV On off), keep at BCM_Powermode = ACC and release after 10s.
- After 10s:
 - if key is in OFF: then BCM_power mode change to OFF
 - if key is in ACC: then BCM_power mode keeps at ACC, relay ACC also ON.
 - if key is in ON/START: then BCM_powermode change to ON/START

SF-25-4 Limp home mode 跛行（故障）运行模式

BCM shall perform the following functional strategy steps: BCM 应执行以下职能战略步骤:

- brake light functionality: BCM shall ensure activation of brake light according read in brake light switch, in case of micro chip of BCM cannot work. 刹车灯功能：BCM 应根据刹车开关上的读数确保刹车灯的激活，以防 BCM 的微芯片无法工作。
- front wiper functionality: BCM shall ensure activation of low speed wiping according read in Stalk Switch , in case of MCU of BCM cannot work. 前雨刷功能：如果 BCM 的 MCU 无法工作，BCM 应确保启动低速雨刮。
- Low beam/position light: BCM shall ensure activation of low beam and position light when the car is start readySTART mode (emotor running), in case of MCU of BCM cannot work. 近光灯/位置灯：当汽车处于启动准备启动模式（电机运行）时，BCM 应确保近光灯和位置灯处于激活状态，以防 BCM 的 MCU 无法工作。
- Horn: BCM shall ensure activation of HORN by pressing relevant horn button, in case of MCU in BCM cannot work. 喇叭：当 BCM 中的 MCU 无法工作时，BCM 应按下相应的喇叭按钮确保喇叭激活。

MF-26 IPC & BCM Display function IPC 和 BCM 显示功能

This function mainly involves BCM, IPCand other components;
It mainly sends fault, light and other information from BCM, and finally displays it to IPCto achieve prompt or warning function.

它主要从 BCM 发送故障、灯光等信息，最后显示给 IPC以实现提示或警告功能。

Output CAN signals

ID Name	Signal Name	Signal Description	Value Matrix
BCM_ODO_0X20D			
	Checksum_0x20D	Checksum	
	Rolling_Counter_0x20D	Rolling_Counter	15 "Invalid"
	BCM_ODO	odograph	1048575 "Invalid"
BCM_LIGHT_0x106			
	Checksum_0x106	Checksum	
	Rolling_Counter_0x106	Rolling_Counter	15 "Invalid"
	BCM_HighBeamSts	High beam output Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_LowBeamSts	Low beam output Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_PosLmpSts	Position light output Status	0 "OFF" 1 "ON"

			2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_TurnIndicatorSts	Turn light output Status	0 "OFF" 1 "Left On" 2 "Right On" 3 "Left and Right On"
	BCM_Overtaking_lamp	Overtaking lamp output Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_ReverseLightSts	Reverse light output Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_StopLightSts	Stop light output Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_Fr_Dimming_dome_lamp Sts	Dimming dome lamp Front Status	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
	BCM_Emergency_brake_light_Sts	双闪灯控制指令（车内面板按键） Emergency brake light output status(Interior panel button)	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_number_plate_light_Sts	牌照灯控制指令 number plate light output status	3 "Signal invalid" 2 "Reserved" 1 "ON"

			0 "OFF"
	BCM_horn_relay_Sts	喇叭继电器控制指令 horn relay output status	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_Doorlock_ledlight_Sts	门锁 led 灯控制（车内面板按键） Door lock led light output status(Interior panel button)	3 "Signal invalid" 2 "Reserved" 1 "ON" 0 "OFF"
	BCM_STAT_InteriorLight	status of backlight illumination	0 "OFF" 1 "ON" 2 "Fault (Reserved)" 3 "Signal Invalid"
BCM_STAT_DOOR_FLAP_0x105			
	Checksum_0x105	Checksum	
	Rolling_Counter_0x105	Rolling_Counter	15 "Invalid"
	BCM_STAT_DriverDoor	Driver's door status 司机车门状态	3 "Signal invalid" 2 "Reserved" 1 "Opened" 0 "Closed"
	BCM_STAT_PassDoor	Passenger's door status 乘客门状态	3 "Signal invalid" 2 "Reserved" 1 "Opened" 0 "Closed"
	BCM_STS_TailDoor	尾门开关状态 Switch status of the tail door	3 "Signal invalid" 2 "Reserved" 1 "Opened" 0 "Closed"

Input CAN signals

ID Name	Signal Name	Signal Description	Value Matrix
IPC_ODO_0x20C			
	Checksum_0x20C	Checksum	
	Rolling_Counter_0x20C	Rolling_Counter	15 "Invalid"
	IPC_ODO	odograph	1048575 "Invalid"

SF-26-1 Pso.lamp working instruction diaplay 位置灯工作指示灯

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	√	√	√

(1)Function on 功能开启:

- ① When the position lampstart working, the BCM sends the position lamp working status signal (BCM_PosLmpSts =0x1 "ON"): 位置灯开始工作时, BCM 发送位置灯工作状态信号
- ② When IPC receives this signal, the driving position lamp working indicator is on;
当 IPC 收到该信号时, 驾驶位置指示灯工作指示灯亮起;

(2) Function off 功能关闭:

- ① When the position lamp stop working, the BCM sends the position lamp stop working status signal (BCM_PosLmpSts =0x0"OFF") :
位置灯停止工作时, BCM 发送位置灯停止工作状态信号 (BCM_PosLmpSts =0x0 "OFF") :
- ② When IPC receives this signal, the driving position lamp working indicator goes out.
当 IPC 接收到该信号时, 驾驶位置指示灯工作指示灯熄灭。

SF-26-2 High beam working instruction diaplay 远光灯工作指示灯显示

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	×	×	√	√

(1) Function ON 功能开启:

- ① When the High beam start working, the BCM sends High beam working status signal (BCM_HighBeamSts =0x1 "ON") 远光灯开始工作时, BCM 发送远光灯工作状态信号 (BCM_HighBeamSts =0x1 "ON") ;
- ① When IPC receives this signal, the driving High beam working indicator is on.

当 IPC 接收到该信号时，行驶远光灯工作指示灯亮起

(2) Function OFF 功能关闭:

① When the High beam stop working, the BCM sends the High beam stop working status signal (BCM_HighBeamSts = 0x0 "OFF");

远光灯停止工作时，BCM 发送远光灯停止工作状态信号 (BCM_HighBeamSts = 0x0 "OFF") ;

② When IPC receives this signal, the driving High beam working indicator goes out.

当 IPC 接收到该信号时，驾驶远光灯工作指示灯熄灭。

(3) Note 注释:

① when the high beam lamp works, the working indicator will be on. This function is also applicable in flash.

当远光灯工作时，工作指示灯将亮起。该功能也适用于超车灯。

SF-26-3 Turn & Hazard lamp working instruction display 转向灯与危险灯工作指示灯显示

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	√	√	√	√	√

(1) Function ON 功能开启:

① Left turn lamp 左转向灯

a. When the left Turn lamp start working, the BCM sends the left Turn lamp working status signal

当左转向灯开始工作时，BCM 发出左转向灯工作状态信号 (BCM_TurnIndicatorSts = 0x1 "Left On") ;

b. The IPC receives the left Turn lamp working status signal sent by BCM, turn on the left Turn lamp working indicator, accompanied by Synchronous sound.

IPC 接收到 BCM 发送的左转向灯工作状态信号后，打开左转向灯工作指示灯，并伴有同步声。

② Right turn lamp 右转向灯

a. When the Right Turn lamp start working, the BCM sends the right Turn lamp working status signal

当右转向灯开始工作时，BCM 发送右转向灯工作状态信号 (BCM_TurnIndicatorSts = 0x2 "Right On") ;

b. the IPC receives the right Turn lamp working status signal sent by BCM, turn on the right Turn lamp working indicator, accompanied by Synchronous sound.

IPC 接收到 BCM 发送的右转向灯工作状态信号后，打开右转向灯工作指示灯，并伴有同步声

③ Hazard lamp 危险灯

a. When the hazard lamp start working, the BCM sends the hazard lamp working status signal (BCM_TurnIndicatorSts = 0x3 "Left and Right On") ; the IPC receives the hazard lamp working status signal sent by BCM, turn on the Left and right Turn lamp working indicator, accompanied by Synchronous sound.

当危险指示灯开始工作时，BCM 发出危险指示灯工作状态信号 (BCM_TurnIndicatorSts = 0x3 "Left and Right On") , IPC 接收到 BCM 发送的危险指示灯工作状态信号后，打开左右转向灯工作指示灯，并伴有同步声。

(2) Function OFF 功能关闭:

① When left or Right Turn lamp or Hazard lamp stop working, the BCM sends the Turn lamp stop working status signal(BCM_TurnIndicatorSts= 0x0 "OFF");

当左、右转向灯或危险指示灯停止工作时，BCM 发送转向灯停止工作状态信号（BCM_TurnIndicatorSts= 0x0 "OFF"）；

② When the IPC receives the lamp stop working status signal sent by BCM, turn off the turn lamp working indicator and prompt sound.

当 IPC 收到 BCM 发送的车灯停止工作状态信号时，关闭转向灯工作指示灯并发出提示音。

(3) Note 注释:

① When IPC receives the working signal of turn signal or hazard light, in addition to flashing indication synchronously, relay analog pull-in sound prompt shall be carried out synchronously.

当 IPC 接收到转向灯或危险指示灯的工作信号时，除同步闪烁指示外，还应同步进行继电器模拟拉入声音提示。

SF-26-4 Open door warning 开门警告

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	√	√	√	√

(1) Function on 功能开启:

① When the BCM detects that any door or multiple doors (front left, front right, back door) are open, the BCM sends the opening signal of the corresponding side door;

当 BCM 检测到任何车门或多个车门（前左、前右、后背门）打开时，BCM 将发送相应门的打开信号

② According to the received BCM status signal,IPC displays the opening reminder of any door or multiple doors.

根据接收到的 BCM 状态信号，IPC 显示任意一扇门或多扇门的开启提示。

(2) Function OFF 功能 关闭:

① When the BCM detects that the three doors are closed, it sends three door closing signal; After IPC receives the signal that all three doors of BCM are closed, Stop the door opening reminder function.

当 BCM 检测到三扇门已关闭时，会发送三扇门关闭信号；当 IPC 收到 BCM 所有三扇门均已关闭的信号后，会停止开门提醒功能。

SF-26-5 Position light not turned off warning 位置灯未关功能

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	√	×	×	×

(1) Function on 功能开启:

① When all the following conditions are met, the BCM sends a Pos.lamp not turn off reminder:

满足以下所有条件时，BCM 将发送 Pos.lamp not turn off 提醒:

a.when The light switch is located in Pos. lamp or headlamp mode and theVehicle power is change to OFF; BCM sent the signal (BCM_Positionlight_reminder= 1 "Active") to IPC .

当车灯开关位于位置灯或大灯模式且车辆电源变为 OFF 时；BCM 向 IPC 发送信号

(2) Function OFF 功能关闭:

① after the Pos.lamp turn on 60Sconds 位置指示灯亮起 60 秒后

SF-26-6 ODO Storage backup ODO 存储备份

Function availability in Vehicle States 车辆状态下的功能可用性

Power status	ASLEEP	OFF	ACC	ON	ON
Vehicle State	Vehicle sleep	Comfort Enable	Accessory	Driver Present	Ready
Applicable	×	√	√	√	√

(1) BCM receive and memory ODO BCM 接收和存储 ODO:

① When all of the following conditions are met,BCM need update and stores the ODO information from the IPC (IPC _Odometer_IC = 1,048,575 "Invalid")

满足下列所有条件时，BCM 需要更新并存储来自 IPC 的 ODO 信息（MHU _Odometer_IC = 1,048,575 “nvalid（无效）”）。

Precondition 前提条件：（A&B）

a.Value received in CAN IPC _Odometer is less than 0xFFFFF.

CAN IPC_Odometer (里程)接收到的值小于 0xFFFFF

b.E2E verification is cosrrect. E2E 验证是正确的

② Trigger condition 触发条件

Value received in CAN IPC _Odometer 从 CAN IPC 收到的里程表值

(2) Note 备注:

① Value received in CAN signal IPC_Odometer is greater than ($\geq$) value stored in BCM+50KM && less than ($\leq$) value stored in BCM+100KM. BCM will stored IPC_Odometer Value

CAN 信号中接收到的 IPC 里程表值大于 BCM+50KM 中存储的值 ($\geq$) 且小于 BCM+100KM 中存储的值 ($\leq$)。BCM 将存储 IPC 里程表值

② Value received in CAN signal IPC_Odometer is higher than ($>$) value stored in **BCM+100KM**. If BCM receive a value longer than **2 minutes** BCM will stored IPC_Odometer Value (For update the BCM odometer when replacing the BCM with a new one)

CAN 信号中接收到的 IPC 里程表值高于 ($>$) BCM+100KM 中存储的值。如果 BCM 接收到的值超过 2 分钟，BCM 将存储 IPC 里程表值（用于在更换新 BCM 时更新 BCM 里程表）

③ Value received in CAN signal IPC_Odometer is less than the value stored in BCM。BCM sends the last BCM_Odometer CAN 信号 IPC 里程中接收到的值小于 BCM 中存储的值。BCM 发送最后一个 BCM 里程表信号

④ Value received in CAN signal IPC_Odometer is greater than value BCM_Odometer,BCM sent new value in BCM_Odometer CAN 信号 IPC 里程表中接收到的值大于 BCM 里程表中的值、BCM 向 BCM 里程表发送新值

MF-27 LV battery top-up strategy 低压电池补电逻辑

1. Functional Description 功能说明

1.1 This function remains a high energy state for the LV battery in different circumstances, thus maintaining the longevity of the LV battery and the ability to start at low temperatures or when the battery is degraded. 该功能可在不同情况下保持低压蓄电池的高能量状态，从而保持低压蓄电池的使用寿命以及在低温或蓄电池性能下降时启动的能力。

1.2 BCM shall measure voltage and take the average value of all nearest measured data in 2 seconds then send it on the Ubatt signal when CAN Network active.

当 CAN 网络激活时，BCM 应测量电压并在 2 秒钟内取所有最近测量数据的平均值，然后将其发送到 Ubatt 信号上。

2. Code Definition 编码定义

There are following coding parameters to be applicable for this function and can be changed according to project requirement. 该功能有以下编码参数，可根据项目要求进行修改。

- T1- Timer for periodically Ubatt check by BCM during sleep mode.

T1- 用于 BCM 在睡眠模式下定期检查 Ubatt 的计时。

- T2- Timer for the HV ON duration due to LV top-up.

T2- 低压补电导致高压接通持续时间的计时

- V1- Voltage threshold for initiation the vehicle CAN wakeup and LV top-up.

V1- 启动车辆 CAN 唤醒和 LV 补电的电压阈值

- C1 – Counter to count the continuous wakeup in every wake up period T1, based on that, the battery health can be determined.

C1 - 计数器，用于计算每个唤醒周期 T1 中的连续唤醒次数，并据此确定电池的健康状况。

- C1_TH is a continuous wake up threshold to disable auto wake up for checking Ubatt.

C1_TH 是连续唤醒阈值，用于禁用自动唤醒以检查 Ubatt。

Below data is preliminary setting for MINIVAN considering current vehicle architecture.

以下数据是 MINIVAN 的初步设置，考虑到了当前的车辆结构。

Parameter	Byte Pos	Bit Pos	Bit Length	Data Type	MethodType	Default Value
V1: voltage threshold range	Byte 3	[0 to 3]	4bits	unsigned	0x0: 12V 0x1: 12.2V 0x2: 12.4V	0x1
T1 : check LV Time range	Byte 3	[4 to 7]	4bits	unsigned	0x0: 1h 0x1: 2h 0x2: 3h 0x3: 4h 0x4: 5h 0x5: 6h 0x6: 7h 0x7: 8h 0x8: 9h 0x9: 10h	0x3
T2: turn on HV Time range	Byte 4	[0 to 3]	4bits	unsigned	0x0: 20 mins 0x1: 30 mins 0x2: 40 mins 0x3: 50 mins 0x4: 60 mins 0x5: 70 mins 0x6: 80 mins 0x7: 90 mins 0x8: 100 mins 0x9: 110 mins 0xA: 120 mins	0x02

3.Network Interface 网络接口

3.1 Input CAN signals CAN 输入信号

Signal Name	Signal Description	Message Name	Signal Length [1-	Start Bit [0-	Value Matrix
-------------	--------------------	--------------	-------------------	---------------	--------------

			64 bit]	63 bit]	
VCU_VehicleHVStat atus1	Vehicle High voltage Ready State (VCU_VehicleHVStat us to be received by all modules,Used as a flash programming precondition (replacement for engine running) 车辆高压就绪状态 (所有模块都将接收到 VCU_VehicleHVStatu s)。 作为闪存编程的先决 条件（替代发动机运 转）。	GW_BCM_Pedals_0 x269	2	25	7 "Reserved" 6 "Reserved" 5 "Reserved" 4 "Error" 3 "Charging" 2 "ReadyToDri ve" 1 "HV On" 0 "Not ready"
BMS_SocActual_E ST	Actual SOC 实际 SOC	GW_BMS_Sts_0x21 5	10	25	1023 "Invalid"

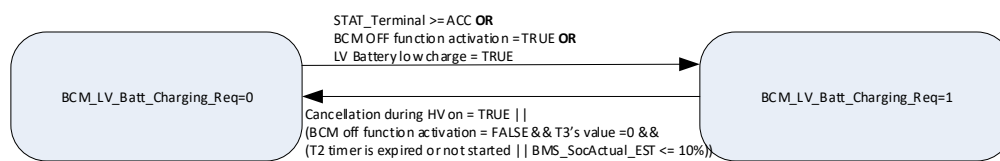
3.2 Output CAN signals

Signal Name	Signal Description	Message Name	Signal Length [1-64 bit]	Sta rt Bit [0- 63 bit]	Value Matrix
Ubatt	Battery reference Voltage(UBatt to be received by all modules,Requir ed for DTC snapshots) 电池 参考电压（所有 模块接收的 UBatt, DTC 快 照所需的电压）	BCM_VOLTAGE_0x 10a	14	23	
BCM_LV_Batt_Charging _Req	LV battery top up requice signal	BCM_VOLTAGE_0x 10a	1	24	0 "stop Top-up" 1 "start Top-up"

	低压蓄电池充电 要求信号				
LV_ContinuousWakeupC nt	Continuous top up counter 连续补电计数器	BCM_VOLTAGE_0x 10a	8		0-255 C1 的次数

4. functionality General 概述

4.1 Enable top-up 启用充值



- 1) BCM shall enable top up (set *BCM_LV_Batt_Charging_Req* = 1) if any of below conditions is satisfied 如果满足以下任一条件，BCM 应启用充值（设置

BCM_LV_Batt_Charging_Req = 1）：

- ◆ *STAT_Terminal* >= ACC OR
- ◆ BCM OFF function activation（功能激活）== TRUE OR
- ◆ LV Battery low charge（LV 电池电量不足）== TRUE

- 2) When BCM set *BCM_LV_Batt_Charging_Req* = 1, it shall also

当 BCM 设置 *BCM_LV_Batt_Charging_Req* = 1 时，它还应：

- ◆ Wake up the network and the vehicle if the network is inactive
如果网络处于非活动状态，则唤醒网络和车辆
- ◆ Start Timer T2 if “LV battery low charge” == TRUE

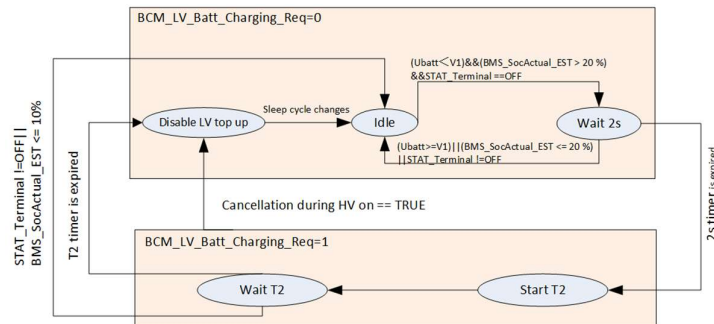
如果 “LV 电池电量不足”==真，则启动计时器 T2

- 3) BCM shall receive signal 0x10A: *BCM_LV_Batt_Charging_Req*, and enable HV if the below conditions are satisfied, no matter *STAT_Terminal*'s value

BCM 应接收信号 0x10A: *BCM_LV_Batt_Charging*（充电）_Req 信号，无论 *STAT_Terminal* 的值如何，如果满足以下条件，则启用 HV：

- ◆ 0x10A: *BCM_LV_Batt_Charging*（充电）_Req == 1 &&
- ◆ HV SOC actual（实际）> 10%
- ◆ No error in HV system, No crash 高压系统无故障，无碰撞事件
- ◆ DCDC is not in error state

4.2 LV Battery low charge evaluation LV 电池电量不足评估



1) BCM shall evaluate “LV battery low charge” == TRUE if all the below conditions are satisfied for 2 seconds BCM 在满足下列所有条件的情况下，应评估 “LV 电池电量低” == TRUE 持续 2 秒钟:

- ◆ STAT_Terminal= 0 "OFF(Comfort Enable)" &&
- ◆ Ubatt<V1 (configurable (可配置)) &&
- ◆ BMS_SocActual_EST > 20 % &&
- ◆ T2 is not started or expired previously in same power cycle
T2 之前未在同一电源周期内启动或过期.

2) Otherwise, “LV battery low charge” is understood as FALSE
 否则, “LV 电池电量低”被理解为“FALSE (假)”

4.3 Disable top-up 禁用补电

1) BCM shall disable top up (BCM LV Batt Charging Req = 0) if

如果出现以下情况，BCM 应禁用智能补电 (BCM LV Batt Charging Req = 0)

- ◆ Cancellation during HV on = TRUE ||
在 HV 操作期间取消=真
- ◆ (BCM off function activation = FALSE && T3's value = 0 &&
BCM OFF 功能激活 = 错误 && T3 值=0
- ◆ (T2 timer is expired or not started || BMS_SocActual_EST $\leq$ 10%))
T2 定时器已过期或未启动 || BMS 实际 SOC $\leq$ 10%

Note:

1) **VCU** shall turn off HV if BCM_LV_Batt_Charging_Req = 0 and there's no other conditions to keep HV ON

如果 BCM_LV_Batt_Charging_Req = 0 且没有其他保持 HV 开启的条件，则 VCU 应关闭 HV ON.

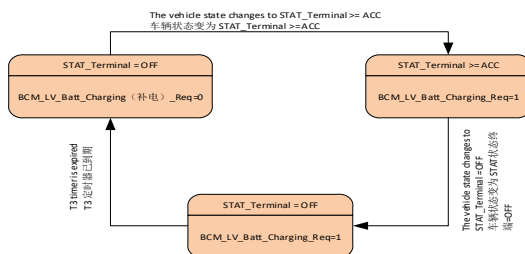
2) During top up, if VCU detects HV SOC actual $\leq 10\%$, it shall turn off HV if there's no other conditions to keep HV ON than top up request.

在充值过程中，如果 VCU 检测到 HV SOC 实际值 $\leq 10\%$ ，则在除充值请求外没有其他条件保持 HV 开启的情况下，应关闭 HV ON。

3) After disabling top-up and STAT_Terminal is in OFF state, BCM shall release NM and vehicle goes to sleep.

停止补电且 STAT_Terminal 处于 OFF 状态后，BCM 应释放 NM，车辆进入休眠状态。

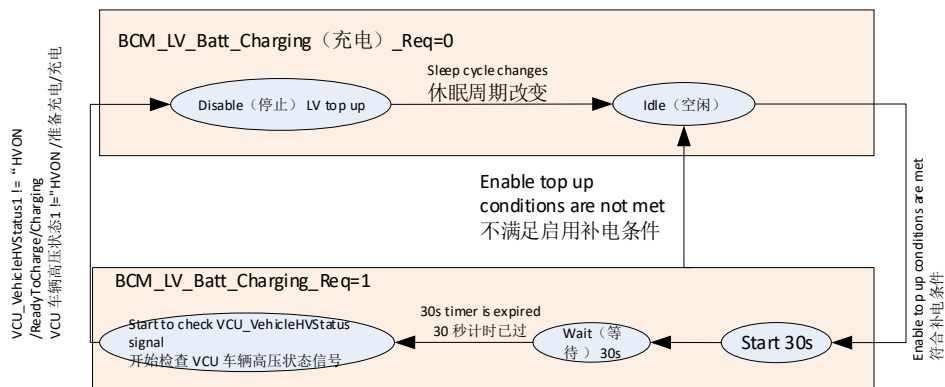
4.4 Delay HV OFF timer (T3) 延迟高压 OFF 计时(T3)



1) Timer T3 is a timer that helps delay turning off the BCM_LV_Batt_Charging_Req when shutting down the vehicle. 定时器 T3 是一个定时器，有助于在关闭车辆时延迟关闭 BCM LV 电池充电请求。

- ◆ T3 is handled by BCM T3 由 BCM 处理
- ◆ T3's value shall be set as 10 minutes when STAT_Terminal switches from ACC or ON to OFF
当 STAT 状态从 ON 或 ACC 切换到 OFF 时，T3 的值应设为 10 分钟。
- ◆ T3's value shall be set as 2 minutes when "Engine start (emotor running)" to OFF.
当 "Engine start (emotor running)" 至 OFF 时，T3 的值应设为 2 分钟。
- ◆ During STAT_Terminal $\neq$ OFF && STAT_Terminal $\neq$ Invalid, the T3 shall not count down. 在 STAT 状态 $\neq$ OFF && STAT 状态 $\neq$ 无效，T3 不进行倒计时
- ◆ T3 shall count down to Zero during ClampState = OFF.
在 STAT 状态 = OFF，T3 应倒计时置零

4.5 Cancellation during HV ON 高电压接通期间的取消



1) Cancellation during HV ON is the state that HV or DCDC error, thus the LV battery cannot be charged, when cancellation is detected, Top-up shall be stopped.

HV ON 期间的取消是指 HV 或 DCDC 出错，导致 LV 蓄电池无法充电的状态，当检测到取消时，应停止充值。

2) After sending BCM_LV_Batt_Charging_Req = 1 for 30s, BCM shall continuously evaluate “Cancellation during HV ON” as TRUE if any of below conditions is satisfied:

发送 BCM_LV_Batt_Charging_Req = 1 30 秒后，如果满足下列任何条件，BCM 应持续将

“HV ON 期间取消”评估为“TRUE”：

- ◆ BCM_VehicleHVStatus1 != “HV ON/ReadyToCharge/Charging”
BCM 车辆 HV 状态 1 != “HV 开启/准备充电/充电”

3) “Cancellation during HV ON” state is reset to FALSE when

当出现以下情况时，“高压接通时取消”状态重置为 FALSE（无效）

- ◆ BCM_VehicleHVStatus1 = “HV ON/ReadyToCharge/Charging”
BCM 车辆高压状态 1 = “高压开启/准备充电/充电”

4) If Cancellation during HV ON is TRUE, Vehicle shall not top up again for whatever reason in the same power cycle

如果“TRUE（真）”（“高电压开启期间取消”），则在同一电源周期内，无论出于何种原因，车辆均不得再次充值。

Note 注释:

1) Timer T1 定时器 T1:

- ◆ T1 is the timer to schedule periodically wake by BCM to check the Ubatt for top-up starting conditions.

T1 是 BCM 定期唤醒以检查 Ubatt 是否处于补电启动状态的计时器。

- ◆ When T1 expired during sleeping, BCM shall measure Ubatt.

当 T1 在睡眠期间到期时，BCM 应测量 Ubatt

- ◆ T1 shall not be permitted to operate in sleep mode if Cancellation during HV ON is TRUE for 3 top-up times continuously in different power cycles, that prevent periodically wake up when HV system is permanently error.

如果在不同电源周期中连续 3 次充值时 HV ON 期间的取消为“TRUE”，则不允许 T1 在睡眠模式下运行，以防止在 HV 系统永久错误时定期唤醒。

- ◆ T1 shall not be permitted to operate in sleep mode if T1's configurable expired value is 0xF.

如果 T1 的可配置为 0xF，则不允许 T1 在睡眠模式下运行。

- ◆ T1 is permitted to operate again in next sleep cycle if

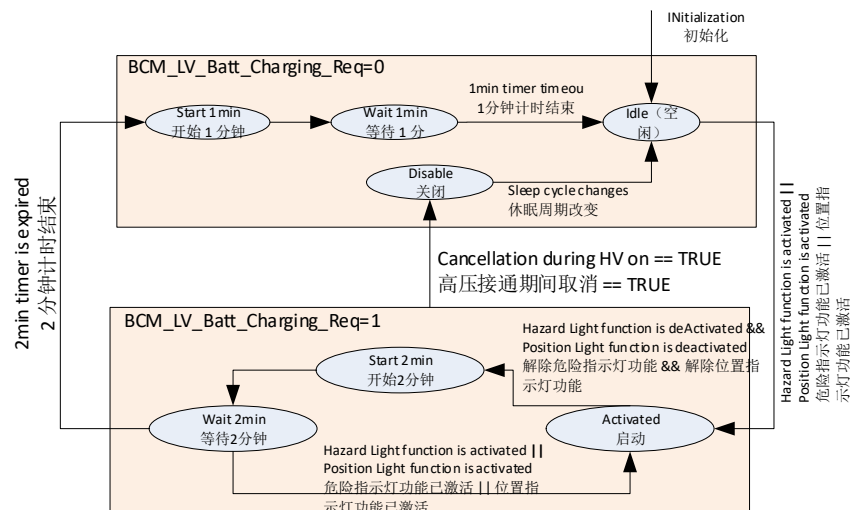
如果出现以下情况，则允许 T1 在下一个睡眠周期再次运行

- BCM_VehicleHVStatus1 == “HV ON/ReadyToCharge/Charging”
- BCM 车辆高压状态 1 == “高压开启/准备充电/充电”&&
- T1's configurable expired value is NOT 0xF. T1 的可配置过期值不是 0xF

2) Timer T2 定时器 T2:

- ◆ T2 is the timer to schedule terminating “top-up due to LV battery low charge”.
- T2 是计时器，用于安排结束“因 LV 电池电量不足而补电”。
- ◆ T2 shall start when BCM enable top-up due to “LV battery low charge” == TRUE.
- 当 BCM 因“LV 电池电量低”== TRUE 而启用充值时，T2 将启动。

4.6 BCM OFF function activation 激活 BCM OFF 功能



- 1) BCM off functions are the function keeps vehicle functioning and not getting to sleep mode, thus LV battery is discharged during BCM OFF functions are active.

BCM OFF 功能是保持车辆正常运行而不进入休眠模式的功能，因此在 BCM OFF 功能激活期间，LV 蓄电池会被放电。

- 2) BCM shall evaluate "BCM off function activation" as FALSE during initialization.

在初始化过程中，BCM 应将 "BCM 关闭功能激活" 评估为 FALSE。

- 3) BCM shall evaluate "BCM off function activation" as TRUE if any of the following conditions is satisfied:

如果满足以下任一条件，BCM 应将 "BCM OFF 功能激活" 评估为 "有效"：

- ◆ Hazard Light function is activated || Position light is activated (' BCM PosLmpSts = ON")
危险指示灯功能已启动||位置指示灯亮起（"BCM PosLmpSts = ON"）。

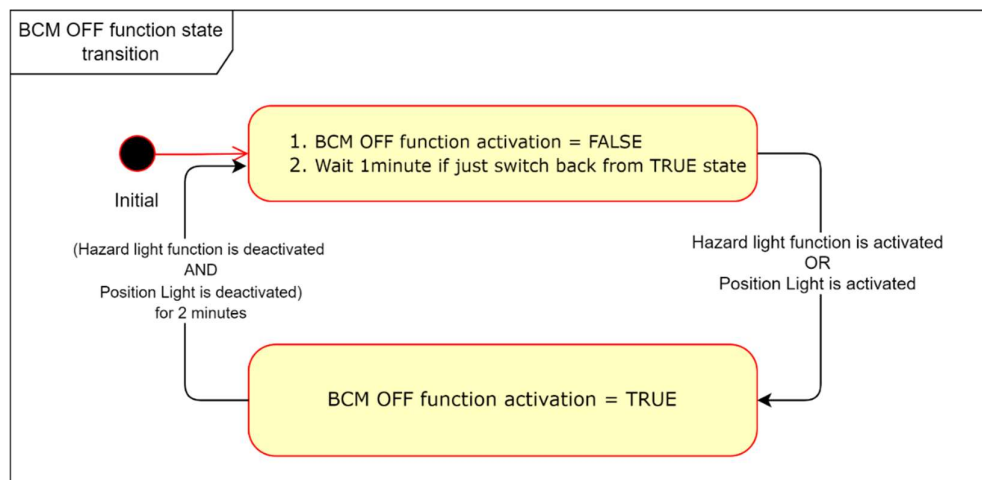
- 4) BCM shall evaluate "BCM off function activation" as FALSE if the below conditions are satisfied in 2 minutes.

如果在 2 分钟内满足以下条件，BCM 将把 "BCM OFF 功能激活" 设置为 "FALSE（无效）”

- ◆ Hazard Light function is deactivated && Position light is not activated (' BCM PosLmpSts = OFF")
危险指示灯功能已关闭位置指示灯不亮（"BCM PosLmpSts = OFF"）。

- ◆ After the "BCM off function activation" switches from TRUE to FALSE, BCM shall wait 1 minute before re-evaluating the condition for switching to TRUE state.

BCM OFF 功能激活 "从" 有效 "切换到 "退出 "后，BCM 应等待 1 分钟再重新评估切换到 "有效 "状态的条件。

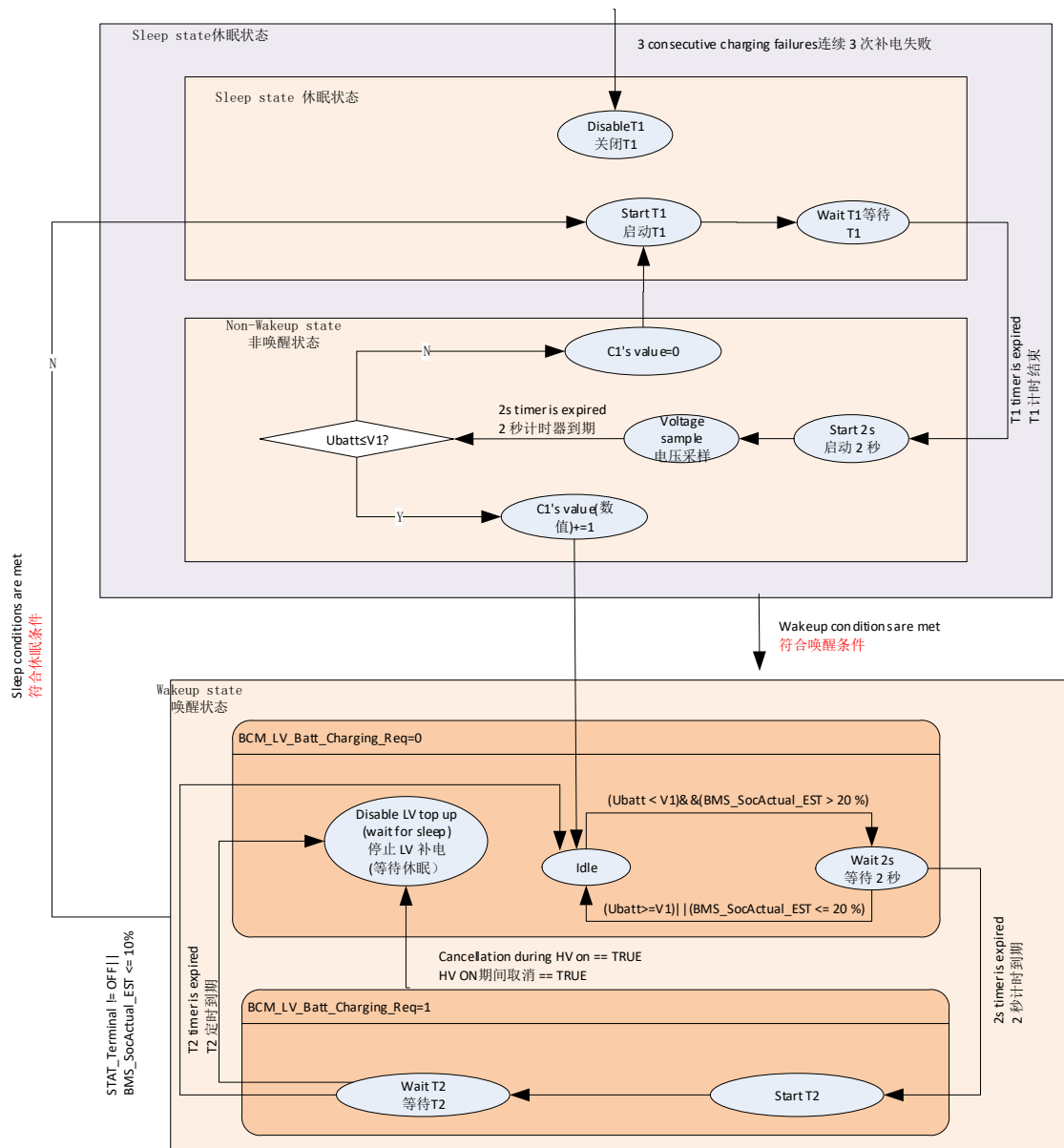


Note 注释:

- 1) This function does not affect changes in BCM power status;

此功能不影响 BCM 电源状态的更改

4.7 The vehicle is at sleep 车辆处于睡眠状态



1) After the Vehicle change to sleep state and T1 is expired BCM shall read C1 previous value from NVM

当车辆转入睡眠状态且 T1 计时结束后，BCM 应从 NVM 中读取 C1 以前的值。

2) IF voltage($Ubatt$) $\leq V1$ for 2s

如果电压 ($Ubatt$) $\leq V1$ ，持续 2 秒:

◆ BCM wakes up and enters the OFF state

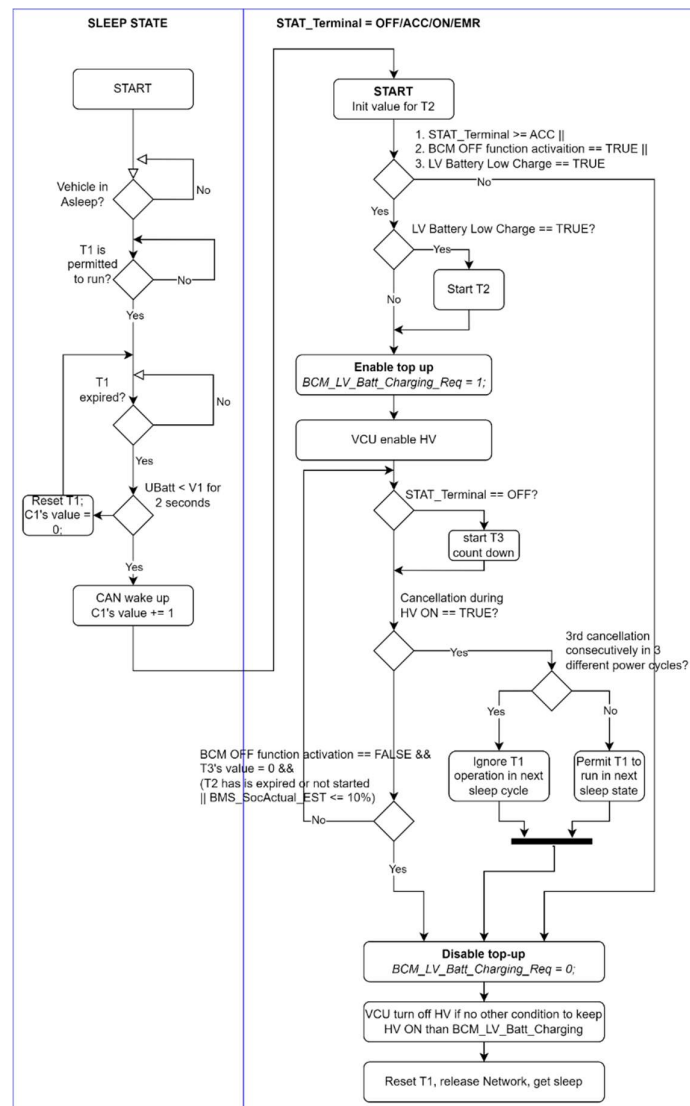
BCM 唤醒并进入 OFF 状态。

- ◆ Counter C1's value shall be increased by 1 (C1'value += 1).

计数器 C1 的值应增加 1 (C1 的值 +=1)

- a) The maximum of C1'value is 255 C1 的最大值为 255
- b) Otherwise, C1's value = 0. 否则, C1 的值 = 0
- c) C1'value shall be stored to NVM before entering sleep mode.
C1 的值应在进入睡眠模式前存储到 NVM 中。

4.8 LV battery top-up flow chart 低压蓄电池补电流程图



Note:

- 1) According to the BCM and low-voltage battery test results, the following information has a certain impact on the accuracy of the strategy, as detailed in the preliminary analysis report:

根据 BCM 和低压电池测试结果，以下信息对策略的准确性有一定影响，详见初步分析报告：

- ◆ BCM has certain deviations in detecting voltage values under different problems;
BCM 在检测不同问题下的电压值时存在一定偏差；
- ◆ BCM detects a certain deviation between the voltage and the actual voltage;
BCM 检测到电压与实际电压之间存在一定偏差；
- ◆ Under different discharge conditions, there is a deviation in the relationship between voltage and quantity of electricity;
在不同的放电条件下，电压和电量之间的关系会出现偏差

MF-28 SBR(Seat Reminder Function 安全带提醒功能)

ID Name	Signal Name	Signal Description	Value Matrix
BCM_STAT_DOOR_FLAP_0x105			
	Checksum_0x105	Checksum	
	Rolling_Counter_0x105	Rolling_Counter	15 "Invalid"
	BCM_Seatbelt_Reminder_Request	BCM request seatbelt reminder (1 signal for whole front and rear seats)	0 "Seatbelt reminder NOT activated" 1 "Seatbelt reminder activated"
	BCM_SBR_Chime_Req	BCM request chime to MHU for seatbelt reminder front row	0 "No Active Chime" 1 "Active Chime"
	BCM_SBR_Blinking_Req	BCM send blinking request to MHU (trigger blinking of seatbelt reminder lamp)	0 "No Blinking" 1 "Blinking 1Hz"
	BCM_Seat_Occupancy_Passenger	Signal to indicate status of occupancy	3 "Failure" 2 "Not_Available" 1 "Seat Occupied" 0 "Seat Not Occupied"
	BCM_Seat_Belt_Status_Passenger	Seatbelt status for seatbelt reminder function	3 "Failure" 2 "Not_Available" 1 "Buckled" 0 "Not_buckled"
	BCM_Seat_Belt_Status_Driver	Seatbelt status for seatbelt reminder function	3 "Failure" 2 "Not_Available" 1 "Buckled" 0 "Not_buckled"

The purpose of the Seat Belt Warning feature is to alert the driver when any of the vehicle occupants including the driver has not buckled the seat belt. The feature is constituted by detection of unbuckled seat belt in an occupied seat and two levels of driver's alert, a first level warning and a second level warning.

安全带警告功能的目的是在包括驾驶员在内的任何车内人员未系安全带时提醒驾驶员。该功能通过检测有人乘坐的座椅上是否有未系安全带的情况，以及两级驾驶员警报（一级警告和二级警告）来实现。

1. First level warning means at least a visual warning activated when the power mode is in HV ON (Driver Present) state and any of the occupant's safety-belt is not fastened.

第一级警告是指当动力模式处于 HV ON（驾驶员在场）状态，且任何乘员未系安全带时，至少启动一个视觉警告。

2. Second level warning means a visual and audible warning activated when the vehicle is running, vehicle speed has reached 15km/h or more, and when the safety-belt is or becomes unfastened.

第二级警告是指在车辆行驶、车速达到或超过 15 公里/小时以及安全带未系或已解开时发出的视觉和听觉警告。

(1) Input 输入

1. Occupant Detection System 占用探测系统

BCM shall provide an input to sense the occupancy status through of ODS as simple switch which is not diagnosable. BCM 应提供一个输入端，通过 ODS 作为不可诊断的简单开关来感知占用状态

In Mini VAN, ODS switch is only installed in front passenger seat.

在 Mini VAN 中，ODS 开关只安装在前排乘客座位上。

2. CAN-Signals CAN 信号

2.1 IGN State IGN 状态

The ignition state or clamp state represented by CAN signal STAT_Terminal.
CAN 信号 STAT 状态所代表的点火状态或钳位状态

IGN state	Definition	Value
OFF	Off	0x0
ACC	Parked	0x1
ON	Driver present	0x2
START	eMotor running	0x3

2.2 Vehicle speed 车辆速度

BCM shall consume the speed signal displayed by the speedometer in IPC. The CAN signal IPC_VehSpd represents this value.

BCM 应接收 IPC 中速度计显示的速度信号。CAN 信号 IPC_VehSpd 表示该值。

(2) Outputs 输出

BCM shall send occupancy state of front passenger seat through the signal

BCM 应通过信号发送前排乘客座椅的占用状态

ACM_Seat_Occupancy_Passenger with the enumeration defined below.

ACM 座位占用乘客，定义如下

Signal value 信号值	Front Passenger Seat Occupancy state 前排乘客座椅占用状态
0x0	Seat_Not_Occupied 未占用座位

0x1	Seat Occupied 占用座位
0x2	Not_Available 不可用
0x3	Failure 故障

The CAN following CAN signals shall be considered as output for Seat Belt Reminder functionality for front seats:

下列 CAN 信号应视为前排座椅安全带提醒功能的输出信号:

BCM_Seatbelt_Reminder_Request BCM 安全带提醒请求

BCM_SBR_Blinking_Request BCM SBR 闪烁请求

BCM_SBR_Chime_Req BCM SBR 蜂鸣要求

(3)Processing 处理过程

1. All occupied seats are buckled. 所有有人的座位都系好安全带。

Event: IGN state changes from OFF or ACC to ON (Driver Present) or START (eMotorRunning).
BCM shall send these signals to IPC for the first-row seats:

事件: IGN 状态从 OFF 或 ACC 变为 ON (驾驶员在场) 或 START (电子电机运行)。BCM 应为第一排座椅向 IPC 发送这些信号:

- BCM_Seatbelt_Reminder_Request = 0 "Seatbelt Reminder Not Activated"
BCM 安全带提醒请求 = 0 "安全带提醒未激活"
- BCM_SBR_Chime_Req = 0 "No Active Chime"
BCM SBR 风鸣请求 = 0 "无风鸣"
- BCM_SBR_Blinking_Req = 0 "No Blinking".
BCM SBR 闪烁要求 = 0 "无闪烁"

2. At least one of the occupied seats is not buckled. 至少一个有人的座位没有系安全带

Event: IGN state changes from OFF or ACC to ON(Driver Present) or START (eMotorRunning).

事件: IGN 状态从 OFF 或 ACC 变为 ON (驾驶员在场) 或 START (电子电机运行)。

If at least one among the occupied unbuckled seats is a first row seat, then BCM shall send these signals to IPC 如果被占用的未系安全带座位中至少有一个是第一排座位, 则 BCM 应向 IPC 发送这些信号:

- BCM_Seatbelt_Reminder_Request = 1 "Seatbelt Reminder Activated"
BCM 安全带提醒请求 = 1 "安全带提醒激活"
- BCM_SBR_Chime_Req = 0 "No Active Chime"
BCM SBR 风鸣请求 = 0 "无风鸣"
- BCM_SBR_Blinking_Req = 0 "No Blinking".
BCM SBR 闪烁要求 = 0 "无闪烁"

Otherwise, BCM shall keep the current values of the above signals

否则, BCM 应保持上述信号的当前值

3. IGN State = ON(Driver Present) or ON(eMotor Running), all occupied seats are buckled, vehicle speed <15km/h. IGN 状态 = ON (驾驶员在场) 或 ON (电子电机运行), 所有乘员座位都系好安全带, 车速 <15km/h.

Event: At least one of the occupied seats becomes unbuckled.

事件: 至少有一个有人的座位没有系安全带。

If at least one among the occupied unbuckled seats is a first row seat, then BCM shall send these signals to IPC 如果被占用的未系安全带座位中至少有一个是第一排座位, 则 BCM 应向 IPC 发送这些信号:

- BCM_Seatbelt_Reminder_Request = 1 "Seatbelt Reminder Activated"

BCM 安全带提醒请求 = 1 "安全带提醒激活"

- BCM_SBR_Chime_Req = 0 "No Active Chime"

BCM SBR 风鸣请求 = 0 "无风鸣"

- BCM_SBR_Blinking_Req = 0 "No Blinking".

BCM SBR 闪烁要求 = 0 "无闪烁"

Otherwise, ACM shall keep the current values of the above signals. 否则, ACM 应保持上述信号的当前值。

4. IGN State = ON(eMotor Running), vehicle speed $\geq 15\text{km/h}$. Event: At least one of the occupied seats becomes unbuckled. IGN 状态 = ON (电机运行), 车速 $\geq 15\text{km/h}$. 事件: 至少有一个有人乘坐的座椅未系安全带

If at least one among the occupied unbuckled seats is a first row seat, then BCM shall send these signals to IPC for 95 sec 如果占用的未系安全带座位中至少有一个是第一排座位, 则 BCM 应向 IPC 发送这些信号, 持续 95 秒

- BCM_Seatbelt_Reminder_Request = 1 "Seatbelt Reminder Activated"

BCM 安全带提醒请求 = 1 "安全带提醒激活"

- BCM_SBR_Chime_Req = 1 "Active Chime"

BCM SBR 风鸣请求 = 1 "风鸣"

- BCM_SBR_Blinking_Req = 1 "Blinking 1Hz".

BCM SBR 闪烁要求 = 1 "闪烁(频率 1Hz)"

Otherwise, BCM shall keep the current values of the above signals 否则, BCM 将保持上述信号的当前值

5. IGN State = ON(Driver Present) or ON(eMotor Running), vehicle speed $< 15\text{km/h}$, at least one of the occupied seats becomes unbuckled. IGN 状态 = ON (驾驶员在场) 或 ON (电子电机正在运行), 车速 $< 15\text{km/h}$, 至少一个有人乘坐的座椅未系安全带。

Event: Vehicle speed rises to 15km/h or rises above 15km/h . 事件: 车速升至 15km/h 或超过 15km/h

If at least one among the occupied seats which changed its status from buckled to unbuckled before vehicle reaches speed of 15km/h is a first row seat, then BCM shall send these signals to IPC for 95 secs: 如果在车速达到 15km/h 之前, 所占座位中至少有一个座位从系安全带状态变为未系安全带状态, 则 BCM 应向 IPC 发送这些信号 95 秒

- BCM_Seatbelt_Reminder_Request = 1 "Seatbelt Reminder Activated"

BCM 安全带提醒请求 = 1 "安全带提醒激活"

- BCM_SBR_Chime_Req = 1 "Active Chime"

BCM SBR 风鸣请求 = 1 "风鸣"

- BCM_SBR_Blinking_Req = 1 "Blinking 1Hz".

BCM SBR 闪烁要求 = 1 "闪烁(频率 1Hz)"

Otherwise, ACM shall keep the current values of the above signals. 否则, ACM 应保持上述信号的当前值

6. IGN state = ON(eMotor Running), vehicle speed $\geq 15\text{km/h}$, SWS is blinking, chime is active, time since chime is activated $< 95\text{s}$, no new occupied seat becomes unbuckled. IGN 状态 = ON (电子电机运行), 车速 $\geq 15\text{km/h}$, SWS 闪烁, 报警激活, 自报警激活以来的时间 < 95 秒, 没有新的有人乘坐的座椅未系安全带。

Event: Vehicle speed falls below 15km/h and it either stays below 15km/h or rises back to/above 15km/h .

事件: 车速降至 15km/h 以下, 并保持在 15km/h 以下或回升至 15km/h 以上。

Logical Output: Once 95s timer is started, it shall not be interrupted by any change in vehicle speed. BCM shall not change the value of these signals: 逻辑输出: 95 秒计时器启动后, 不得因车速的任何变化而中断。BCM 不得更改这些信号的值:

BCM_Seatbelt_Reminder_Request BCM 安全带提醒请求

BCM_SBR_Blinking_Req BCM SBR 闪烁请求

BCM_SBR_Chime_Req BCM SBR 凤鸣要求

7. IGN state = ON(eMotor Running), the SWS is blinking, the chime is active.

IGN 状态 = ON (电机运行), SWS 闪烁, 报警激活。

Event: Time since chime is activated > 95s, but the seat which triggered the warning remains unbuckled.

事件: 警报响起后的时间 > 95 秒, 但触发警报的座椅仍未系安全带。

If at least one among the seats which triggered the warning is a first row seat, then BCM shall send these signals to IPC:

如果触发警告的座椅中至少有一个是第一排座椅, 则 BCM 将向 IPC 发送这些信号:

- BCM_Seatbelt_Reminder_Request = 1 "Seatbelt Reminder Activated"
- BCM 安全带提醒请求 = 1 "安全带提醒激活"
- BCM_SBR_Chime_Req = 0 "No Active Chime"
- BCM SBR 凤鸣请求 = 0 "无凤鸣"
- BCM_SBR_Blinking_Req = 0 "No Blinking".
- BCM SBR 闪烁要求 = 0 "无闪烁"

Otherwise, ACM shall keep the current values of the above signals 否则, ACM 应保持上述信号的当前值

8. Vehicle speed $\geq$ 15km/h, a seat belt warning is ongoing. 车速 $\geq$ 15km/h, 安全带警告持续。

Event: At least one of the other occupied seat becomes unbuckled during the warning.

事件: 在警告期间, 至少有一个座位上的乘客未系安全带。

If at least one among the newly unbuckled occupied seats is a first row seat, then BCM shall send these signals to IPC for 95 sec:

如果新解开安全带的座位中至少有一个是第一排座位, 则 BCM 将向 IPC 发送这些信号 95 秒:

- BCM_Seatbelt_Reminder_Request = 1 "Seatbelt Reminder Activated"
- BCM 安全带提醒请求 = 1 "安全带提醒激活"
- BCM_SBR_Chime_Req = 1 "Active Chime"
- BCM SBR 凤鸣请求 = 1 "凤鸣"
- BCM_SBR_Blinking_Req = 1 "Blinking 1Hz".
- BCM SBR 闪烁要求 = 1 "闪烁(频率 1Hz)"

The 95 secs timer shall be restarted by this event. 该事件将重新启动 95 秒计时。

Otherwise, ACM shall keep the current values of the above signals.

否则, ACM 应保持上述信号的当前值

9. IGN State = ON(Driver Present) or ON(eMotor Running), a seat belt warning is going on.

IGN 状态 = ON (驾驶员在场) 或 ON (电子电机运行), 表示安全带警告正在响起。

Event: All of the seats which triggered the warning change from unbuckled to buckled.

事件: 所有触发警告的座椅都从未系安全带变为系好安全带。

BCM shall send these signals to IPC BCM 将向 IPC 发送这些信号:

- BCM_Seatbelt_Reminder_Request = 0 "Seatbelt Reminder Not Activated"
- BCM 安全带提醒请求 = 0 "安全带提醒未激活"
- BCM_SBR_Chime_Req = 0 "No Active Chime"
- BCM SBR 凤鸣请求 = 0 "无凤鸣"
- BCM_SBR_Blinking_Req = 0 "No Blinking".
- BCM SBR 闪烁要求 = 0 "无闪烁"

10. IGN State = ON(Driver Present) or ON(eMotor Running), a seat belt warning is going on.

IGN 状态 = ON (驾驶员在场) 或 ON (电子电机运行), 表示安全带警告正在响起。

Event: All of the seats which triggered the warning are no longer occupied

事件: 所有触发警告的座位都已空出

BCM shall send these signals to IPC BCM 应向 IPC 发送这些信号

- BCM_Seatbelt_Reminder_Request = 0 "Seatbelt Reminder Not Activated"

BCM 安全带提醒请求 = 0 "安全带提醒未激活"

- BCM_SBR_Chime_Req = 0 "No Active Chime"

BCM SBR 风鸣请求 = 0 "无风鸣"

- BCM_SBR_Blinking_Req = 0 "No Blinking".

BCM SBR 闪烁要求 = 0 "无闪烁"

11. IGN State = ON(Driver Present) or ON(eMotor Running).

IGN 状态 = ON (驱动程序存在) 或 ON (电机运行)

Event: Ignition is turned off (that is, when IGN State $\neq$ ON(Driver Present) and ON(eMotorRunning)).

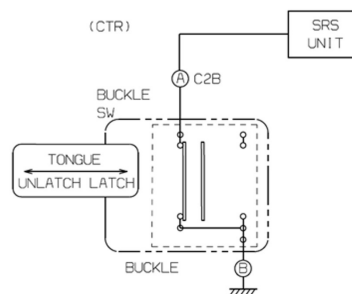
事件: 点火关闭 (即 IGN 状态 $\neq$ ON (驾驶员在场) 和 ON (电机运行))。

BCM need not send any signals to IPC for seat belt warning.

BCM 无需向 IPC 发送任何信号来发出安全带警告。

Note 注释:

- 1) When the SBR resistance range of 0-100 Ω is detected, the seat is considered occupied and the BCM sends the BCM_Seat_Occupancy_Passenger= 1 "Seat Occupied 当检测到 SBR 电阻范围在 0-100 Ω 内, 认为座位有人
- 2) When the BCM detects that the main driver seatbelt changes from low to suspended, the BCM sends the BCM_Seat_Belt_Status_Driver =1 "Buckled" BCM 检测到主驾安全带由低电平变为悬空
- 3) When the BCM detects that the main passenger seatbelt changes from low to suspended, the BCM sends the BCM_Seat_Belt_Status_Passenger=1 "Buckled" BCM 检测到乘客安全带由低电平变为悬空
- 4) Seat Belt Latch Working Principle Diagram



1.1.1 Faulty Operation 运行故障

N/A

1.1.2 Emergency Operation 应急行动

N/A

1.1.3 Diagnostics 诊断

To be able to fault trace and verify function and system. The goal is to provide a fast and accurate way of finding the root cause of the fault symptom. It is also important to be able to verify functions with help of diagnostics. Diagnostic fault detection is also used for quality follow up/improvement purposes.

能够对功能和系统进行故障跟踪和验证。目的是提供一种快速、准确的方法, 找出故障症状的根本原因。同样重要的是, 能够在诊断的帮助下验证功能。诊断性故障检测还可用于质量跟踪/改进目的。

Conditions when to set a DTC shall be clear and documented. Values specified in technical regulations shall be “real values” that is read out or measured from the vehicle with Aftermarket tools. For fault tracing purposes, it shall be possible to activate all ECU outputs and read inputs via the off-board service tool independent of system mode (e.g. failure mode). 设置 DTC 的条件应明确并记录在案。技术规范中规定的数值应为“真实值”，即使用售后工具从车辆中读出或测量的数值。出于故障追踪目的，应能通过车外服务工具激活所有 ECU 输出并读取输入，而与系统模式（如故障模式）无关。

Examples of diagnostics 诊断实例:

There shall be a diagnostic function for electrical I/O of the component. The diagnostics function shall be able to detect the following faults and preferably be able to discriminate between them:

组件的电气 I/O 应具有诊断功能。诊断功能应能检测出以下故障，最好还能对故障进行区分：

- Short circuit to ground, short circuit to battery and open circuit. 接地短路、蓄电池短路和开路。
- Hardware malfunctions on connected components. 连接组件的硬件故障
- Supply voltage too high & too low. 电源电压过高或过低。
- ECU Internal faults. The supplier shall give suggestions of which functions blocks that could have fault codes. ECU 内部故障。供应商应就可能出现故障代码的功能块提出建议。
- Functional related faults. 与功能相关的故障。

The diagnosis shall work both in on-state and off-state. Each load must be investigated in order to decide the kind of diagnosis which is possible. The supplier shall measure and test all diagnostic codes and write a report. 诊断应在通电和断电状态下进行。必须对每种负载进行调查，以决定可以进行哪种诊断。供应商应测量和测试所有诊断代码并撰写报告。

Detailed diagnostic information please refer to Component diagnostics specification. 详细诊断信息请参阅组件诊断规范。

Refer to 参考

Basic Diagnostics Specification 基本诊断规范 [REF_02]

Basic DID Definition Specification 基本 DID 定义规范 [REF_03]

Flash Specification 闪存规格 [REF_04]

Coding Specification 编码规范 [REF_05]

Security Access Algorithm Specification 安全访问算法规范 [REF_06]

Implement DID F194 in ECU, data format is (VFyymmdd. X) 在 ECU 中执行 DID F194，数据格式为 (VFyymmdd. X)

VFyymmdd.X => 10 bytes data length and decoded by ASCII. VFyymmdd.X => 10 字节数据长度，用 ASCII 解码。

- "VF" is VINFAST

- "yy" is last 2 characters of the year 年份
- "mm" is 2 characters for month 月份
- "dd" is 2 characters for date 时间
- "." is for separation 分
- "X" is for numbering start from 0 (in case >2 SW delivered in 1 day) X "表示编号从 0 开始 (如果 1 天内交付的 SW 超过 2 件

F194 captures supplier software versions ext: 56 46 32 34 30 37 30 34 2E 30 (VF240704.0)

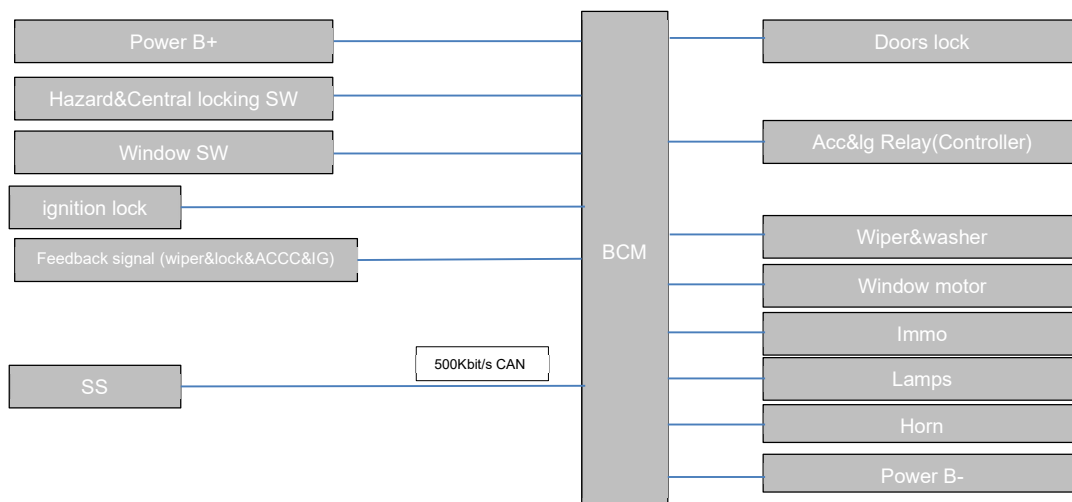
F194 捕捉供应商软件版本扩展: 56 46 32 32 34 30 37 30 34 34 2E 30 (VF240704.0)

F195 Applied to internal software versions of suppliers

F195 适用于供应商的内部软件版本

d E/E System Schematic E/E 系统示意图

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e Control Unit Concept 控制单元概念

The unit shall be a combination BCM,PS, and APM and contain all functionality of those sub-components

1.1.4 Hardware 硬件

1. PWM interface circuit PWM 接口电路

o Description 说明

The PWM signal generally outputs different voltage signals through the regulator chip by adjusting the duty ratio. The common PWM interface circuit has a backlight input, and the

matched interface circuit usually includes a voltage dividing resistor, a pull-down resistor, and a capacitor.

PWM 信号一般通过调整占空比，通过稳压芯片输出不同的电压信号。常见的 PWM 接口电路有一个背光输入端，匹配的接口电路通常包括一个分压电阻、一个下拉电阻和一个电容。

- 电容器。
- Design requirements 设计要求
 - The interface circuit must meet the PWM signal duty cycle adjustment range requirements; 接口电路必须满足 PWM 信号占空比调整范围的要求
 - The interface circuit must meet the PWM signal frequency definition requirements; 接口电路必须满足 PWM 信号频率定义要求
 - The input operating current must meet the definition of the interface circuit 输入工作电流必须符合接口电路的定义;
 - The interface circuit shall have a reverse polarity protection electrical function, and the 接口电路应具有极性反接保护电气功能，并且 reverse connection will not cause damage to the interface circuit or the electrical device; 反向连接不会损坏接口电路或电气设备;
 - The interface circuit should have EMC protection function to meet the EMC performance requirements of parts and vehicles. 接口电路应具有 EMC 保护功能，以满足部件和车辆的 EMC 性能要求。
- 2. CAN interface circuit CAN 接口电路
 - Description 描述

The signal is sent to the vehicle CAN network in the form of binary data, and the controller receives the data from the CAN network by using the CAN transceiver.

信号以二进制数据的形式发送到车辆 CAN 网络，控制器通过 CAN 收发器从 CAN 网络接收数据

- 3. Analog signal interface circuit 模拟信号接口电路
 - Description 描述
 - The analog signal interface circuit input is generally grounded through a resistor. The interface circuit generally has the same structure, and usually includes a pull-up/pull-down resistor, an EMC capacitor, a filter circuit, and an anti-reverse diode protection circuit. There are differences in interface circuit parameter settings and circuit configurations for different functions.

模拟信号接口电路输入一般通过一个电阻接地。接口电路的结构一般相同，通常包括一个上拉/下拉电阻、一个 EMC 电容、一个滤波电路和一个防反二极管保护电路。不同功能的接口电路参数设置和电路配置有所不同。

- Design requirements 设计要求
 - The analog signal input resistance must meet the definition of the interface circuit;

模拟信号输入电阻必须符合接口电路的定义

- The input operating current must meet the definition of the interface circuit; The interface shall have reverse polarity protection function, and the reverse connection will not cause damage to the interface circuit and electrical components;

输入工作电流必须符合接口电路的定义；接口应具有反极性保护功能，反接不会对接口电路和电气元件造成损坏；

- The interface circuit should have EMC protection function to meet the EMC performance requirements of parts and vehicles

接口电路应具有 EMC 保护功能，以满足部件和车辆的 EMC 性能要求

1.1.5 Software 软件

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The underlying driver software is generally integrated in the microkernel. It is the bottom software layer of the basic software of the MHU panel cluster. It mainly completes the actual hardware-related hardware operations, and provides all the specific hardware operation related interfaces required by the abstraction layer to improve portability. The underlying driver generally writes/reads data using I2C bus timing and splits it into functions to form kernel modules according to specific structural requirements.

底层驱动软件一般集成在微内核中。它是 MHU 面板集群基础软件的底层软件。它主要完成与硬件相关的实际硬件操作，并提供抽象层所需的所有与具体硬件操作相关的接口，以提高可移植性。底层驱动程序一般利用 I2C 总线时序写入/读取数据，并根据具体的结构要求将其拆分成功能，形成内核模块。

The main content of the underlying driver module needs to include the following parts 底层驱动模块的主要内容需要包括以下部分:

1. Microprocessor driver (MCU driver, watchdog driver); 微处理器驱动程序（MCU 驱动程序、看门狗驱动程序）；
2. Storage driver (RAM test, FLASH module driver, E2PROM driver); 存储驱动程序（RAM 测试、FLASH 模块驱动程序、E2PROM 驱动程序）；
3. Communication driver (SPI driver, FLEX driver, CAN driver); 通信驱动程序（SPI 驱动程序、FLEX 驱动程序、CAN 驱动程序）；
4. I/O driver (ADC conversion module, ADC comparison module, PWM driver, port module driver)

I/O 驱动器（ADC 转换模块、ADC 比较模块、PWM 驱动器、端口模块驱动器）

f Electrical Interfaces 电气接口

N/A

1.1.6 Signal Characteristics 信号特征

The BCM uses CAN bus and Lin to communicate between vehicle ECUs. Refer to CAN Matrix [REF_12].

BCM 使用 CAN 总线和 Lin 在车辆 ECU 之间进行通信。请参阅 CAN 矩阵 [REF_12]。

1.1.7 Diagnostics 诊断

Refer to Basic Diagnostics Specification [REF_02] & Basic DID PID Specification [REF_03].

请参阅基本诊断规范 [REF 02] 和基本 DID PID 规范 [REF 03]。

1.1.8 Network Management 网络管理

The BCM needs to meet the CAN network management wake-up and sleep. When receiving the wakeup request of the network management messages, it needs to be woken up and send corresponding messages. When receiving the sleep command, it can enter the sleep state normally.

BCM 需要满足 CAN 网络管理唤醒和休眠的要求。当接收到网络管理报文的唤醒请求时，它需要被唤醒并发送相应的报文。收到休眠命令时，可正常进入休眠状态。

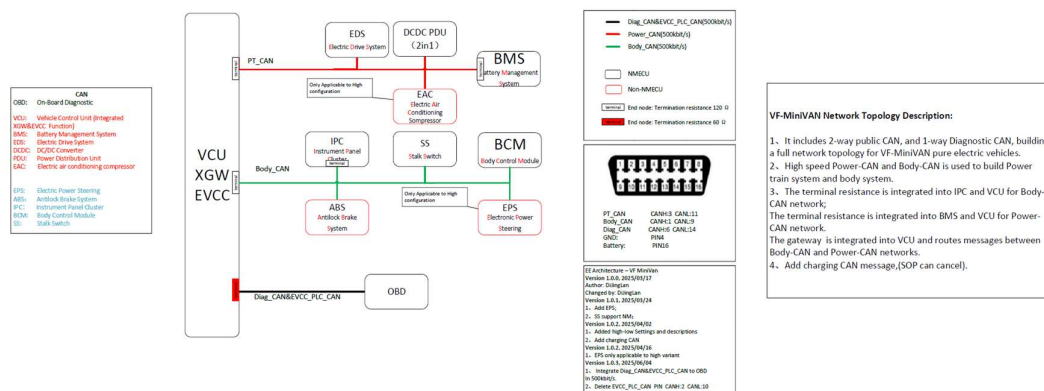
BCM shall have Lin communication function BCM 应具有 Lin 通信功能

Refer to 参考:

CAN Specification (CAN 规范) [REF 01].

Lin Specification [REF 11].

BCM shall following the AutoSAR NM state transition as following BCM 应遵循以下 AutoSAR NM 状态转换:



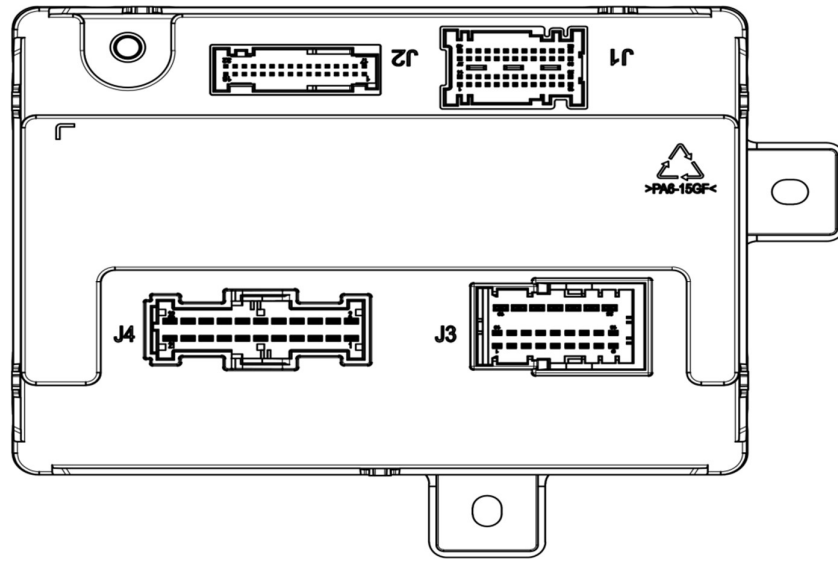
1.1.9 Flashing and Coding 刷写和编码

Refer to 参考

- Flash Specification（刷写规范）[REF_04]
- Coding Specification（编码规范）[REF_05]
- Security Access Algorithm（安全访问算法）[REF_06]

1.1.10 Pin Assignment and Connector 引脚分配和连接器

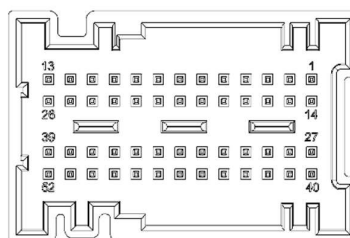
Bellow illustrates locations of each connectors on the backside of the BCM 下图说明了 BCM 背面各连接器的位置:



1.1.10.1 Harness end matching plug-in information 线束端匹配插件信息:

No.	supplier	Connector Information (EE part side)	Connector Information (Wiring Harness side)	Remark
J1	Tyco	284975-1	284972-1	Harness plug-in
J2	Tyco	966658-1	1420000-1+1534222-1+1420001-1	Harness plug-in
J3	Tyco	171849-1	2359439-1	Harness plug-in
J4	Tyco	self opening die	2356718-1	Harness plug-in

1.1.10.2 J1 Pin Assignment

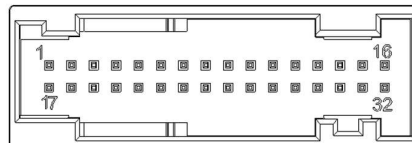


BCM plug-in pin view

Socket Number	Pin Number	I/O Type	Clamp Definition	Signal Definition	Drive Definition - HighSide - LowSide - int. Relay - PWM - ...	Typical/ Nominal Current [A]
J1	1	SGND	J1-1	SGND	SGND	
	2	I-D-L	J1-02	FR_Door_ajar_SW	Lo Side Sw	
	3	I-D-L	J1-03	Seat_Belt_Detection (Driver)	Lo Side Sw	
	4	I-D-L	J1-04	Seat_Belt_Detection (Passenger)	Lo Side Sw	
	5	I-D-L	J1-05	NC	Lo Side Sw	
	6	I-D-L	J1-06	FR_WIN_UP_SW	Lo Side Sw	
	7	I-D-L	J1-07	FR_WIN_DN_SW	Lo Side Sw	
	8	I-D-L	J1-08	NC	Lo Side Sw	
	9	I-D-L	J1-09	FL_WIN_UP_SW	Lo Side Sw	
	10	I-D-L	J1-10	FL_WIN_DN_SW	Lo Side Sw	
	11	I-D-L	J1-11	Passenger_Seat_Pass_SW	Lo Side Sw	
	12	I-D-L	J1-12	FRONT_WIPER_PARK_SW	Lo Side Sw	
	13	O-D-L	J1-13	HORN_RLY_OUTPUT	Low_ExRelay_output	0.17A
	14	I-D-L	J1-14	Mechanical Key_UNLOCK_SW	Lo Side Sw	
	15	I-D-L	J1-15	Mechanical Key_LOCK_SW	Lo Side Sw	
	16	I-D-L	J1-16	FL_Door_ajar_SW	Lo Side Sw	
	17	I-D-L	J1-17	DR_DOOR_LOCK_STATUS_SW	Lo Side Sw	
	18	I-D-L	J1-18	NC	Lo Side Sw	
	19	I-D-L	J1-19	Rear_Door_ajar_SW	Lo Side Sw	
	20	I-D-L	J1-20	CDL_UNLOCK_SW	Lo Side Sw	
	21	I-D-L	J1-21	CDL_LOCK_SW	Lo Side Sw	
	22	I-D-L	J1-22	NC	Lo Side Sw	
	23	I-D-L	J1-23	NC	Lo Side Sw	
	24	I-D-L	J1-24	NC	Lo Side Sw	
	25	I-D-L	J1-25	HAZARD_SW	Lo Side Sw	
	26	O-D-L	J1-26	NC	Low_ExRelay_output	/
	27	I-D-L	J1-27	NC	Lo Side Sw	/
	28	I-D-PWM	J1-28	CRASH_SIGNAL (Reserved)	PWM	

29	I-D-L	J1-29	HS_RESERVE_SW3	Hi Side Sw	
30	I-D-H	J1-30	BRAKE_PEDAL_SW	Hi Side Sw	
31	I-D-H	J1-31	ON_SIGNAL_INPUT	Hi Side Sw	
32	I-D-L	J1-32	NC	Lo Side Sw	
33	I-D-L	J1-33	NC	Lo Side Sw	
34	I-D-L	J1-34	NC	Lo Side Sw	
35	I-D-L	J1-35	NC	Lo Side Sw	
36	I-D-L	J1-36	HORN_SW	Lo Side Sw	
37	O-D-H	J1-37	NC	High_MOS_output	/
38	O-D-H	J1-38	NC	High_MOS_output	/
39	CAN	J1-39	CAN1_H	Comm	
40	I-D-L	J1-40	NC	Lo Side Sw	
41	I-D-L	J1-41	NC	Lo Side Sw	
42	I-D-L	J1-42	NC	Lo Side Sw	/
43	/	J1-43	/	/	
44	SGND	J1-44	SGND	GND	
45	/	J1-45	/		
46	/	J1-46	/		
47	O-D-L	J1-47	NC	Low_MOS_output	
48	I-D-H	J1-48	IGN_Relay_FB	Hi Side Sw	
49	I-D-H	J1-49	ACC_Relay_FB	Hi Side Sw	
50	I-D-H	J1-50	ACC_SIGNAL_INPUT /	Hi Side Sw	
51	I-D-H	J1-51	START_SIGNAL_INPUT	Hi Side Sw	
52	CAN	J1-52	CAN1_L	Comm	

1.1.10.3 J2 Pin Assignment

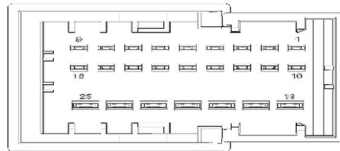


BCM plug-in pin view

Socket Number	Pin Number	I/O Type	Clamp Definition	Signal Definition	Drive Definition - HighSide - LowSide - int. Relay - PWM - ...	Typical/ Nominal Current [A]
J2	1	/	J2-01	/		
	2	I-D-L	J2-02	NC	Lo Side Sw	
	3	/	J2-03	/		
	4	/	J2-04	/		
	5	/	J2-05	/		
	6	/	J2-06	/		
	7	SGND	J2-07	SGND	GND	
	8	CAN	J2-08	NC	Comm	
	9	CAN	J2-09	NC	Comm	
	10	LIN	J2-10	LIN1	Comm	
	11	LIN	J2-11	NC	Comm	
	12	O-D-H	J2-12	NC	PWM_output	/
	13	O-D-H	J2-13	CDL_STATUS_LED_OUTPUT	Hi_LED_output	0.02A
	14	O-D-H	J2-14	HAZARD_LED_OUTPUT	Hi_LED_output	0.02A
	15	COM	J2-15	NC	Comm	
	16	COM	J2-16	NC	Comm	
	17	/	J2-17	/		
	18	/	J2-18	/		
	19	/	J2-19	/		
	20	/	J2-20	/		
	21	/	J2-21	/		
	22	/	J2-22	/		
	23	/	J2-23	/		
	24	/	J2-24	/		
	25	/	J2-25	/		
	26	SGND	J2-26	SGND	GND	1A
	27	/	J2-27	/		
	28	/	J2-28	/	/	
	29	COM	J2-29	NC	Comm	

	30	COM	J2-30	NC	Comm	
	31	COM	J2-31	NC	Comm	
	32	COM	J2-32	NC	Comm	

1.1.10.4.J3 Pin Assignment

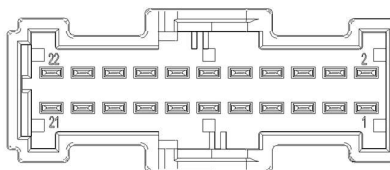


BCM plug-in pin view

Socket Number	Pin Number	I/O Type	Clamp Definition	Signal Definition	Drive Definition - HighSide - LowSide - int. Relay - PWM - ...	Typical/ Nominal Current [A]
J3	1	O-D-H	J3-01	LEFT_FLASHER_OUTPUT	High_MOS_output	21W+21W
	2	POWER	J3-02	TURN_LIGHTS_PWR	PWR	8A
	3	O-D-PWM	J3-03	Backlight_Adjust_OUTPUT	High_MOS_output	6MA
	4	O-D-H	J3-04	NC	High_MOS_output	NA
	5	PGND	J3-05	PGND	GND	10A
	6	O-D-H	J3-06	STOP_LAMP_OUTPUT	High_MOS_output	5W+5W
	7	O-D-H	J3-07	LICENSE_PLATE_LAMP_OUTPUT	High_MOS_output	2*0.5W
	8	O-D-H	J3-08	REVERSE_LAMP_OUTPUT	High_MOS_output	21W+21W
	9	O-D-H	J3-09	POSITION_LAMP_OUTPUT	High_MOS_output	5W+5W+21W+21W
	10	O-D-H	J3-10	RIGHT_FLASHER_OUTPUT	High_MOS_output	21W+21W
	11	O-D-H	J3-11	NC	High_MOS_output	/
	12	O-D-H	J3-12	IGN1_RLY_OUTPUT	Hi_ExRelay_output	0.17A

	13	O-D-H	J3-13	ACC_RLY_OUTPUT	Hi_ExRelay_out put	0.17A
	14	O-D-H	J3-14	NC	Hi_ExRelay_out put	0.17A
	15	O-D-H	J3-15	NC	Hi_ExRelay_out put	0.17A
	16	/	J3-16	/		
	17	/	J3-17	/		
	18	POWER	J3-18	STOP_LAMP_PWR	PWR	
	19	O-D-H	J3-19	NC	High_MOS_outp ut	/
	20	O-D-H	J3-20	Right_LOW_BEAM_OUTPUT	High_MOS_outp ut	55W
	21	O-D-H	J3-21	LEFT_LOW_BEAM_OUTPUT	High_MOS_outp ut	55W
	22	PGND	J3-22	PGND	GND	
	23	O-D-H	J3-23	Right_HIGH_BEAM_OUTPUT	High_MOS_outp ut	8W（电磁阀反向 电压 50V）
	24	O-D-H	J3-24	LEFT_HIGH_BEAM_OUTPUT	High_MOS_outp ut	8W（电磁阀反向 电压 50V）
	25	POWER	J3-25	BEAM_LAMP_PWR	PWR	

1.1.10.5 J4 Pin Assignment



BCM plug-in pin view

Socket Number	Pin Number	I/O Type	Clamp Definition	Signal Definition	Drive Definition - HighSide - LowSide - int. Relay - PWM - ...	Typical/ Nominal Current [A]
J4	1	POWER	J4-01	CDL_DOOR_PWR	PWR	
	2	O-D-H	J4-02	DOOR_UNLOCK_OUTPUT	High_Relay_output	0.1A*3
	3	O-D-PWM	J4-03	Dimming_Dome_LAMP_OUTPU T	Low_MOS_output	2W
	4	O-D-H	J4-04	DOOR_LOCK_OUTPUT	High_Relay_output	0.1A*3
	5	O-D-L	J4-05	NC	Low_ExRelay_output t	
	6	O-D-H	J4-06	DR_DOOR_UNLOCK_OUTPUT	High_Relay_output	0.1A
	7	O-D-L	J4-07	NC	Low_ExRelay_output t	/
	8	PGND	J4-08	CDL_DOOR_GND	GND	
	9	O-D-H	J4-09	FR_WASHER_OUTPUT	High_Relay_output	3A
	10	O-D-H	J4-10	NC	High_Relay_output	
	11	POWER	J4-11	FR_WIPER_PWR	PWR	20A
	12	PGND	J4-12	FL_WINDOW_PWR	PWR	20A
	13	/	J4-13	/		
	14	O-D-H	J4-14	FL_WIN_UP_OUTPUT	High_Relay_output	10A
	15	PGND	J4-15	FR_WIPR_GND	GND	
	16	O-D-H	J4-16	FL_WIN_DN_OUTPUT	High_Relay_output	10A
	17	O-D-H	J4-17	FR_WIPER_HI_OUTPUT	High_Relay_output	4.5A
	18	PGND	J4-18	PGND	PGND	
	19	O-D-H	J4-19	FR_WIPER_LO_OUTPUT	High_Relay_output	4.5A
	20	O-D-H	J4-20	FR_WIN_UP_OUTPUT	High_Relay_output	10A
	21	POWER	J4-21	FR_WINDOW_PWR	PWR	
	22	O-D-H	J4-22	FR_WIN_DN_OUTPUT	High_Relay_output	10A

1.1.11 Component Connected to Terminal 30 连接至端子 30 的组件

The BCM shall connect to Cl.30 for main power supply ; At the same time, BCM shall have can network sleep wake-up management function. BCM 应连接 Cl.30 作为主电源；同时，BCM 应具备网络休眠唤醒管理功能。

g Specific Characteristics 具体特征

N/A

h Safety Requirements 安全要求

N/A

1.1.12 Person and Passenger Protection 人员和乘客保护

The supplier is responsible to actively support the integration of vehicle safety functions. 供应商有责任积极支持车辆安全功能的集成。

Lifetime relevant laws and ratings of selling markets as well as OEM requirements must be considered.

必须考虑销售市场的终身相关法律和评级以及原始设备制造商的要求。

Vehicle system reliable operation is achieved by all sub-component integration. VinFast in partnership with suppliers guarantee the vehicle's intended operating function is free of electronic malfunction by following the process outlined in ISO26262. 车辆系统的可靠运行是通过所有子组件的集成来实现的。VinFast 与供应商合作，按照 ISO26262 规定的流程，确保车辆的预期运行功能不会出现电子故障。

The BCM submodule of this unit has been reviewed as being associated with ASIL-B classified risks as per ISO26262. Supplier is to conduct assessments as needed including but not limited to FTA, DFMEA, PFMEA, STPA, HAZOP etc. It is expected that functional safety risks should be reviewed with actions identified or confirmed in design (including inspection of hardware and software design principles) and manufacturing (manufacturing processes, validation of design processes, and verification of conformance of production processes). 根据 ISO26262，本单元的业连管子模块已被审查为与 ASIL-B 级风险相关。供应商应根据需要进行评估，包括但不限于 FTA、DFMEA、PFMEA、STPA、HAZOP 等。预计功能安全风险应与设计（包括硬件和软件设计原则检查）和制造（制造流程、设计流程验证和生产流程一致性验证）中确定或确认的行动一起进行审查。

1.1.13 Vehicle Safety 汽车安全

N/A

i Alternative and Future Variants 替代方案和未来方案

N/A

k Mechanical Energy Management 机械能量管理

Static current $\leq 3.5\text{mA}$

1.1.14 Weight Targets 重量目标

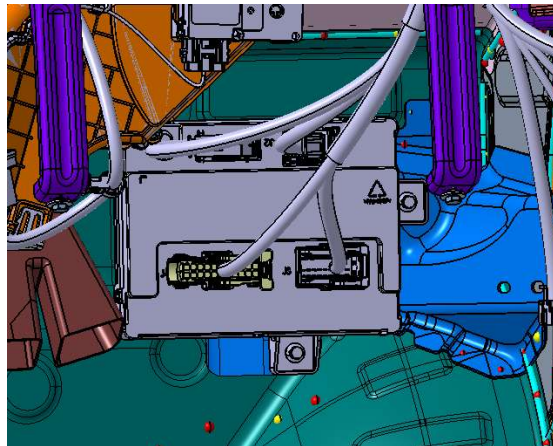
Weight 450g

1.1.15 Fuel Consumption & Driving Performance 燃料消耗与驾驶性能

N/A

1.2 Installation 安装

1.2.1 Installation Location 安装位置



1.2.2 Assembly Concept and Requirements from Production 装配概念和生产要求

Installation of the unit in any of their normal vehicle mounting applications (within design tolerances) shall not distort them or cause a change in any operating characteristics such that they would not meet the requirements of this standard. 在任何正常的车辆安装应用中（在设计公差范围内），安装该装置不得使其变形或导致任何操作特性的改变，从而使其不符合本标准的要求。

Ensure ergonomic access to all commands. Ensure visibility cone of all keys, dashboard and display ideograms according to the eye - points and the vehicle illumination requirements. 确保所有指令的使用都符合人体工程学。根据视点和车辆照明要求，确保所有按键、仪表盘和显示表意图形的可视锥度。

The specific pin-out of the connection between the unit and the vehicle system identified in this standard shall be collectively developed by VinFast and the Supplier of the unit. The connection will be designed in such a way as to optimize the layout of the overall system. 本标准中确定的设备

与车辆系统之间的具体连接引脚应由 VinFast 和设备供应商共同制定。连接的设计应优化整个系统的布局。

1.2.3 Geometry 几何结构

1.2.4 Tolerances 公差

See 2D Drawings 见 2D 图纸

1.3 Styling and Design 造型设计

N/A

1.4 Ergonomics 人体工学

1.4.1 Optical Characteristic and Surface Feel 光学特性和表面触感

Outline dimension of the unit housing and cover shall conform to the applicable VinFast and Supplier assembly drawings. 设备外壳和盖子的外形尺寸应符合适用的 VinFast 和供应商装配图。

- Surfaces must be free of sharp edges, burrs, etc. to protect wiring assembly and service personnel.

表面必须没有尖锐边缘、毛刺等，以保护接线装配和维修人员。

- Shall meet all other dimensional and finish requirements identified in the performance standard and on offer drawing. 应符合性能标准和报价图纸中确定的所有其他尺寸和表面处理要求
- Graphics shall match master letter pattern provided by VinFast Engineering and Design office. 图案应与 VinFast 工程设计办公室提供的字母图案相匹配。

Reference standard GB11552-2009 参考标准 GB11552-2009

1.4.2 Acoustics 声音

N/A

1.4.3 Handling 处理

Shall meet all other dimensional and finish requirements identified in the performance standard and on offer drawing. 应符合性能标准和报价图纸中确定的所有其他尺寸和表面处理要求。

1.5 Technical Material Requirements 技术材料要求

See 2D Drawings

1.6 Resistance to Media and Chemical Requirements 耐介质和化学要求

1.6.1 Contamination and Cleaning 污染与清洁

The component shall be able to maintain functionality in surrounding air (clean and polluted) and in its application on vehicle, during its entire lifetime.

The operation, function and surface of the component must not be adversely affected by accidental liquid splash or spray from liquids used in the cab when mounted in the vehicle. Any type of detergent agents or liquids used in vehicle must, not adversely affect the operation, function or surface of the component when used in vehicle. 安装在车辆上时，部件的操作、功能和表面不得受到驾驶室内使用的液体意外飞溅或喷射的不利影响。车内使用的任何类型的清洁剂或液体都不得对车内组件的操作、功能或表面造成不利影响。

1.6.2 Corrosion Protection 防腐保护

Generally 一般而言：

VFDSAAEEP0202-Requirements for prohibited substances on automobiles

VFDSAAEEP0202-对汽车禁用物质的要求

For determination of corrosion requirements, the corrosivity position needs to be classified for each component in the system 为确定腐蚀要求，需要对系统中每个部件的腐蚀性位置进行分类：

1. Corrosivity Low: Used when the position is not affected by macro wetness, mud/ dirt, road (deicing) salts 低腐蚀性：用于不受宏观潮湿、泥浆/污垢、道路（除冰）盐分影响的位置。.
2. Corrosivity Medium: Used when the position is at most moderately affected by macro wetness, mud/ dirt, road (deicing) salts 中等腐蚀性：用于受宏观湿度、泥浆/污垢、道路（除冰）盐类影响最轻微的位置。.
3. High corrosivity: Default for all exterior positions directly affected by road spray.

高腐蚀性：默认用于所有直接受路面喷雾影响的外部位置。

4. Very high corrosivity: Default for all positions directly affected by road spray, stone chipping and dirt accumulation. 腐蚀性极强：默认用于直接受路面喷洒物、碎石和污垢堆积影响的所有位置

Electrically functional units in casings and boxes with enclosed air volumes, such as different types of control units shall be ventilated via a moisture/ wetness resistant membrane, when in wet locations. 在潮湿场所，应通过防潮/防湿膜对外壳和箱体中的电气功能单元（如不同类型的控制单元）进行通风

Material combinations shall be chosen so that galvanic corrosion does not occur. This includes also fasteners and attached brackets. 材料组合的选择应避免发生电化学腐蚀。这也包括紧固件和连接支架。

Avoid direct contact to more noble metals like steel, stainless steels, copper, brass, nickel and conductive or anti-static rubber in environments affected by wetness. Use surface treatment on steel fasteners etc. that are long term corrosion resistant and fully compatible to other materials. 在受潮湿影响的环境中，避免直接接触钢、不锈钢、铜、黄铜、镍等贵金属以及导电或防静电橡胶。钢制紧固件等的表面处理应具有长期耐腐蚀性，并与其他材料完全兼容。

The electrical components must be installed so that vibrations do not generate fretting corrosion. 电气元件的安装必须避免振动产生摩擦腐蚀。

Terminals, connectors: Don't place Terminals on lowest points. Corrosion protection of Terminals is first a requirement for tightness: Electrical connections must be protected from humidity. 端子、连接器：不要将端子放在最低点。端子的防腐蚀保护是密封性的首要条件：电气连接必须防潮。

Groundings: Connecting surfaces of grounding must be water-tight, if placed in a corrosive environment.

接地：如果放置在腐蚀性环境中，接地的连接面必须防水。

Corrosion tests 腐蚀试验：

The parts must pass System/component tests as well as a complete vehicle corrosion test. 部件必须通过系统/组件测试以及整车腐蚀测试

Before and after corrosion testing a functional testing is to be carried out. Before and after corrosion testing a functional testing is to be carried out. 在腐蚀测试之前和之后都要进行功能测试。在腐蚀测试之前和之后都要进行功能测试。

1.6.3 Tightness and Protection Classes 密封性和防护等级

If possible or available, the specific stipulations of protection classes that the component has to fulfil in terms of certain characteristics (e.g. contact and dust protection, splash water, surface, and IP classes) are to take place. 如果可能或有条件，则应根据组件的某些特性（如接触和防尘、溅水、表面和 IP 等级），对其保护等级做出具体规定。

Refer to DVP 参考 DVP

1.7 Environmental Compatibility 环境兼容性

1.7.1 Material Selection 选材

The following materials shall be avoided since there is a risk, they can impair the functionality of the component itself or vehicle. 应避免使用下列材料，因为它们有可能损害部件本身或车辆的功能。

Sulphur free rubber requirement 无硫橡胶要求

No sulfur cured rubber material is allowed inside of the component. Use peroxide cured material or silicone rubber materials. This requirement also applies to electrolytic capacitors rubber sealing. 组件内部不得使用硫磺硫化橡胶材料。请使用过氧化物固化材料或硅橡胶材料。这一要求也适用于电解电容器的橡胶密封

Silicone free requirement 无硅要求

The use of silicone as cleaning agent, release agent, lubricant, paste, adhesive, inside dampers or any other non-cross-linked or uncured silicon where risk of leakage of volatile silicone to surroundings occur is not permitted. 不允许将硅胶用作清洁剂、脱模剂、润滑剂、膏状物、粘合剂、风门内部或任何其他非交联或未固化的硅胶，以免挥发性硅胶泄漏到周围环境中。

In general, cross-linked silicon such as silicone rubber shall not emit any low molecular agents that can be harmful or have any functional impact for electrical appliances or in paint shop. 一般来说，硅橡胶等交联硅不会释放出对电器或油漆车间有害或对功能有任何影响的任何低分子物质。

There are no “safe” silicone materials regarding electrical arcing failures. On units with electrical functionality including mechanical switches, micro switches and relays controlling inductive loads there shall be no silicone material used in the design. This includes silicone oil, silicone grease, and silicone rubber parts. The evaporation of silicone molecules on surfaces that can see electrical arcing will cause open circuit failures over time. 对于电弧故障，没有“安全”的硅胶材料。对于具有电气功能的设备，包括机械开关、微动开关和控制电感负载的继电器，在设计中不得使用硅胶材料。这包括硅油、硅脂和硅橡胶部件。硅胶分子在可产生电弧的表面上蒸发，时间一长就会造成开路故障。

Lead-Free requirements 无铅要求

Electrical and Electronic Assemblies shall have a lead-free design. 电气和电子组件应采用无铅设计。

1.7.2 Recycling Scheme 循环利用计划

Dismantling scheme 拆卸方案

The dismantling scheme describes the requirements to the contractor in terms of a simple, unmixed dismantling of the component as well as its accessibility when assembled. 拆卸计划说明了对承包商的要求，即简单、无混杂地拆卸部件以及组装后的可接近性。

Dismantling scheme has to be provided to environmental expert's team at VF before milestone PTO Production Try-Out. 在里程碑式的 PTO 试生产之前，必须向 VF 的环境专家小组提供拆卸方案。

Recycling scheme 循环利用计划

This chapter establishes the general project-wide requirements on the recycling and repair scheme in the development of a component. 本章规定了整个项目在组件开发过程中对回收和维修计划的一般要求。

Design for recycling of parts and components are required in order to enable cost-effective recycling at end-of life. 需要对零部件进行回收设计，以便在报废时进行具有成本效益的回收。

The recycling scheme provided by the contractor has to show a simple layout for the possible recycling process of the component including information about preferred recycling partners and cost estimation.

承包商提供的回收计划必须显示部件可能回收过程的简单布局，包括首选回收合作伙伴的信息和成本估算。

Recycling scheme has to be provided to environmental expert's team at VF before milestone PTO Production Try-Out. 在里程碑式的 PTO 生产试运行之前，必须向 VF 的环境专家团队提供回收计划。

1.7.3 Emissions & Toxicology 排放与毒理学

Harmful substances in components and materials

The emission behavior of the E/E product with direct contact to interior compartment has to be checked on representative components as soon as possible. Emissions of volatile organic compounds in the passenger compartment have to be minimized accordingly. Required measurements for confirmation of the emission characteristics have to be done by the supplier in agreement with the specialist department of VF. The measurements have to be done at testing institutes approved by VF. It is recommended to obtain an overview of all approved testing institutes from VF. VFDSAAEEP0202-Requirements for prohibited substances on automobiles 必须尽快在有代表性的部件上检查直接接触车厢的 E/E 产品的排放情况。必须相应地尽量减少乘客舱内挥发性有机化合物的排放。确认排放特性所需的测量必须由供应商与 VF 的专业部门协商完成。测量必须在 VF 认可的测试机构进行。建议向 VF 索取所有经批准的测试机构的概览。

VFDSAAEEP0202-汽车禁用物质要求

1.8 Mechanical Requirements 机械要求

1.8.1 Load 负载

Define load requirement as DVP or related standard

负载将负载要求定义为 DVP 或相关标准

1.8.2 Vibration Behaviour 振动行为

Meet relevant marking requirements. 符合相关标识要求。

1.8.3 Deformation 变形

The component must be stable over time. No deformation or shrinkage over time, due to temperature, moisture, dirt, UV radiation allowed. 部件必须长期稳定。不允许因温度、湿度、灰尘、紫外线辐射而变形或收缩。

1.8.4 NVH & Acoustic Performance NVH 和声学性能

The assembly shall be installed and fixed reliably to prevent noise caused by looseness. 组件的安装和固定必须可靠，以防止因松动而产生噪音。

1.9 Service Life 使用寿命

3 years or 100,000 km, whichever comes first. 3 年或 10 万公里，以先到者为准。

1.10 Electrical Requirements 电气要求

1.10.1 Description of the Requirements 要求说明

ECU Components shall meet the requirements of the Technical Specification for the specific vehicle(s) that may include the following ECU 组件应符合特定车辆的技术规范要求，其中可能包括以下内容：

- Component Electrical Technical Specification 组件电气技术规格
- Component Functional Description 组件功能说明
- Network Transmission Matrix 网络传输矩阵
- CAN bus communications requirements CAN 总线通信要求
- Diagnostic requirements 诊断要求
- Other requirements documents 其他要求文件

1.10.2 Power Consumption 功耗

Details Power Consumption 详细信息 功耗

1.10.3 Electromagnetic Compatibility 电磁兼容性

Refer to DVP specification 参见 DVP 规格

Follows standards 按照标准

Name	Standard
EMC Requirements and Testing for Component_V3	VFDST00026307

1.10.4 Electrostatic Discharge 静电放电

Refer to DVP specification 参见 DVP 规格

1.10.5 Power Consumption 功耗

Please refer to VFDSP50000005_Energy_Management [REF_07] and VFDSP50000004_General_Electric [REF_08] in the latest version. 请参阅最新版本的 VFDSP50000005_能源管理 [REF_07] 和 VFDSP50000004_通用电气 [REF_08]。

1.10.6 Low Voltage Energy Management 低电压能源管理

Refer Properties, Condition and Durability 参考特性、状况和耐用性

1.10.7 High Voltage Energy Management 高压能源管理

Refer Properties, Condition and Durability 参考特性、状况和耐用性

1.11 Climatic Requirements 气候要求

The ambient temperature, relative humidity and altitude shall meet the vehicle performance operating environment. 环境温度、相对湿度和海拔高度应满足车辆性能运行环境的要求。

The development and validation of the heat protection in the complete vehicle ensures that all relevant component temperatures remain below the maximum allowed limits which are derived from vehicle operation within the environmental operating temperatures. 整车防热系统的开发和验证可确保所有相关部件的温度保持在最大允许温度以下，而最大允许温度是根据车辆在环境工作温度下的运行情况得出的。

The heat protection validation tests for the components are performed in the climatic wind tunnel (CWT) and on-road. 部件的热保护验证测试在气候风洞（CWT）和道路上进行。

All relevant component temperatures and measurements points must be defined in accordance with the responsible modules and will be specified in respective component specifications. Vehicle tests must be completed and followed VF Thermal Testing Standard in the MULE, Super-MULE and PTO Vehicle Generation. 所有相关部件的温度和测量点必须根据负责模块进行定义，并在相应的部件规格中进行说明。车辆测试必须完成，并遵循 MULE、Super-MULE 和 PTO 车辆系列中的 VF 热测试标准。

Please refer to CTS_VINFAST_EE_VFDSP50000009_Thermal_Management [REF_09] in the latest version.

请参阅最新版本的 CTS_VINFAST_EE_VFDSP50000009_热管理 [REF_09]。

1.12 Service Requirements 服务要求

Support Aftersales and is serviceable, with easy installation and back-up battery replacement

支持售后服务，易于安装和更换备用电池

1.13 Transport Protection 运输保护

Should meet Guidelines for transport package 应符合运输包件准则

1.14 Logistics Requirements 后勤要求

N/A

1.15 Quality Assurance Requirements 质量保证要求

Meet the requirements of relevant standards. 符合相关标准的要求。

LV battery top-up strategy 低压电池补充战略

This function remains a high energy state for the LV battery in different circumstances, thus maintaining the longevity of the LV battery and the ability to start at low temperatures or when the battery is degraded. 该功能可在不同情况下保持低压蓄电池的高能量状态，从而保持低压蓄电池的使用寿命以及在低温或蓄电池性能下降时启动的能力。

6 Testing and Validation and DVP&R 测试和验证以及 DVP&R

DVP and PC test responsibility follow RASIC. DVP (virtual and physical testing) for System, Sub-system & Component Level inline with Complete Vehicle Specification book in the concept phase. Detailed test plan for hardware tests in file DVP&R and need work with Testing & Validation department: DVP 和 PC 测试责任遵循 RASIC。系统、子系统和组件级的 DVP（虚拟和物理测试）与概念阶段的整车规格书保持一致。在 DVP&R 文件中提供硬件测试的详细测试计划，并需要与测试和验证部门合作：

- Testing with declaration of standard, norm, legal requirement 根据标准、规范和法律要求进行测试
- How and with which devices the testing was performed (description of testing set-up with pictures)

测试是如何进行的，使用了哪些设备（测试装置说明，附图片

- Where the testing was performed (internal/external) 进行测试的地点（内部/外部）
- Duration and time flow of the testing (in form of project plan) 测试的持续时间和时间流程（以项目计划的形式）
- Part variant, description of part 部件变体，部件描述
- No. of tested components/parts 测试组件/部件数量

Ensure Mileage Target Life: 160000 km or 7 years (whichever comes first) 确保里程目标寿命：160000 公里或 7 年（以先到者为准）

1.16 Test Equipment and Test Mules 测试设备和测试骡子

Here the customer is to define the test equipment and facilities to be used and establish the possible provision for the contractor (e.g. test mules [engines, vehicles], cable harnesses, plugs, add-on parts, wear and tear parts). 在此，客户应确定要使用的测试设备和设施，并确定可能为承包商提供的设备（如测试骡子[发动机、车辆]、电缆线束、插头、附加部件、损耗部件）。

1.17 Proof of Compliance 合规证明

In this chapter, binding regulations are to be made as to which proof of compliance through tests carried out is to be provided by the contractor. The information given here should only define the universal requirements (minimum data scope of the test documentation, proof of compliance, deadlines, customer's departments that are to be involved, etc). Specific proof of compliance (e.g. separate parameters) should be dealt with in the respective chapters. 在本章中，将对承包商通过测试提供的合规性证明做出有约束力的规定。此处提供的信息应仅定义通用要求（测试文件的最小数据范围、符合性证明、截止日期、应涉及的客户部门等）。具体的合规性证明（如单独参数）应在相应章节中阐述。

1.18 Testing Plan 测试计划

Refer to DVP plan and FRS plan and SOR Sheet DVP 参见 DVP 计划和 FRS 计划以及 SOR 表 DVP

It is necessary to have DVPR template so that suppliers have to follow 有必要制定 DVPR 模板，以便供应商遵循：

Testing plan, named DVPR (Design Validation Plan and Report), shall be prepared well by supplier in reasonable with overall project timeline. This DVPR should be reviewed by Vinfast EE Engineering prior to upload on Team center system for official releasing. 测试计划应由供应商根据整个项目的时间表编制，并命名为 DVPR（设计验证计划和报告）。在上传到团队中心系统正式发布之前，该 DVPR 应由 Vinfast EE Engineering 审查。

DVPR's content including test report have to follow a template which had been defined by Vinfast and released on Teamcenter as document number "VFDST00000005/03-DVP_DVP&R template"包括测试报告在内的 DVPR 内容必须遵循 Vinfast 定义并在 Teamcenter 上发布的模板，文件编号为“VFDST00000005/03-DVP_DVP&R 模板”。

Validation items shall be included but not limit to Electromagnetic, Electrical and Environment testing in order to verify whether functions work properly and operation of safety, stability and reliability. The service life test should be carried out in order to verify whether component/system work normally in durability of 7 years or 160,000 Km (which ever come first)

验证项目应包括但不限于电磁、电气和环境测试，以验证功能是否正常工作以及运行的安全性、稳定性和可靠性。应进行使用寿命测试，以验证组件/系统在 7 年或 160,000 Km（以先到者为准）的耐久性内是否正常工作。

Refer to DVPR number DVPRADEEP0006 for detail test plan as well as validation items 有关详细测试计划和验证项目，请参阅编号为 DVPRADEEP0006 的 DVPR。

1.18.1 Electrical Systems & Functions 电气系统和功能

Several validation cycles for electrical systems and functions are planned to guarantee correct implementation and will be carried out according to mentioned standards. 为确保正确实施，计划对电气系统和功能进行若干次验证，并将根据上述标准进行验证。

It is understand that the validation items should be agreed by both parties and will be carried out in accordance with plan as stated in DVPR. 据了解，验证项目应经双方同意，并将按照 DVPR 所述计划执行。

1.18.2 Electromagnetic Compatibility (EMC) requirements and tests 电磁兼容性 (EMC) 要求和测试:

'Electromagnetic compatibility (EMC) testing measures the ability of equipment or systems to function satisfactorily in their electromagnetic environment without introducing intolerable electromagnetic disturbance to anything in that environment. 电磁兼容性 (EMC) 测试测量设备或系统在其电磁环境中令人满意地运行，而不对环境中的任何事物造成不可容忍的电磁干扰的能力。

It is understand that the validation items should be agreed by both parties and will be carried out in accordance with plan as stated in DVPR. 据了解，验证项目应经双方同意，并将按照 DVPR 所述计划执行

1.18.3 Environmental requirements and tests: 环境要求和试验

Environmental test applies to electric and electronic systems/component for vehicles. Device shall be tested under potential environmental stress and specifies test and requirements recommended for specific mounting location on/in the vehicle. 环境试验适用于车辆的电气和电子系统/组件。设备应在潜在的环境压力下进行测试，并规定针对车辆特定安装位置建议的测试和要求。

It is understand that the validation items should be agreed by both parties and will be carried out in accordance with plan as stated in DVPR. 据了解，验证项目应经双方同意，并将按照 DVPR 所述计划执行。

1.19 Tests 试验

A method for how the test results are to be determined should be defined for all relevant characteristics. Furthermore, the number of parts to be tested, the frequency of the tests and who is responsible for testing is to be established. By means of which criteria and what the limits are for a part to be evaluated as meeting the specification are also to be established. 应针对所有相关特性确定如何确定测试结果的方法。此外，还应确定需要测试的部件数量、测试频率以及测试责任人。还应确定通过何种标准和限制来评估零件是否符合规范。

1.20 Test Parameters and Cycles 试验参数和周期

Establish all relevant environmental factors (temperature, pressure, operating condition of the component to be tested, type and angle of the force effect, etc.), the results to be determined (deformation, absorbed forces, etc.), and other relevant data (cycle progression, number of cycles, etc). 确定所有相关的环境因素（温度、压力、待测部件的工作条件、力作用的类型和角度等）、待测结果（变形、吸收力等）以及其他相关数据（循环进度、循环次数等）。

1.21 Properties and Condition of the Prototype Status to be Tested 待测原型状态的特性和条件

Define the properties and condition of the components or parts to be tested (if necessary with reference to standardised prototypes statuses in the CRS Module I). 确定要测试的组件或部件的属性和状态（必要时参考 CRS 模块 I 中的标准原型状态）。

1.22 Operating Conditions 运行条件

If the component has different operating conditions (e.g. normal operation and emergency operation), the requirements that exist for the intentional bringing about of an operating condition in order to be able to carry out certain tests at all (e.g. in a service garage) are to be listed here. Which requirements are to be tested in which operating condition are also to be listed here. 如果部件有不同的工作条件（如正常工作和紧急工作），则应在此列出为能进行某些测试而有意造成工作条件的要求（如在维修车间）。哪些要求在哪种运行状态下进行测试，也应在此列出

1.23 Virtual Testing and Simulation 虚拟测试和模拟

The requirements concerning type and scope of virtual tests and simulations are to be defined in this chapter. If the customer provides the data and models which allow the contractor to, for example, represent the system environment during simulations, this is also to be stated here. 关于虚拟试验和模拟的类型和范围的要求应在本章中确定。如果客户提供的数据和模型允许承包商在模拟过程中表现系统环境等，也应在此说明。

1.24 Vehicle Testing 车辆测试

In this chapter the specific requirements on the testing of the component in the vehicle are to be defined.

本章规定了车辆部件测试的具体要求。

1.25 Content Amendments 内容修订

If the given content structure does not suffice to assign all requirements on level 2 to the superordinated subject on level 1, the requirements can be assigned to this chapter. If this chapter is not needed, you should delete it! 如果给定的内容结构不足以将第 2 层的所有要求分配给第 1 层的上位主题，则可将这些要求分配给本章。如果不需要这一章，则应将其删除！

7 Function Safety, Homologation and safety regulations 功能 安全、认证和安全法规

Safety is one of the key issues in the development and integration of automotive functionalities. 安全是汽车功能开发和集成的关键问题之一。

Functional safety is concerned with E/E systems. 功能安全涉及 E/E 系统

1.26 Functional Safety Requirements 功能安全要求

N/A

1.27 Deliverable for Functional Safety 功能安全交付品

N/A

1.28 Homologation and safety regulation requirements 认证和安全规定要求

-- This component shall ensure that related sub system and complete vehicle will comply with destination country's regulation 这部分应确保相关子系统和整车符合目的地国家的规定

1.28.1 Vietnam 越南

Vietnam market 越南市场

Vietnam 越南:

a. Homologation and safety regulation activities 认证和安全监管活动

The Supplier must ensure that their part(s), product(s) or system(s) comply with applicable regulations in Vietnam 供应商必须确保其部件、产品或系统符合越南的适用法规

b. Test report(s) and Certificate(s) 测试报告和证书

For part (s), system (s): The suppliers are responsible for providing test samples and test reports that meet Vietnam regulation. 部件、系统: 供应商负责提供符合越南规定的测试样品和测试报告。

VinFast also request Supplier must provide supporting documentation, technical information, including drawings to be submitted to the relevant authorities in order to get certification for the vehicle type-approval (if any). VinFast 还要求供应商必须提供证明文件、技术信息, 包括提交给相关当局的图纸, 以便获得车辆类型批准认证 (如有)

c. Duty to notify VinFast of a non-compliance with any Vietnam regulations 通知 VinFast 不遵守越南任何法规的义务

Notwithstanding its certification of a product, a supplier may subsequently determine that a non-compliance with Vietnam regulations or a safety-related defect exists in a motor vehicle equipment item it has produced.

The Supplier has a duty to notify VinFast on such issue within five working days after determining the existence of a non-compliance or a safety-related defect. 尽管供应商对产品进行了认证, 但其随后可能会确定其生产的机动车辆设备项目存在不符合越南法规的情况或与安全相关的缺陷。

供应商有义务在确定存在不合规或与安全相关的缺陷后五个工作日内就该问题通知 VinFast

The Supplier must propose appropriate risk assessment and resolution plan. 供应商必须提出适当的风险评估和解决计划。

d. With telecommunication/ short range device: ICT certificate is mandatory

配备电信/短程设备: 必须持有 ICT 证书

1.29 Privacy Requirement 隐私要求

N/A

1.30 Security Requirement 安全要求

N/A

8 List of Referencing Documents 参考文件清单

Req 123471	ID:	Type: Information	ASIL: --	Modified date: 17-08-2021 17:23	Status: New
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ID	Name	Version	Source
REF_01	CES_VINFAST_EE_VFDSXVEEP0047_CAN_Specification-V2.0.0	V2.0.0	CES_VINFAST_EE_VFDSXVEEP0047_CAN_Specification-V2.0.0
REF_02	CES_VINFAST_EE_VFDSP50000012_LIN_Specification_V3.1.2	V3.1.2	CES_VINFAST_EE_VFDSP50000012_LIN_Specification_V3.1.2
REF_03	VFDSP50000017 Coding Specification_V1.0.4	V1.0.4	VFDSP50000017 Coding Specification_V1.0.4
REF_04	VFDSVAEEP0003_Flash_Specification_V3.0	V3.0	VFDSVAEEP0003_Flash_Specification_V3.0
REF_05	CES_VINFAST_EE_VFDSP50000020_Security_Access_Algorithm_V1.0.3	V1.0.3	CES_VINFAST_EE_VFDSP50000020_Security_Access_Algorithm_V1.0.3
REF_06	VFDSVAEEP0002_Basic_DID_PID_Specification_V1.6	V1.6	VFDSVAEEP0002_Basic_DID_PID_Specification_V1.6
REF_07	VFCSMCSS00110-VinFast-Cybersecurity-Requirements- V3.0	V3.0	VFCSMCSS00110-VinFast-Cybersecurity-Requirements- V3.0
REF_08	VinFast Addendum Flash Specification for Cybersecurity_V1.1	V1.1	VinFast Addendum Flash Specification for Cybersecurity_v1.1
REF_09	VFDSVAEEP0001_Generic_Diagnostic_Specification_V1.7	V1.7	VFDSVAEEP0001_Generic_Diagnostic_Specification_V1.7
REF_10	VFDST00026307 EMC Requirements and Testing for Component_V3	V3	VFDST00026307 EMC Requirements and Testing for Component_V3
REF_11	VFDST00028121 Electrical and electronic components in motor vehicles - Electrical requirements and testings_V2	V2	VFDST00028121 Electrical and electronic components in motor vehicles - Electrical requirements and testings_V2
REF_12	VFDST00028131_Electrical and electronic components in vehicles - Environmental requirements and testing_V4	V4	VFDST00028131_Electrical and electronic components in vehicles - Environmental requirements and testing_V4

9Definitions and List of Abbreviations 定义和缩略语表

Req ID: 123511	Type: Folder	ASIL: --	Modified date: 09-04-2021 13:20	Status: New
Abbreviation		Explanation		
AA		Attack Angle		

ABG	Airbag 碰撞传感器
ABS	Anti-lock braking system 防抱死制动系统
AC	Alternating Current 交流电流
ACM	Airbag control module 安全气囊控制模块
ADAS	Advanced Driver Assistance Systems 高级驾驶员辅助系统
ADC	Analog to Digital Converter 模拟数字转换器
AI	Articulation Index 衔接索引
AIA	Activity Interface Agreement = Work Interface Chart (RASi) 活动界面协议 = 工作界面图 (RASi)
AKS	Aktiver Kurzschluss (Active short circuit) 活跃的短期主动短路)
APQP	Advanced Product Quality Planning 先进的产品质量规划
ASIL	Automotive Safety Integrity Level 汽车安全完整性等级
ASM	Asynchronous Machine 异步设备
AWD	All wheel drive 全轮驱动
BCM	Body control module 车身控制模块
BDU	Battery disconnect unit 电池断电装置
BEV	Battery Electric Vehicle 电池电动汽车
BFR	Basic Function Requirement 基本功能要求
BIW	Body in White 白车身
BMU	Battery management unit 电池管理单元
BOM	Bill Of Material 材料清单
BSM	Blind Spot Monitoring 盲点监控
CAD	Computer-Aided Design 计算机辅助设计
CAE	Computer-Aided Engineering 计算机辅助工程
CAM	Computer-Aided Manufacturing 计算机辅助制造
CAN	Controller Area Network 控制器局域网
MSC	Manufacturing Strategy Confirmation 生产策略确认
TS	Theme selection Model 主题选择 模式
MFC	Manufacturing Feasibility Confirmation 生产可行性确认
MSI	Major Suppliers Identified 已确定的主要供应商
SST	Sourcing Strategy 采购战略
LLS	Long Lead Suppliers on board 长线供应商加入

SLS	Short Lead Suppliers on board 短期供应商加入
LLT	Long Lead Tooling 长周期模具
PTP	Prototype Parts out of Serial tooling (Hand finished) 系列模具中的原型部件 (手工加工)
OTS	Off Tools Samples 关闭工具样品
ESO	Engineering Sign Off 工程签字
MSO	Manufacturing Sign Off 生产签字
SOP-LM	SOP Launch Model (1 specific Model -„Edition“) SOP 发射模型 (1 个特定模型--"版本)
SOP	Start of Production 生产开始
SOP-ID	SOP-ID...SOP implementation date SOP-ID...SOP 实施日期
SOB	Supplier on Board 加盟供应商
LOI	Letter of Intend 意向书
LON	Letter of Nominated 提名信
DVP&R	Design Verification Plan & Report 设计验证计划和报告
GTC	General Term and Condition 一般条款和条件
RFQ	Request For Quotation 要求报价
FUSA	Function Safety 功能安全
T&V	Testing & Validation 测试与验证